Supplementary material for: More Answers, or Better Thinking? What a Fixed Budget Buys Across Claude 5.5 Models on Synthetic Multi-Step Puzzles

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S1 About this supplement

This supplement accompanies the paper. It holds the full version of the paper's appendix, with every table in full, and the records an auditor needs: the run log, the pre-registration timeline, the data stamps' fields, the exact intervals that replace collapsed ones (Table S8), and every analysis file's SHA-256 and code commit. Nothing here changes a result, a test reading or a label in the paper, and the paper's own appendix is shorter, drawn from it, with a few notes for the paper's readers. Sections of the paper are cited as "Section N of the paper"; tables and figures are numbered within this supplement, and every table and figure the paper prints appears here in full, in Section S2 or in the section that discusses it.

S2 The paper's tables and figures, in full

Every table and figure the paper prints appears in this supplement in full, here or in the section that discusses it. This section holds Tables S1, S2 and S3, Figures S1, S2, S3, S4, S5, S6, S7 and S8, and Table S4; the paper's other tables are with the sections that discuss them.

Table S1: None of the three medium-effort designs in the top-up run came near one Opus-medium answer. The top-up run's settings and comparisons (exploratory‡, with no Holm correction and no verdict; 6 steps, 80 puzzles): the Haiku team, the Haiku picker and step voting on pieces. Their comparisons with main-run settings span the two runs, with the drift check between runs not fired; the Haiku team against step voting on pieces compares two top-up settings within one run. The main-run comparator rows that Study 1 also reports (Opus-medium, and step voting on chains) print Study 1's own values and are pre-specified descriptive. Accuracy intervals from a cluster bootstrap over puzzles (2,000 resamples), except those rows, which keep Study 1's intervals (exact Clopper–Pearson for Opus-medium); differences from a paired BCa bootstrap (10,000 resamples). Cost per puzzle at list price.

Setting	Shape	Run	Accuracy (%)	Accuracy 95% interval	Format compliance (%)	Readable picks (%)
Haiku team	chains (1×6)	top-up run	52.50	[41.25, 62.50]	97.50	
Haiku team	pieces (6×1)	top-up run	72.50	[62.50, 82.50]	100.00	
Haiku team	pooled	top-up run	62.50	[55.63, 68.75]		
Haiku picker over 8 samples	chains (1×6)	top-up run	62.50	[52.50, 72.50]		100.00
Haiku picker over 8 samples	pieces (6×1)	top-up run	67.50	[57.50, 78.75]		100.00
Haiku picker over 8 samples	pooled	top-up run	65.00	[57.50, 72.50]		
Step voting on pieces	pieces (6×1)	top-up run	51.25	[40.00, 62.50]		
One Haiku-medium sample (the first 8 samples)	chains (1×6)	main run, rebuilt from its log	43.59	[36.72, 50.78]		
One Haiku-medium sample (the first 8 samples)	pieces (6×1)	main run, rebuilt from its log	52.97	[45.78, 60.63]		
One Haiku-medium sample (the first 8 samples)	pooled	main run, rebuilt from its log	48.28	[42.81, 53.91]		
Haiku-medium vote of the first 8 samples	chains (1×6)	main run, rebuilt from its log	52.86	[42.24, 63.49]		
Haiku-medium vote of the first 8 samples	pieces (6×1)	main run, rebuilt from its log	65.42	[55.00, 75.00]		
Haiku-medium vote of the first 8 samples	pooled	main run, rebuilt from its log	59.14	[52.21, 65.73]		
Any of the first 8 Haiku-medium samples right (the ceiling for a picker)	chains (1×6)	main run, rebuilt from its log	87.50	[80.00, 95.00]		
Any of the first 8 Haiku-medium samples right (the ceiling for a picker)	pieces (6×1)	main run, rebuilt from its log	86.25	[78.75, 92.50]		
Any of the first 8 Haiku-medium samples right (the ceiling for a picker)	pooled	main run, rebuilt from its log	86.88	[81.25, 92.50]		
Opus-medium, main run	chains (1×6)	main run (Study 1)	96.25	[89.43, 99.22] (exact)	97.50	
Opus-medium, main run	pieces (6×1)	main run (Study 1)	100.00	[95.49, 100.00] (exact)	100.00	
Opus-medium, main run	pooled	main run, rebuilt from its log	98.13	[95.63, 100.00]		
Step voting, main run	chains (1×6)	main run (Study 1)	53.75	[42.50, 63.75]		

Table S1: (continued)

Setting	Shape	List cost per puzzle ($)	Extra list cost of the picker call alone ($)	List cost per correct answer ($)	Label
Haiku team	chains (1×6)	0.00623		0.0119	exploratory (no Holm, no verdict)
Haiku team	pieces (6×1)	0.00649		0.00895	exploratory (no Holm, no verdict)
Haiku team	pooled	0.00636			exploratory (no Holm, no verdict)
Haiku picker over 8 samples	chains (1×6)	0.0103	0.00154	0.0165	exploratory (no Holm, no verdict)
Haiku picker over 8 samples	pieces (6×1)	0.0101	0.00142	0.0150	exploratory (no Holm, no verdict)
Haiku picker over 8 samples	pooled	0.0102	0.00148		exploratory (no Holm, no verdict)
Step voting on pieces	pieces (6×1)	0.00351		0.00684	unadjusted, exploratory, outside the pre-registered family
One Haiku-medium sample (the first 8 samples)	chains (1×6)				comparator; exploratory (no Holm, no verdict)
One Haiku-medium sample (the first 8 samples)	pieces (6×1)				comparator; exploratory (no Holm, no verdict)
One Haiku-medium sample (the first 8 samples)	pooled				comparator; exploratory (no Holm, no verdict)
Haiku-medium vote of the first 8 samples	chains (1×6)	0.00879			comparator; exploratory (no Holm, no verdict)
Haiku-medium vote of the first 8 samples	pieces (6×1)	0.00873			comparator; exploratory (no Holm, no verdict)
Haiku-medium vote of the first 8 samples	pooled	0.00876			comparator; exploratory (no Holm, no verdict)
Any of the first 8 Haiku-medium samples right (the ceiling for a picker)	chains (1×6)				comparator; exploratory (no Holm, no verdict)
Any of the first 8 Haiku-medium samples right (the ceiling for a picker)	pieces (6×1)				comparator; exploratory (no Holm, no verdict)
Any of the first 8 Haiku-medium samples right (the ceiling for a picker)	pooled				comparator; exploratory (no Holm, no verdict)
Opus-medium, main run	chains (1×6)	0.0402		0.0418	comparator; pre-specified descriptive (Study 1, as Table S14 prints it)
Opus-medium, main run	pieces (6×1)	0.0481		0.0481	comparator; pre-specified descriptive (Study 1, as Table S14 prints it)
Opus-medium, main run	pooled				comparator; exploratory (no Holm, no verdict)
Step voting, main run	chains (1×6)	0.00356		0.00662	comparator; pre-specified descriptive (Study 1, as Table S14 prints it)

Table S1: (continued)

Panel 1: Comparisons (d = challenger minus comparator, points)

Challenger	Comparator	Shape	d (points)	d 95% interval	Across runs	Drift check (top-up)	Label
Haiku team	Step voting, main run	chains (1×6)	−1.25	[−15.00, 12.50]	yes	not fired	exploratory (no Holm, no verdict)
Haiku team	Step voting on pieces	pieces (6×1)	+21.25	[6.25, 35.00]	no	not needed (same run)	exploratory (no Holm, no verdict)
Haiku team	Opus-medium, main run	chains (1×6)	−43.75	[−55.00, −31.25]	yes	not fired	exploratory (no Holm, no verdict)
Haiku team	Opus-medium, main run	pieces (6×1)	−27.50	[−37.50, −18.75]	yes	not fired	exploratory (no Holm, no verdict)
Haiku picker over 8 samples	Any of the first 8 Haiku-medium samples right (the ceiling for a picker)	chains (1×6)	−25.00	[−36.25, −16.25]	yes	not fired	exploratory (no Holm, no verdict)
Haiku picker over 8 samples	Any of the first 8 Haiku-medium samples right (the ceiling for a picker)	pieces (6×1)	−18.75	[−28.75, −11.25]	yes	not fired	exploratory (no Holm, no verdict)
Haiku picker over 8 samples	One Haiku-medium sample (the first 8 samples)	chains (1×6)	+18.91	[9.53, 27.97]	yes	not fired	exploratory (no Holm, no verdict)
Haiku picker over 8 samples	One Haiku-medium sample (the first 8 samples)	pieces (6×1)	+14.53	[6.41, 22.50]	yes	not fired	exploratory (no Holm, no verdict)
Haiku picker over 8 samples	Haiku-medium vote of the first 8 samples	chains (1×6)	+9.64	[−1.88, 21.35]	yes	not fired	exploratory (no Holm, no verdict)
Haiku picker over 8 samples	Haiku-medium vote of the first 8 samples	pieces (6×1)	+2.08	[−6.88, 10.83]	yes	not fired	exploratory (no Holm, no verdict)
Step voting on pieces, top-up run	Opus-medium, main run	pieces (6×1)	−48.75	[−60.00, −37.50]	yes	not fired	unadjusted, exploratory, outside the pre-registered family

On pieces the Haiku team is compared with step voting on pieces, both in the top-up run (across runs: no); in the main run, step voting ran on chains only. The top-up is exploratory (no Holm, no verdict). Step voting on pieces against one Opus-medium answer from the main run is the pieces counterpart of Test 4, not one of Tests 1–5: it is labelled unadjusted (exploratory, outside Holm) and is outside the pre-registered family; the drift check between runs: not fired. A comparison across runs sets a top-up setting against a main-run setting run at another time, and carries the status of the drift check between runs. The drift check, set by the top-up run's specification (frozen with Lock 4), re-ran Haiku-medium at the main run's Haiku-medium settings on 20 puzzles in both shapes (40 puzzle-shape pairs, 5 samples each, 200 calls) and set it against the main run's first 5 Haiku-medium samples on each of the same puzzles: pooled d (top-up minus main run) = +1.50 points (95% interval −8.00 to +10.00); mean output tokens per call changed by +0.29%; status: not fired (a flag, not a gate; these answers are never mixed into Study 1). Rule for the drift check between runs: fires if the pooled 95% interval excludes 0 and $|d| > 10$ points, or mean output tokens move by more than 25%; a flag, not a gate. Picker cost: the list cost per puzzle is the total, the 8 main-run samples plus the picker call; the extra list cost is the picker call alone, on top of the 8 samples that the vote of 8 also uses. Cost per correct answer is given for the total only. A Label cell that starts with "comparator" marks a main-run comparator. The rows of the first 8 Haiku-medium samples and the pooled Opus-medium row are rebuilt by the top-up analysis from the main run's log; the Opus-medium rows per shape and step voting on chains print Study 1's own stamped values (the rebuild gives the same accuracies). The vote of 8 is priced at the list cost of its 8 samples, the picker's stamped total minus its stamped extra. The one-sample row has no stamped cost. The any-of row is a ceiling that needs the answer key, not a setup one could run, so it is given no cost. Format compliance is the share of answered replies whose format was ok, where a file stamps it: the Haiku team's (top-up file) and Opus-medium's (Study 1). Readable picks is the share of picker replies that named one of the samples. Other empty cells are not stamped. "(exact)" marks an exact (Clopper–Pearson) 95% interval. The main-run rows that Study 1 also reports print the interval Study 1 stamps for the same result (as Table S14 prints it), so the paper gives one interval per result: Opus-medium's one sample on chains and on pieces (exact) and step voting on chains (Study 1's cluster bootstrap). The top-up file's own bootstraps for these rows are not printed; for Opus-medium on pieces it collapsed to one value. The same exact interval for the Opus-medium pieces cell is stamped by free analysis 4 (post hoc): [95.49, 100.00]. Every other row keeps the top-up file's cluster bootstrap; Study 1's pre-registered settings table (Table S14) has no row for the top-up designs, the first 8 Haiku-medium samples or pooled shapes. Every comparison is paired over 80 puzzles at its shape.

Table S2: Within each study, cost-matched on the same puzzles, one Haiku-max sample scored higher than every lower-effort Haiku vote we compared. These comparisons are post hoc† and paired over puzzles: exploratory, with no Holm correction and no verdict, each within one study, and the two studies never pooled. The main table is Study 1 at 6 steps (80 puzzles per shape); its two panels give Study 1b at 16 steps (80 puzzles) and at 6 steps (40), against Haiku-medium, Haiku-high and Haiku-xhigh votes. d from a paired BCa bootstrap over puzzles (10,000 resamples); accuracies from a cluster bootstrap, with Clopper–Pearson for the one-sample Opus-medium setting; Study 1b's vote accuracies have no stamped interval. A vote of k samples costs k samples at list price. The main table also gives one Haiku-max sample against one Opus-medium answer in Study 1 (exploratory‡), with its paired d from free analysis 7 (post hoc†, exploratory).

Shape	Setting	Samples voted (k)	Accuracy (%)	Accuracy 95% interval	Interval type	Format compliance of the setting's samples (%)
1×6	One Haiku-max sample	1	97.50	[95.42, 99.17]	cluster bootstrap	100.00
1×6	One Haiku-medium sample	1	43.72	[36.44, 50.72]	cluster bootstrap	98.59
1×6	Haiku-medium vote, cost-matched	4	49.36	[41.15, 57.41]	cluster bootstrap	98.59
1×6	One Opus-medium answer	1	96.25	[89.43, 99.22]	Clopper–Pearson (exact)	97.50
6×1	One Haiku-max sample	1	95.00	[91.25, 97.92]	cluster bootstrap	99.58
6×1	One Haiku-medium sample	1	51.00	[44.28, 57.25]	cluster bootstrap	99.72
6×1	Haiku-medium vote, cost-matched	3	56.26	[48.56, 63.97]	cluster bootstrap	99.72
6×1	Haiku-medium vote at the chains' cost-matched k	4	58.95	[50.85, 66.94]	cluster bootstrap	99.72
6×1	One Opus-medium answer	1	100.00	[95.49, 100.00]	Clopper–Pearson (exact)	100.00

Shape	Setting	List cost per puzzle ($)	Samples per Haiku-max sample (Haiku-max cost per puzzle / one sample of this setting)	d: one Haiku-max sample minus this row (points)	d 95% interval (paired)
1×6	One Haiku-max sample	0.00441	—	—	—
1×6	One Haiku-medium sample	0.00110	—	—	—
1×6	Haiku-medium vote, cost-matched	0.00440	4.00	+48.14	[39.82, 56.22]
1×6	One Opus-medium answer	0.0402	—	+1.25	[−2.50, 7.50]
6×1	One Haiku-max sample	0.00407	—	—	—
6×1	One Haiku-medium sample	0.00109	—	—	—
6×1	Haiku-medium vote, cost-matched	0.00328	3.72	+38.74	[31.16, 46.24]
6×1	Haiku-medium vote at the chains' cost-matched k	0.00437	—	—	—
6×1	One Opus-medium answer	0.0481	—	−5.00	[−9.58, −2.50]

Table S2: (continued)

Shape	Setting	Label
1×6	One Haiku-max sample	exploratory setting; point pre-specified descriptive (Study 1)
1×6	One Haiku-medium sample	pre-specified descriptive (Study 1)
1×6	Haiku-medium vote, cost-matched	post hoc paired d and samples per Haiku-max sample (free analysis 1); vote point pre-specified descriptive (Study 1)
1×6	One Opus-medium answer	paired d (free analysis 7; post hoc, exploratory); point pre-specified descriptive (Study 1)
6×1	One Haiku-max sample	exploratory setting; point pre-specified descriptive (Study 1)
6×1	One Haiku-medium sample	pre-specified descriptive (Study 1)
6×1	Haiku-medium vote, cost-matched	post hoc paired d and samples per Haiku-max sample (free analysis 1); vote point pre-specified descriptive (Study 1)
6×1	Haiku-medium vote at the chains' cost-matched k	pre-specified descriptive (Study 1); above the cost match on this shape, no paired interval
6×1	One Opus-medium answer	paired d (free analysis 7; post hoc, exploratory); point pre-specified descriptive (Study 1)

Panel 1: Study 1b, 16 steps: one Haiku-max sample against cost-matched lower-effort Haiku votes

Setting	Samples voted (k)	Accuracy, pooled (%)	Accuracy 95% interval	Interval type	Format compliance of the setting's samples, 1×16 / 16×1 (%)
One Haiku-max sample	1	92.63	[90.38, 94.63]	cluster bootstrap	99.50 / 99.88
One Haiku-medium sample	1	17.75	[13.75, 21.75]	cluster bootstrap	95.49 / 100.00
Haiku-medium vote, cost-matched	4	22.49	—	—	95.49 / 100.00
One Haiku-high sample	1	31.25	[26.88, 36.00]	cluster bootstrap	93.75 / 100.00
Haiku-high vote, cost-matched	3	37.34	—	—	93.75 / 100.00
One Haiku-xhigh sample	1	58.38	[53.88, 62.88]	cluster bootstrap	97.50 / 100.00
Haiku-xhigh vote, cost-matched	2	58.94	—	—	97.50 / 100.00

Panel 1: (continued)

Setting	List cost per answer ($)	Samples per Haiku-max sample (Haiku-max cost per answer / one sample of this setting)	d pooled: one Haiku-max sample minus this row (points)	d pooled 95% interval (paired)
One Haiku-max sample	0.0112	—	—	—
One Haiku-medium sample	0.00254	—	—	—
Haiku-medium vote, cost-matched	0.0102	4.40	+70.14	[63.79, 75.57]
One Haiku-high sample	0.00334	—	—	—
Haiku-high vote, cost-matched	0.0100	3.35	+55.28	[49.57, 60.50]
One Haiku-xhigh sample	0.00507	—	—	—
Haiku-xhigh vote, cost-matched	0.0101	2.20	+33.69	[29.56, 37.63]

Table S2: (continued)

Panel 1: (continued)

Setting	d, 1×16 (points)	d, 1×16: 95% interval	d, 16×1 (points)	d, 16×1: 95% interval
One Haiku-max sample	—	—	—	—
One Haiku-medium sample	—	—	—	—
Haiku-medium vote, cost-matched	+78.40	[71.23, 83.96]	+61.88	[52.63, 70.38]
One Haiku-high sample	—	—	—	—
Haiku-high vote, cost-matched	+55.31	[48.21, 62.33]	+55.25	[47.13, 62.75]
One Haiku-xhigh sample	—	—	—	—
Haiku-xhigh vote, cost-matched	+33.88	[28.06, 40.06]	+33.50	[27.50, 39.50]

Panel 1: (continued)

Setting	Label
One Haiku-max sample	pre-specified descriptive (effort sweep)
One Haiku-medium sample	pre-specified descriptive (effort sweep)
Haiku-medium vote, cost-matched	post hoc (free analysis 1): k, accuracy, samples per Haiku-max sample and d; cost derived (k times the one-sample cost); format compliance pre-specified descriptive
One Haiku-high sample	pre-specified descriptive (effort sweep)
Haiku-high vote, cost-matched	post hoc (free analysis 1): k, accuracy, samples per Haiku-max sample and d; cost derived (k times the one-sample cost); format compliance pre-specified descriptive
One Haiku-xhigh sample	pre-specified descriptive (effort sweep)
Haiku-xhigh vote, cost-matched	post hoc (free analysis 1): k, accuracy, samples per Haiku-max sample and d; cost derived (k times the one-sample cost); format compliance pre-specified descriptive

Panel 2: Study 1b, 6 steps: one Haiku-max sample against cost-matched lower-effort Haiku votes

Setting	Samples voted (k)	Accuracy, pooled (%)	Accuracy 95% interval	Interval type	Format compliance of the setting's samples, 1×6 / 6×1 (%)
One Haiku-max sample	1	95.75	[93.00, 97.88]	cluster bootstrap	100.00 / 100.00
One Haiku-medium sample	1	54.50	[46.00, 63.25]	cluster bootstrap	99.00 / 99.00
Haiku-medium vote, cost-matched	3	60.63	—	—	99.00 / 99.00
One Haiku-high sample	1	60.75	[52.25, 69.50]	cluster bootstrap	100.00 / 100.00
Haiku-high vote, cost-matched	3	65.38	—	—	100.00 / 100.00
One Haiku-xhigh sample	1	76.75	[70.00, 83.00]	cluster bootstrap	98.50 / 99.50
Haiku-xhigh vote, cost-matched	2	77.69	—	—	98.50 / 99.50

Table S2: (continued)

Panel 2: (continued)

Setting	List cost per answer ($)	Samples per Haiku-max sample (Haiku-max cost per answer / one sample of this setting)	d pooled: one Haiku-max sample minus this row (points)	d pooled 95% interval (paired)
One Haiku-max sample	0.00419	—	—	—
One Haiku-medium sample	0.00110	—	—	—
Haiku-medium vote, cost-matched	0.00331	3.80	+35.13	[25.79, 45.02]
One Haiku-high sample	0.00125	—	—	—
Haiku-high vote, cost-matched	0.00376	3.34	+30.38	[21.50, 39.54]
One Haiku-xhigh sample	0.00172	—	—	—
Haiku-xhigh vote, cost-matched	0.00344	2.43	+18.06	[12.50, 25.25]

Panel 2: (continued)

Setting	d, 1×6 (points)	d, 1×6: 95% interval	d, 6×1 (points)	d, 6×1: 95% interval
One Haiku-max sample	—	—	—	—
One Haiku-medium sample	—	—	—	—
Haiku-medium vote, cost-matched	+40.96	[27.88, 54.92]	+29.29	[18.92, 41.29]
One Haiku-high sample	—	—	—	—
Haiku-high vote, cost-matched	+33.75	[22.25, 47.33]	+27.00	[16.75, 39.42]
One Haiku-xhigh sample	—	—	—	—
Haiku-xhigh vote, cost-matched	+15.38	[9.50, 22.63]	+20.75	[13.00, 31.00]

Table S2: (continued)

Panel 2: (continued)

Setting	Label
One Haiku-max sample	pre-specified descriptive (effort sweep)
One Haiku-medium sample	pre-specified descriptive (effort sweep)
Haiku-medium vote, cost-matched	post hoc (free analysis 1): k, accuracy, samples per Haiku-max sample and d; cost derived (k times the one-sample cost); format compliance pre-specified descriptive
One Haiku-high sample	pre-specified descriptive (effort sweep)
Haiku-high vote, cost-matched	post hoc (free analysis 1): k, accuracy, samples per Haiku-max sample and d; cost derived (k times the one-sample cost); format compliance pre-specified descriptive
One Haiku-xhigh sample	pre-specified descriptive (effort sweep)
Haiku-xhigh vote, cost-matched	post hoc (free analysis 1): k, accuracy, samples per Haiku-max sample and d; cost derived (k times the one-sample cost); format compliance pre-specified descriptive

Stamped note of free analysis 1, in its own words: "$d > 0$ means one Haiku-max try scored higher than the lower-effort vote at the same money. Each row is within one study and one set; the two studies are never pooled. Under the $1/m$ tie rule a 2-try vote scores about its single try." Stamped note of free analysis 7, in its own words: "Exploratory arm, post hoc interval; $d > 0$ means the Haiku-max try scored higher." In those two stamped notes, a "try" is a sample, an "arm" is a setting and "at the same money" is cost-matched. Samples per Haiku-max sample: one Haiku-max sample's list cost divided by the list cost of one sample of the setting in the row. Cost-matched: the vote's k is the floor of the samples per Haiku-max sample, held to the setting's designed samples, so the vote costs no more than one Haiku-max sample. Study 1 costs are in US dollars per puzzle and Study 1b costs in US dollars per answer, as each study stamps them. On pieces (6×1) the chains' cost-matched vote, $k = 4$, costs more than one Haiku-max sample (cost column), so it is shown without a paired interval; the cost-matched vote on this shape has $k = 3$. Format compliance (pre-specified descriptive) is that of the setting's own samples; a vote is taken over the same samples. Haiku-max was an exploratory setting in Study 1. No row is a test: the paired differences carry no p-value, no Holm correction and no verdict. Each row stays within one study. Panels 1 and 2 (Study 1b): vote rows: free analysis 1 (post hoc) stamps the vote's k, pooled accuracy, samples per Haiku-max sample and paired d, with no accuracy interval and no per-shape vote accuracy; Study 1b's own file stamps no lower-effort vote; the vote's list cost per answer is k times the setting's stamped one-sample cost per answer (derived); one-sample rows: Study 1b's pre-specified effort sweep (descriptive), as stamped.

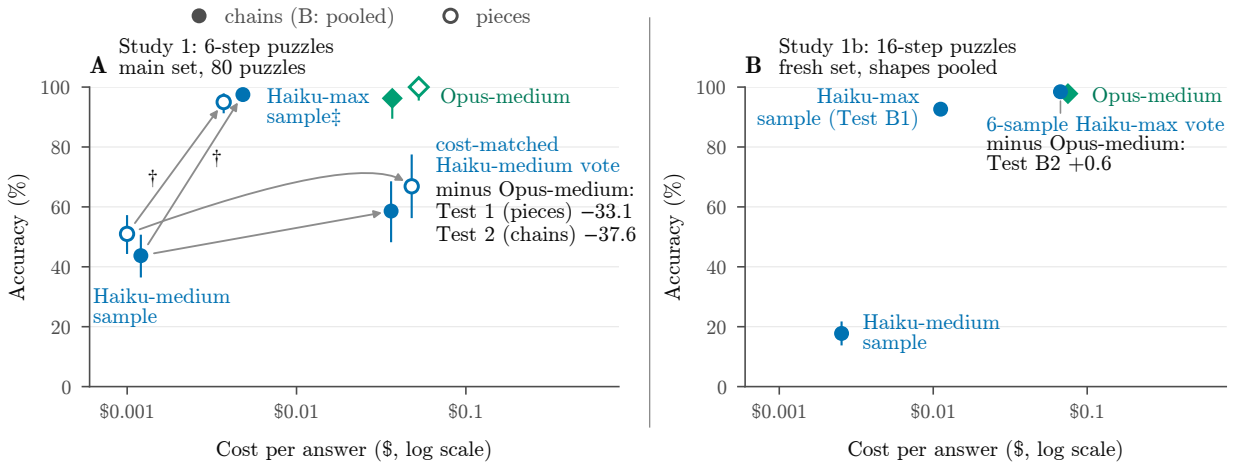


Figure S1: A cost-matched medium-effort Haiku vote was worse than one Opus-medium answer by more than the 10-point margin, while a small max-effort Haiku vote was shown no worse by more than 5 points at about its price. A: Study 1, 6 steps (chains filled, pieces hollow, each pair set slightly apart sideways); Tests 1–2, pre-registered decision rule, not Holm-adjusted. B: Study 1b, 16 steps, shapes pooled; Test B2, Holm-adjusted, its vote estimated from 10 samples, costing more than Opus-medium on chains and less on pieces. Values are d in points. Arrows in A to the Haiku-max sample are post hoc† (our reading, not a test); that sample is exploratory‡ in A and Test B1's challenger in B. Bars are 95% intervals, not drawn where shorter than the marker (A: Haiku-max on chains; B: every point but Haiku-medium). The panels come from different studies, so any comparison between them is descriptive. Costs at list price; Batch, which we used, bills half. This is Figure 1 of the paper.

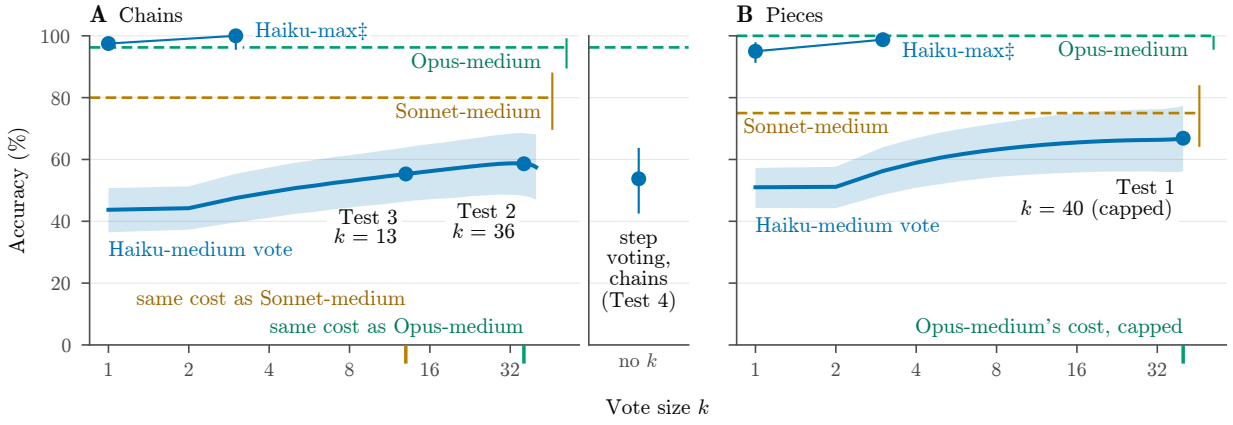


Figure S2: Up to 40 samples, a medium-effort Haiku vote levels off below one Opus-medium answer on both shapes and below one Sonnet-medium answer on chains; on pieces it stays below Sonnet-medium only on point estimates. Study 1's vote curves at 6 steps (pre-specified descriptive). Labelled markers are Tests 1–3's cost-matched votes and the separate strip is step voting (Test 4), all read by the pre-registered decision rule, not Holm-adjusted (Test 3 depends on the marking rule); the joined pair is Haiku-max, one sample and a small vote (exploratory‡). Bands and bars are 95% intervals; Haiku-max's vote on chains has an exact interval (post hoc†); bars shorter than their marker are not drawn (Haiku-max's one sample on chains and its 3-sample vote on pieces). Costs at list price. This is Figure 5 of the paper.

Table S3: On Study 1b's 6-step puzzles, Sonnet at its default effort was not shown to differ from Opus-low, at under half its cost per answer. Sonnet-high paired against Opus-low, Opus-medium, one Haiku-max sample and Sonnet-medium on that set (40 puzzles), with its cost against each (post hoc† exploratory; against Test 5, a reading across studies, not a test). d from a paired BCa bootstrap (10,000 resamples); list price.

Comparator	Shape	Comparator accuracy (%)	Comparator format compliance (%)	d (points), Sonnet-high minus comparator	95% interval (paired bootstrap)	List cost ratio per answer (Sonnet-high / comparator)	Cost-per-correct ratio (Sonnet-high / comparator)
Opus-low	1×6	97.50	97.50	0.00	[−7.50, 7.50]	0.46	0.46
Opus-low	6×1	100.00	100.00	−2.50	[−15.00, 0.00]	0.50	0.51
Opus-low	pooled	98.75	97.50 / 100.00	−1.25	[−6.25, 2.50]	0.48	0.48
Opus-medium	1×6	98.75	100.00	−1.25	[−10.00, 2.50]	0.40	0.40
Opus-medium	6×1	100.00	100.00	−2.50	[−12.50, 0.00]	0.45	0.46
Opus-medium	pooled	99.38	100.00 / 100.00	−1.88	[−7.50, 0.63]	0.43	0.44
Haiku-max, one sample	1×6	97.25	100.00	+0.25	[−9.00, 3.75]	3.72	3.71
Haiku-max, one sample	6×1	94.25	100.00	+3.25	[−6.00, 7.75]	5.32	5.15
Haiku-max, one sample	pooled	95.75	100.00 / 100.00	+1.75	[−4.00, 5.25]	4.50	4.41
Sonnet-medium	1×6	77.50	97.50	+20.00	[7.50, 35.00]	1.08	0.86
Sonnet-medium	6×1	82.50	100.00	+15.00	[2.50, 30.00]	1.36	1.15
Sonnet-medium	pooled	80.00	97.50 / 100.00	+17.50	[10.00, 25.00]	1.23	1.01

Label: post hoc exploratory; against Test 5 a cross-study reading (Study 1b's 6-step set, not Study 1's). Every row is on the 40 puzzles of Study 1b's 6-step set, as a chain and as pieces. In every row, Sonnet-high's accuracy is 97.50% and its format compliance, in both shapes, is 100.00%. The 6-step set includes 2 puzzles for which parts of the answer keys were shown in our own work sessions; those parts were never shown to a model under test, so no sensitivity run was needed. 95% interval: the paired BCa (bias-corrected and accelerated) bootstrap over puzzles, $B = 10{,}000$. Sonnet's default effort is high; Test 5 compared Opus-low with Sonnet-medium on Study 1's puzzles. These rows are on Study 1b's 6-step set, so set beside Test 5 they are a reading across studies: our reading, not a test. Format compliance is the share of answered samples in the required answer format; in the pooled rows it is given per shape, 1×6 / 6×1.

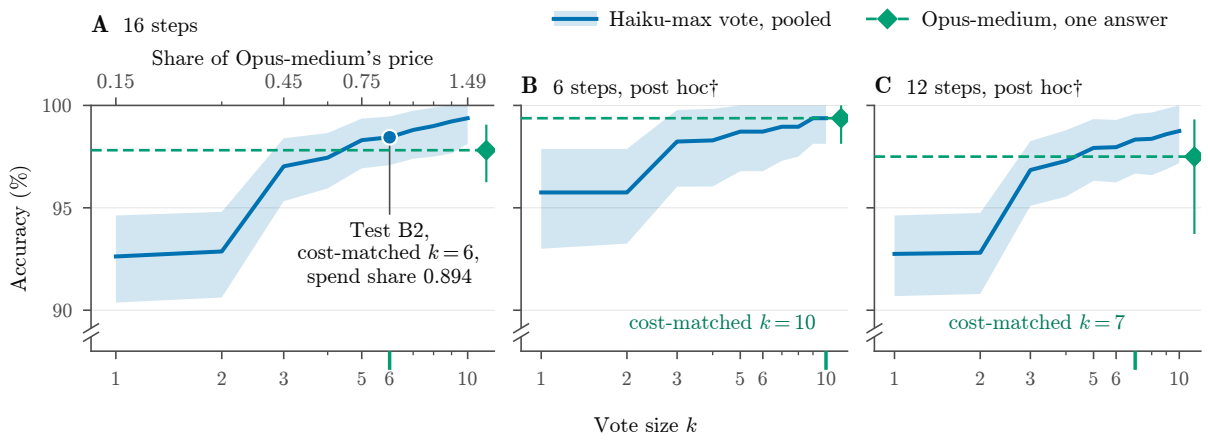


Figure S3: The Haiku-max vote gains most in its first few samples; only the vote of Test B2 was tested. The Haiku-max vote at 16 steps (Study 1b, fresh puzzles; 80 puzzles): pooled accuracy of a vote of k samples, each the expected accuracy estimated from the 10 samples drawn per puzzle, against one Opus-medium answer; the axis above panel A gives the vote's list cost as a share of Opus-medium's price. Pre-specified descriptive; Test B2's vote is marked. Panel A is at 16 steps; panels B and C repeat the curve at 6 steps (40 puzzles) and 12 steps (80), post hoc†, each with its own cost-matched vote size; the lengths use different puzzle sets and are not compared. The band (from one sample, the setting's stamped interval) and the bars on Opus-medium are cluster-bootstrap 95% intervals (Opus-medium's at 12 steps exact Clopper–Pearson); chains and pieces are not drawn separately (Table S22 gives Test B2's vote by shape). The accuracy axis does not start at zero. This is Figure 6 of the paper.

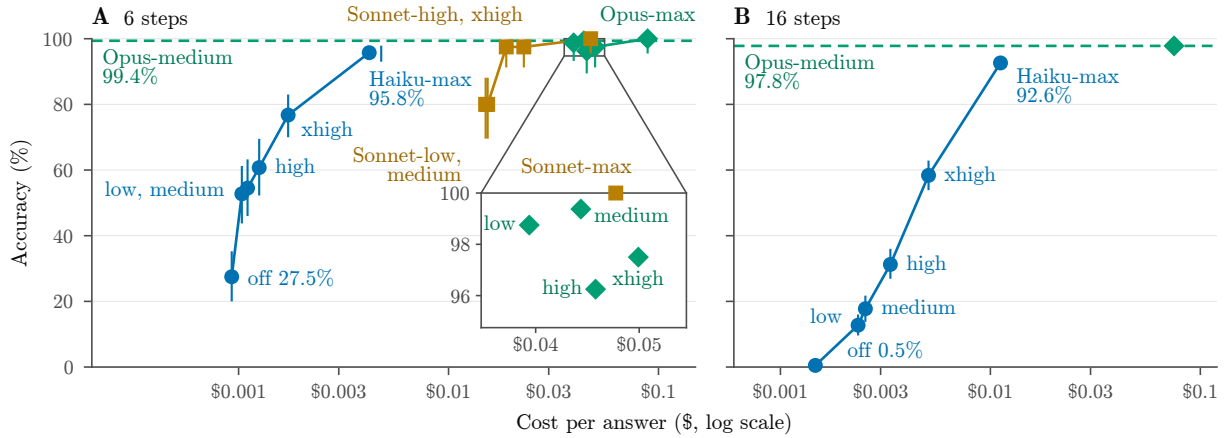


Figure S4: Haiku's accuracy climbs steeply with effort, below Opus's cost; Opus stays near its ceiling at every effort. Study 1b's effort sweep (pre-specified descriptive, no tests) at 6 steps (A) and 16 steps (B). Each point pools chains and pieces, the words are effort levels (off: thinking off), and the dashed line is Opus-medium. Bars are 95% intervals (cluster bootstrap for Haiku and Opus-medium, exact for the one-sample Sonnet and Opus settings), not drawn where shorter than the marker with neither end reaching a printed point past its edge (A: Opus-medium; B: Haiku, thinking off; Haiku-max; and Opus-medium). An inset in A enlarges five overlapping markers; Table S25 lists every value and format flag. Cost per answer at list price. This is Figure 3 of the paper.

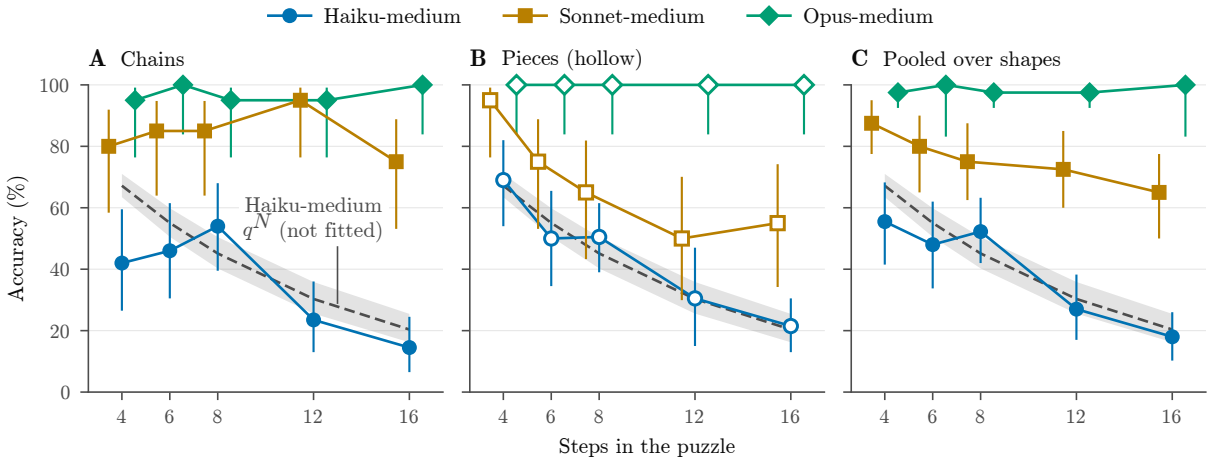


Figure S5: Haiku-medium's accuracy falls steeply as puzzles get longer, while Opus-medium stays near its ceiling at every length we ran. The length sweep (pre-registered descriptive: fixed in a lock before any of its data existed, and no test; Deviation 2's marking rule): A chains, B pieces (hollow), C both shapes pooled; each length has its own puzzle set, and Sonnet-medium and Opus-medium are shifted slightly sideways. The dashed line is Haiku-medium's per-piece rate compounded over the steps, with its band (not fitted, not a test). Bars are 95% intervals (Wilson for Sonnet- and Opus-medium by shape, cluster bootstrap otherwise); Opus-medium's pooled bars at 6 steps and 16 steps are exact (post hoc†). The format check flags, on chains, Haiku-medium at 16 steps, Sonnet-medium at 6 steps and Opus-medium at 4 steps and 12 steps. Three flags sit at exactly 5.0% non-compliance; the locked check raises them through floating-point rounding, though its rule reads "more than 5%"; we report them as stamped. Study 1b's and Pilot 1's points are in Figure S11. This is Figure 7 of the paper.

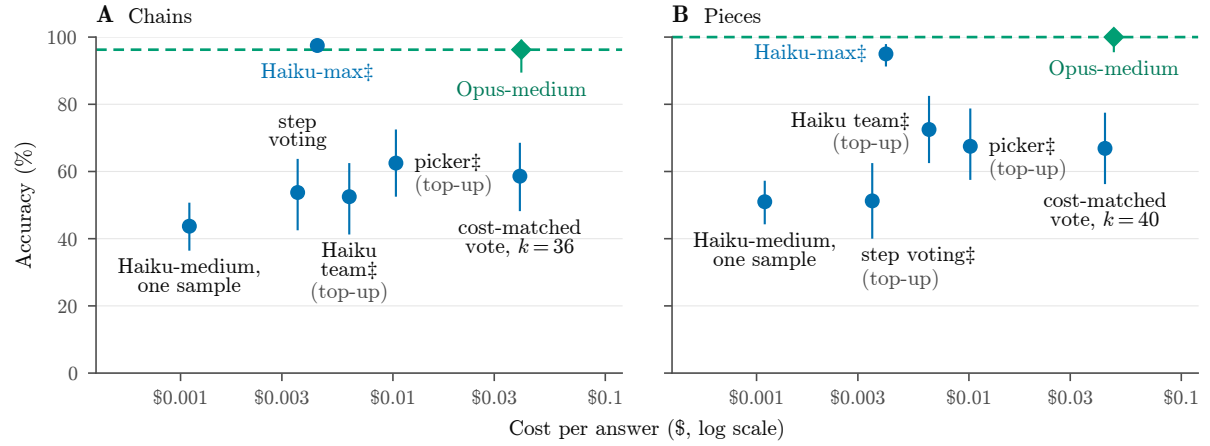


Figure S6: Every Haiku design at medium effort sits far below Opus-medium's accuracy, while a single Haiku-max sample (exploratory‡) scores far higher than any of them at a fraction of Opus-medium's cost. Where each design landed at 6 steps: accuracy against list cost per answer (descriptive and exploratory‡; no frontier or fitted line; 80 puzzles per shape). Designs marked top-up ran in the later top-up run, so their comparisons span the two runs; the drift check between runs did not fire. The dashed line is Opus-medium. Bars are 95% intervals (cluster bootstrap; exact for the one-sample Opus-medium answers), not drawn where shorter than the marker with neither end reaching a printed point past its edge (Haiku-max on chains). This is Figure 8 of the paper.

Chain: rules B, A, D, C in turn

	Rent	Payroll	Supplies	Reserve	Travel	Training
start	1,227	1,050	1,955	2,860	7,438	7,316
after B	**2,324** +1,097	**1,450** +400	1,955	2,860	**7,038** −400	**6,219** −1,097
after A	2,324	**1,850** +400	**2,265** +310	2,860	**6,638** −400	**5,909** −310
after D	2,324	**2,750** +900	**2,715** +450	2,860	**5,738** −900	**5,459** −450
after C	2,324	2,750	**3,151** +436	2,860	5,738	**5,023** −436

Pieces: each step from its own start

	Rent	Payroll	Supplies	Reserve	Travel	Training
start	981	1,165	2,287	3,060	8,107	7,681
after B	**1,381** +400	**2,317** +1,152	2,287	3,060	**7,707** −400	**6,529** −1,152
start	2,254	1,102	2,111	2,259	7,249	6,405
after A	2,254	**1,502** +400	**2,431** +320	2,259	**6,849** −400	**6,085** −320
start	2,835	1,535	1,879	3,174	6,173	6,913
after D	2,835	**2,435** +900	**2,329** +450	3,174	**5,723** −450	**6,013** −900
start	2,602	3,410	2,280	3,374	6,139	5,895
after C	**3,073** +471	3,410	2,280	3,374	6,139	**5,424** −471

Bold on tint: changed by the step (rule, then house rule).
Grey: unchanged by the step. Under a change: its move.
Arrow: piece starts near this state (each line within ±25%).

Rules used. B: the second-largest line gives 15% of its balance (rounded down) to the second-smallest. **A**: the second-largest line gives 5% of its balance (rounded down) to the second-smallest. **D**: three times in a row, move 450 from the largest line to the smallest (or all of it), re-finding the largest and smallest each time. **C**: the second-largest line gives 8% of its balance (rounded down) to the second-smallest. House rule, after each step's own rule: if the smallest line is below 1,900, 400 moves to it from the largest (or all of it).

Figure S7: A chain carries each step's result into the next, so an early slip carries forward; a piece starts afresh and tests one step alone. One illustrative 4-step puzzle, made for this figure and never in a test set; each piece starts from the chain's reference state before its step, every line moved by up to 25% either way. This is Figure 2 of the paper.

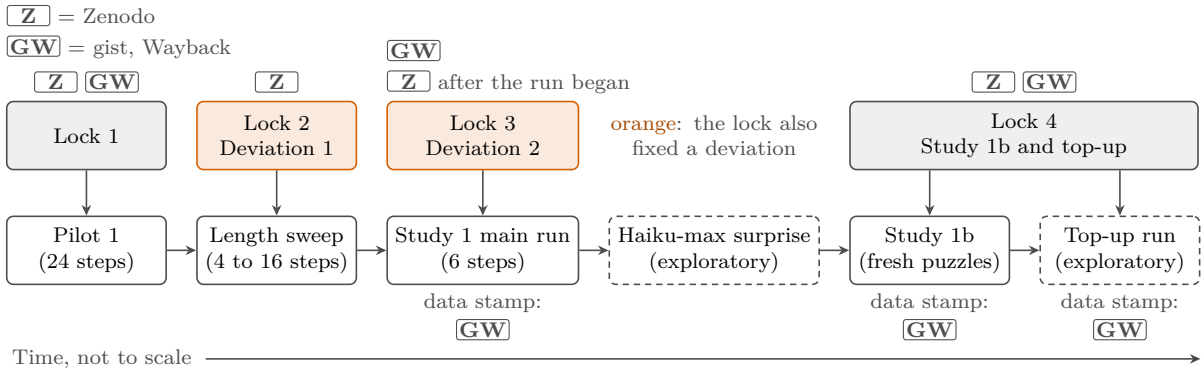


Figure S8: Each study's tests were fixed in a time-stamped lock, its fingerprint posted publicly before any of its data existed; analysis details written later were stamped before any result was opened. A fingerprint sits above each lock; three runs carry a data stamp. Lock 2 also fixed Deviation 1 (24 steps to the swept length). Dashed boxes are exploratory: the Haiku-max surprise, found after Study 1's results were opened, and the top-up run. Figure S9 and Table S10 give the dates, setup checks and an unlocked effort diagnostic. This is Figure 4 of the paper.

Table S4: Every setting we ran, by study (Table 1 of the paper): model and effort, samples per puzzle, puzzles (each in two shapes), steps and role. A test setting is used by a pre-registered confirmatory test; a description is pre-specified with no test; exploratory is neither.

Study	Setting (model, effort)	Samples per puzzle	Puzzles	Steps	Role
Length sweep	Haiku-medium; Opus-medium; Sonnet-medium	10; 1; 1	20	4, 6, 8, 12, 16	description (chose the length)
Study 1	Haiku-medium	40	80	6	test (Tests 1–3, as votes)
Study 1	Opus-medium; Sonnet-medium; Opus-low	1 each	80	6	test (the comparators; Test 5)
Study 1	Step voting: Haiku-medium, one step a call	up to 9 per step	80	6	test (Test 4)
Study 1	Haiku-low; Haiku, thinking off; Haiku-max; Sonnet-low	40; 40; 3; 1	80	6	exploratory
Study 1	Haiku-medium, reworded prompt	10	20	6	description (wording check)
Study 1b	Haiku-max; Opus-medium	10; 2	80	16	test (Tests B1, B2)
Study 1b	Haiku-max; Opus-medium	10; 2	40	6	test (Test B3)
Study 1b	Haiku, thinking off; Haiku at low, medium, high and xhigh effort	5 each	80 and 40	16 and 6	description
Study 1b	Sonnet at low, medium, high, xhigh and max effort; Opus at low, high, xhigh and max effort	1 each	40	6	description
Study 1b	Haiku-max; Opus-medium	10; 1	80	12	description (the 12-step point)
Top-up	Haiku-medium designs: a team, a picker, step voting on pieces; and the drift check between runs	see Table S1	80 (drift check: 20)	6	exploratory

Each puzzle is sent in two shapes, one chain and separate pieces; samples are per puzzle and shape. Role: test, a pre-registered confirmatory test uses the setting; description, pre-specified with no test; exploratory, neither.

S3 Method details

S3.1 The generator and the frozen sets

We built the puzzles with our own seeded generator (version ledger-0.5). It accepts a rulebook only if every step changes the state, on the chain and on each piece; if no 1, 2 or 3 steps and no stretch of 4 or more steps can be left out; and if no pick turns on a tie or on a value exactly at a threshold. It also caps at 5% the share of slips that later steps hide (for 1, 10 or 100 units put on the wrong line, on the reference chain), which bounds the gap between the official chain score and a score on the final answer alone. Before any call of the length sweep, we froze 15 puzzle sets with 700 puzzles (1,400 prompts, one per shape): at each candidate length, a length-sweep set, a main set and a reserve set. No two sets share a set id, seed or prompt, and each is disjoint from every set frozen in Lock 1. A separate re-solver, written apart from the generator, reads only the printed prompt and must reproduce every stored state, and a verify step rebuilds every frozen item from its seed and compares every stored field.

The house rule's window at every length, as a share of the reference chain's steps: 5 to 8 of 24 steps, 1 to 2 of 4 steps, 1 to 2 of 6 steps, 1 to 3 of 8 steps, 2 to 4 of 12 steps and 3 to 6 of 16 steps.

S3.2 Further detail on the method

These paragraphs give details that the Method (Section 3 of the paper) leaves out.

S3.2.1 The puzzles

The rule set. Every puzzle in the length sweep, Study 1, Study 1b and the top-up run used one fixed rule set: 6 budget lines and no decoy rules (other presets print near-duplicates of the used rules that no sequence applies). Its rules come from the look-again, repeat, average, rank, count-then-act, two-clause and compare-and-move kinds. The repeat rule, for example, moves a fixed amount from the largest line to the smallest 3 times in a row, finding both lines again each time.

The house rule. Every step also includes a house rule, printed once in the prompt's introduction. Straight after each rule, as part of the same step, if the line with the smallest balance is below a set threshold, a set amount moves to it from the line with the largest balance, or all of that line if it holds less. The house rule acts at most once per step, after that step's own rule and never on its own result, and the balances written for a step are those after it, so each step is two operations. Its threshold and amount were tuned for each puzzle so that it acts on a set share of the reference chain's steps: 1 to 2 of 6 steps, 2 to 4 of 12 steps and 3 to 6 of 16 steps (given for every length above).

The refusals. The first design's "digit-rule" puzzles drew their refusals in the second of two small initial test runs at the start: 3 of its 9 calls were refused (both Opus calls and one of the two Sonnet calls; none of the five Haiku calls). After the switch to the new family there were no refusals at all: 0 in the 25,007 paid calls from the refusal check through the main run, 0 in Study 1b's 11,640 samples and 0 in the top-up run's 3,820 calls. That the digit puzzles looked like ciphers to the classifier is our untested guess.

S3.2.2 Models, effort and settings

Effort. We define effort as the API's effort value. The vendor calls it "a behavioral signal, not a strict token budget", and Sonnet 5.5's levels are recalibrated, so effort labels are not a common unit across models; measured tokens can be compared because the models share one tokeniser.

S3.2.3 The runs

The length rule's medians. The rule's median Haiku-medium single-sample score, over both shapes, was 55.00, 50.00, 50.00, 10.00 and 10.00% at 4, 6, 8, 12 and 16 steps, so only 6 and 8 steps lay inside the 25–50% band.

Study 1b's effort sweep and 12-step point. The effort sweep, descriptive with no tests, added at both lengths Haiku, thinking off (at medium effort), and at low, medium, high and xhigh effort, 5 samples each, and at 6 steps only Sonnet at low, medium, high, xhigh and max and Opus at low, high, xhigh and max, one sample each. The 12-step point, also descriptive, added Haiku-max with 10 samples and Opus-medium with one sample on 80 further puzzles in both shapes.

The top-up run (exploratory). The top-up run sent every call to Haiku-medium, through Batch only, on the main run's 80 puzzles in both shapes, with a limit of 48,000 tokens on every call of the three new designs; the drift check between runs kept the main run's 5,000-token limit. It ran three designs and one check:

- the **Haiku team:** one planner call per puzzle wrote an instruction for each of the 6 tasks; helper calls carried out each task, drawing until one answer led by 2 votes, up to 9 draws, in order on chains and all at once on pieces; and one aggregator call, which could redo a step, wrote the final answer;
- the **Haiku picker:** one call per puzzle read the first 8 of Haiku-medium's main-run answers to that puzzle, always in sample order, and returned one of them;
- the **step voting on pieces:** the pieces counterpart of Test 4, the main run's step-voting code with each piece as one step (480 step positions), outside the Holm family;
- the **drift check between runs:** Haiku-medium re-run 5 times per puzzle at the main run's exact settings on 20 puzzles in both shapes (200 calls), to measure whether the model's behaviour moved between the two runs.

S3.2.4 Cost-matched votes and scoring

Exact votes. The plan allows exact votes, over all k-sample subsets, wherever they are feasible, and they were feasible for every vote, so each vote accuracy is the value a sampled estimate would aim at.

Deviation 2's other numbers. Under the rule as run, Sonnet-medium's chain format compliance was 97.50% against 65.00% under the original rule; Sonnet-low's chain score was 71.25% against 52.50%, and Opus-low's 97.50% against 96.25%. Under the original rule, Test 3 would read inconclusive (+0.30 points, 95% interval −13.58 to +14.95), and Test 5's difference would be +41.25 points (95% interval +28.75 to +52.50) instead of +17.50. We report both, as Deviation 2 set out.

S3.2.5 Statistics

Tests. The difference d is the challenger's score minus the comparator's, averaged over puzzles, and each test's null hypothesis was that the challenger is worse by its margin or more: the 10-point margin in Study 1, and in Study 1b the 5-point primary and 10-point secondary margins. Study 1b's plan averaged each puzzle's two shapes because near the ceiling it halves the variance of each puzzle's difference and keeps the family at three tests; the per-shape results are description only.

Views and completeness. The as-deployed view was primary. Two further views left refusals out and truncations out. In both studies, a "no worse" reading could head the abstract or a headline only if it held in all three views; otherwise it was reported as holding as deployed only. Study 1 also reports a secondary 5-point reading of its five tests, as deployed only, set by its reporting definitions; it sits beside the 10-point verdict without replacing it. Study 1b fixed a completeness rule before its run: a test counted as completed only if every designed sample of both its settings had a final record, and a test that was not completed would read "not completed" and keep its Holm place with $p = 1$.

Intervals. The paired BCa bootstrap resampled the per-puzzle differences. Every confirmatory p-value comes from a bootstrap of $B = 10{,}000$ resamples, and the share of resamples at or beyond the margin is never taken below 1/10,001, about 0.0001, before the BCa correction. Where no resample reached the margin, the reported p is therefore the smallest the test can return, and we write it as $p \leq$ that value, rounded up at five decimal places: Test 5 (0.00004), B2 at both margins (0.00011), B2 without the partly shown

puzzles (0.00009) and B3 at the 10-point margin (0.00108), as well as the inferiority p-values of Tests 1, 2 and 4 in Table S11. Accuracies and other descriptive numbers carried a cluster bootstrap over puzzles instead, in which each draw keeps all of a puzzle's shapes, samples and settings together (2,000 draws, percentile 95% interval); in Study 1 and Study 1b, a setting with one sample per puzzle got an exact two-sided 95% interval (Clopper–Pearson) instead. Study 1's reporting definitions, which set these conventions, were fixed after Lock 3 and before any main-run result was opened. Item numbers in the appendix tables refer to these reporting definitions, which Table S34 lists.

Comparisons that are not tests. Study 1 and Study 1b differ in their numbers of puzzles and in day: 80 puzzles on 2026-10-12 against 40 on 2026-10-13 at 6 steps. Each top-up comparison reports d and its 95% interval. The vote curves plot accuracy against the number of samples in the vote.

S3.3 The effort diagnostic's counts

On identical prompt text, Haiku in free Claude Code sessions got 141 of 144 single pieces right (97.92%), API Haiku-medium 110 of 144 (76.39%) and API Haiku-high 127 of 144 (88.19%). High scored above medium on all 6 prompts; if the two were equally good, that would happen by chance 1 time in 64. The diagnostic cost $0.416, was exploratory and ran under no lock. Claude Code's effort and thinking budget were not recorded, so the gap is unexplained.

S3.4 Output-token limits

Each setting's output limit was set by a rule fixed before the length sweep. A setting the length sweep had measured got about 1.5 times its largest output there, rounded up to the next 1,000 tokens: 5,000 for Haiku-medium, 4,000 for Opus-medium and 3,000 for Sonnet-medium. Every setting the length sweep had not measured kept the locked pilot limit, so that the limit could not bind: 48,000 for Haiku-low; Haiku, thinking off; Haiku-max; and the step-voting calls, and 32,000 for Opus-low and Sonnet-low.

Study 1b's output limits followed Study 1's rule: a setting already measured, in Study 1 or the length sweep, got 1.5 times its largest output there, rounded up to 1,000 (for example 9,000 for Opus-medium at 16 steps and 30,000 for Haiku-max at 6 steps). Each setting not yet measured first ran a trial round on already-used length-sweep puzzles, never the test sets (10 puzzles in both shapes at each length, one sample each, with a limit of 64,000, or 128,000 for Haiku-max at 12 or 16 steps; its answers were never scored). Its limit then became twice its largest trial-round output, rounded up to 1,000, but no lower than its floor and no higher than 128,000; the floor was the same model's medium-effort limit at that length (for example 14,000 for Haiku at 16 steps), or 96,000 for Haiku-max at 12 or 16 steps. As run, no trial-round call reached its limit, and the cost-only cut order, applied once after the trial round, removed no setting.

S3.5 Models, prices and prompts

Three fixed model IDs, `claude-haiku-5-5`, `claude-sonnet-5-5` and `claude-opus-5-5`, served every setting, at an effort level from low to max, with adaptive thinking on every setting except Haiku's thinking-off settings (one in Study 1, one in Study 1b), which disable it. The file `data/paper-outputs/reproduction/arms.json` gives each of the 43 settings of the length sweep, the main run, Study 1b and the top-up run its model, effort, thinking setting and output limit, with its request as built.

The pinned price table, in US dollars per million tokens and dated 2026-10-12 in the code, is: Haiku 5.5 $0.10 for input and $0.50 for output, Sonnet 5.5 $2 and $10, and Opus 5.5 $4 and $20. A Haiku request whose prompt exceeds 100,000 tokens is billed whole at $0.50 and $2.50, and Batch halves every rate. The table is `data/paper-outputs/reproduction/prices.json`.

The same folder holds 18 further prompts, renders of the prompts the runs sent beyond the chain and pieces prompts (the rewording check's variant, step voting, the team's and the picker's), made by the run code on the illustrative puzzle of Figure S7, never on a test item, each with its request as built; the chain and pieces prompts are with the illustrative puzzle. They show each prompt's form, not complete sent prompts: a labelled placeholder stands in for any text a model wrote (the planner's instructions, the picker's 8 replies),

and the aggregator is shown with a correct helper's results. The file `parsers.json` points to the 13 functions and patterns that read the replies, with each one's docstring or pattern.

S3.6 Holm's thresholds

Holm's step-down correction compares the smallest one-sided p-value with the level divided by the number of tests, then the next with a looser threshold, and stops at the first test that fails. In Study 1 the thresholds were 0.005, 0.00625, 0.00833, 0.0125 and 0.025; in Study 1b they were 0.00833, 0.0125 and 0.025. The correction ran within each study only: Study 1 and Study 1b were never pooled in one family.

S3.7 The brief's four draft hypotheses

The pre-registration's background section lists the project brief's four draft hypotheses, labelled H1 to H4 there. They were drafted, not ruled; only Tests 1–5 and B1–B3 are tests. Here is what our data say about each.

H1: at the price of one Opus-medium call, is a Haiku vote no worse than Opus-medium by more than 10 points?

- It became Tests 1–2, and its Sonnet-medium version became Test 3.
- Study 1b asked the vote question again with max-effort Haiku (Test B2).

H2: does splitting the task by script (step voting) remove the shape effect on dollars, but keep it on time?

- **Time:** not examined. Step voting ran only by Batch, and its timing analysis was dropped before the main run began because it was not fully defined.
- **Dollars:** step voting cost about the same per puzzle on both shapes, at list price: $0.00356 on chains in the main run and $0.00351 on pieces in the top-up run.
- **Accuracy:** 53.75% on chains and 51.25% on pieces.

Both comparisons are descriptive and not tested. Both span the two runs, and the drift check between runs did not fire. Test 4 compared step voting with Opus-medium on chains; it is not a test of H2.

H3: do teams that plan for themselves get worse on deep chains?

- **What we can look at:** only the Haiku team, because the other planned team was dropped in the budget plan, and only at 6 steps, so no change with depth was measured.
- **Chains:** the team was not shown to differ from step voting (exploratory; $d = -1.25$ points, 95% interval -15.00 to $+12.50$; across the two runs, with the drift check not fired).
- **Pieces:** it scored above step voting (exploratory; $+21.25$ points, 95% interval $+6.25$ to $+35.00$; same run).
- **Scores:** the team scored 52.50% on chains and 72.50% on pieces (exploratory). One Haiku-medium sample scored 43.72% and 51.00%.

The team's lower chain score is in H3's direction at this one length (our reading, not a test); the shape difference itself was not tested.

H4: is a picker worth its cost over a plain vote?

- **What we can look at:** only the Haiku picker over 8 Haiku-medium answers. The Opus-low picker and the picker over 16 were dropped in the budget plan.

2026-10-11		all times HH:MM UTC
10:16–10:37		API refusal check (setup), the same day as Lock 1 and before it.
19:09–19:18		Second setup check, before Lock 1.
19:32		**Lock 1** cut: Study 1's plan for 24-step puzzles. Public: gist 19:35, Wayback, Zenodo DOI registered 20:20.
20:48–21:27		Pilot 1 (24 steps), which showed 24 steps too hard for Haiku.
21:57–22:55		Haiku effort diagnostic (exploratory), the same evening.
2026-10-12		
07:11		**Lock 2** cut, with Deviation 1 and the length rule. Public: Zenodo DOI registered 07:20.
07:28–07:53		Length sweep (4 to 16 steps); the rule chose 6 steps.
09:11		**Lock 3** cut: Study 1 at 6 steps, with Deviation 2's marking rule. Public: gist 10:25, Wayback; Zenodo DOI registered 13:28, after the run began.
13:01–13:54		Study 1 main run. Addendum C4 committed at 13:12, before any result was opened.
14:18		Main run's data stamp (gist, Wayback).
2026-10-13		
07:33		**Lock 4** cut: Study 1b and the top-up run. Public: gist 07:34, Wayback, Zenodo record created 07:43.
08:35–11:19		Study 1b on fresh puzzles.
11:26		Study 1b's data stamp (gist, Wayback).
11:50–14:09		Top-up run (exploratory), paused 11:52–13:26 with nothing re-sent.
14:15		Top-up's data stamp (gist, Wayback).

ran under no lock

Figure S9: The study flow of Figure S8 with its dates and times (UTC), from the run logs and the public records. Each lock was cut before any data of the runs it governs existed, and each data stamp was posted before any result of its run was opened. Study 1's reporting definitions were written after Lock 3; the last part of their Addendum C was committed after the main run began and before any result was opened. The top-up run started earlier than Lock 4 had set, a disclosed change, and paused with nothing sent twice. The setup checks before Lock 1 and a small Haiku effort diagnostic between Pilot 1 and Lock 2 ran under no lock and are shaded; the two smoke runs of 10 October 2026, with no logged time, are listed in Table S10.

- **The result:** the picker was not shown to differ from a plain vote over the same 8 answers: +9.64 points (95% interval −1.88 to +21.35) on chains and +2.08 (−6.88 to +10.83) on pieces (exploratory; across the two runs, with the drift check not fired).
- **The cost:** the picker call added about $0.00148 per puzzle at list price.

S3.8 Locks and public fingerprints

Figure S9 sets out the whole sequence with its dates and times.

We cut **Lock 1** at 2026-10-11 19:32:37 UTC, before Pilot 1.

- **What it fixed:** Study 1's plan for 24-step puzzles (Tests 1–5 and their exact text, the headline rule, the bias table, the fairness check, and the rules for draws, extra puzzles and cuts); the puzzle generator and its frozen puzzle sets; the preset-choice and vote-size scripts; the analysis code with fixed seeds; the tests on simulated data with known answers; the power simulator; and the pilot plan.
- **Public proofs:** gist f8da61f9185b5f28bd5780ea55e810a5, created 2026-10-11T19:35:19Z; a Wayback snapshot of it (20261011193547); and Zenodo v1 (DOI `10.5281/zenodo.23304164`), registered with DataCite at 2026-10-11T20:20:40Z.

We cut **Lock 2** at 2026-10-12 07:11:26 UTC, after Pilot 1 and before any length-sweep call.

- **What it fixed:**
  - Deviation 1, which moved the study from 24 steps to N steps, with N to be chosen from 4, 6, 8, 12 and 16 by a Haiku-medium-only rule set in advance;
  - the puzzle generator (ledger-0.5) and new frozen puzzle sets at each length, disjoint from earlier sets;
  - the length-sweep plan, pick rule, runner and curve code;
  - the tests and analysis code with N as a parameter, with their simulated-data checks.
- **Public proof:** Zenodo v2 only (DOI `10.5281/zenodo.23311831`), registered at 2026-10-12T07:20:23Z, before the length sweep's first call was booked at 07:29:27 UTC. The lock procedure had also called for a gist and a Wayback snapshot; none was made for Lock 2.

We cut **Lock 3** at 2026-10-12 09:11:59 UTC, after the length sweep and before any main-run data.

- **What it fixed:**
  - Amendment 1, the one dated amendment to Lock 2, which filled in only the values the pre-set rules chose: the number of steps, 6; the inputs to the vote size; the output-token limits; no extra puzzles; no cuts; the near-ceiling label; and the price window;
  - Deviation 2, with its code and test cases.
- **Public proofs:** Zenodo was unavailable at first, so Lock 3 was first made public as gist eb3ec6a6a9900df0560a4e5a81e07b93, created 2026-10-12T10:25:49Z, with a Wayback snapshot (20261012102601). Zenodo v3 (DOI `10.5281/zenodo.23313537`) was registered at 2026-10-12T13:28:29Z, after the main run began at 13:01 UTC and before any main-run result was opened.

Study 1's reporting definitions were written after Lock 3. Items 1–24 and the first three parts of Addendum C, a section of technical details for those items, were committed by 12:24:52 UTC on 12 October 2026, before the main run began at 13:01:21 UTC. The last part, Addendum C4 (four edge-case pins), was committed at 13:12:58 UTC, about 12 minutes after the main run began and before any result was opened; the main run's public analysis stamp fixes the commit that holds it. We count Addendum C, C4 included, as part of the reporting definitions. It touches none of the five tests.

We cut **Lock 4** on 2026-10-13 at 07:33:17 UTC, before either of the last two runs sent a call.

- **What it fixed:** Study 1b's spec, run plan, runner, vote-size script and analysis driver, and the top-up run's spec, prompt wording, run plan, runner and analysis code.
- **The puzzles:** Study 1b's test puzzles had been frozen in Lock 2 and were never sent to any model under test before the tag (parts of three answer keys were shown in our own work sessions; see Section S5, under "Without the partly shown puzzles").
- **The run commits:** Study 1b ran at the tag's only first-parent child commit, which changed only the price table's date. The top-up ran at a later commit that also moved its start earlier, as a disclosed change (below).
- **Public proofs:** gist f0ec2ea88da93788511b21c9ab036d94, created 2026-10-13T07:34:10Z; a Wayback snapshot (20261013073439); and Zenodo v4 (DOI `10.5281/zenodo.23335207`), created 2026-10-13T07:43:34Z.

Table S5: Each lock's commit ID and tar SHA-256.

Lock	Commit	Tar SHA-256
Lock 1	`b4b6b8474f39538055b3e358b48243d51aca5e3b`	`37454153435ae927ccd5aa3a5f61e4f7033e53a35875b830a3f9ccfd2c610adf`
Lock 2	`95fb5f81b9a3a131da77f5a776af200c8e60477d`	`93a2381571d0eb12ba27bd9084701ecc4ca1e72536bc0300e509ac583e6c615a`
Lock 3	`086fbfd04063a10e2c64f2a737e5771b0aa70dc8`	`1221e8394529d1f5e81bdfebc7827366b32e3fb3aa24cfa39bfe71e98144deb5`
Lock 4	`d1c565fc9188d56e970bbb15ce908271627ccf0a`	`47e40b3597b108088961602267001503 7d356914d0dc379d0ee669acb1b595ad`

The fingerprints. Each lock is a git tag in a repository that stayed private. Its public fingerprint is the tag's commit ID and the SHA-256 of the uncompressed tar from `git -c tar.umask=0002 archive --format=tar <tag>`, made with git 2.39.5, because another git version can write a different tar for the same tree. Table S5 lists each lock's commit ID and tar SHA-256.

Data stamps and later analyses. Besides the locks, we posted three public data stamps, each a gist with a Wayback snapshot, holding only fingerprints and posted before any result of its run was opened. The main run's stamp fixed Study 1's analysis driver and reporting definitions, both written after Lock 3, together with the main run's raw-file fingerprints; the Study 1b and top-up stamps fixed their raw-file fingerprints, and their analysis code was already fixed in Lock 4. The analysis scripts, fixed and time-stamped this way before any result was opened, fix the scorer and the pre-committed readings. New analyses are allowed: each gets its own labelled script, a check of its inputs against the public data stamps, a claims-ledger entry and an exploratory or post hoc label, and none can replace or rescue a test reading.

S3.9 Runs, completeness and cost

All three runs behind the results completed:

- **The main run:** 21,876 calls, with no refusals, truncations or missing answers.
- **Study 1b:** all 11,640 planned samples (280 of them the unscored trial round), with one truncation (Haiku-medium, a 16-step chain) and no refusals.
- **The top-up run:** all 3,820 calls, with no refusals, truncations or missing answers.

In all, the API calls cost $104.14 of the $200 budget, as billed; Table S6 lists every logged run with its cost.

This total is our cost records under the vendor's rule for billing refusals ($104.1353). It agreed with the Anthropic Console's spend figure ($104.14) within one cent, as it did at each earlier Console reading. Our records' raw total, $104.16, also prices two refusals in a setup check that the vendor does not bill ($0.0268).

S3.10 A post hoc analysis corrected after its first run

One post hoc analysis was corrected after its first run. In the Haiku-max vote curves at 6 and 12 steps, a coding error passed each puzzle to the cluster bootstrap as a stratum of its own, so every resample equalled the sample and every band had zero width (for one sample at 6 steps, pooled: 95.75% to 95.75%). A separate checking re-run had reproduced that output byte for byte, which showed that the run was repeatable, not that it was right. We fixed the call to use one stratum holding every puzzle, as Study 1b's pre-registered bootstrap does, and ran the analysis again from a clean checkout as revision 2, with the reason written into its output; the same band became 93.125% to 97.875%. The means, paired differences and costs come from unchanged code. The first output is kept as the record, and the paper uses only the corrected one.

Table S6: Every logged run, in time order, including one the API rejected unbilled and a bookkeeping row: calls as logged, completeness, and spend in US dollars, with the raw spend, the billing-rule total (refusals before any output at zero under the vendor's rule) and the console's readings before and after (Batch at half of list price). A panel gives completeness by setting. Descriptive (operations).

Run	Part	Date	Calls	Refusals	Truncations	Missing	Raw spend (USD)	Raw total to date (USD)	Billing-rule total to date (USD)	Console used before (USD)	Console used after (USD)
smoke-1	Setup check	2026-10-10	1	—	—	—	0.00	—	—	—	—
smoke-2	Setup check	2026-10-10	9	3	—	—	0.07	0.07	0.04	—	0.04
smoke-2 (markers)	Setup check	2026-10-11	0	—	—	—	0.00	—	0.04	—	—
check-ledger-1	Setup check	2026-10-11	20	0	—	—	2.30	—	—	—	—
s3-ledger-1	Setup check	2026-10-11	5	0	—	—	0.29	—	—	—	—
pilot-ledger-1	Pilot 1	2026-10-11	670	0	1	—	8.73	—	—	2.64	—
diag-haiku-effort-1	Effort diagnostic	2026-10-11	36	0	5	—	0.42	—	—	—	—
ladder-l14n-1	Length sweep	2026-10-12	2,400	0	0	—	8.95	—	—	11.78	—
main-l14n6-1	Main run	2026-10-12	21,876	0	0	0	20.97	41.73	41.70	20.72	41.70
study1b-1	Study 1b	2026-10-13	11,640	0	1	0	61.56	103.29	103.26	41.70	103.26
topup-1	Top-up	2026-10-13	3,820	0	0	0	0.88	104.16	104.14	103.26	104.14
All runs	Total							104.16	104.14		

Table S6: (continued)

Panel 1: Completeness by setting (analysis files, item 16)

Run	Setting	Shape	Designed calls (calls made where marked)	Missing	Truncations	Refusals
Main run	Haiku, thinking off	1×6	3,200	0	0	0
Main run	Haiku, thinking off	6×1	3,200	0	0	0
Main run	Haiku-low	1×6	3,200	0	0	0
Main run	Haiku-low	6×1	3,200	0	0	0
Main run	Haiku-medium	1×6	3,200	0	0	0
Main run	Haiku-medium	6×1	3,200	0	0	0
Main run	Haiku-max	1×6	240	0	0	0
Main run	Haiku-max	6×1	240	0	0	0
Main run	Sonnet-low	1×6	80	0	0	0
Main run	Sonnet-low	6×1	80	0	0	0
Main run	Sonnet-medium	1×6	80	0	0	0
Main run	Sonnet-medium	6×1	80	0	0	0
Main run	Opus-low	1×6	80	0	0	0
Main run	Opus-low	6×1	80	0	0	0
Main run	Opus-medium	1×6	80	0	0	0
Main run	Opus-medium	6×1	80	0	0	0
Main run	Haiku-medium (rewording check)	1×6	200	0	0	0
Main run	Haiku-medium (rewording check)	6×1	200	0	0	0
Main run	Haiku-medium, real time	1×6	25	0	0	0
Main run	Haiku-medium, real time	6×1	25	0	0	0
Main run	Sonnet-medium, real time	1×6	5	0	0	0
Main run	Sonnet-medium, real time	6×1	5	0	0	0
Main run	Opus-medium, real time	1×6	5	0	0	0
Main run	Opus-medium, real time	6×1	5	0	0	0
Main run	Step voting*	1×6	1,086	0	0	0
Study 1b	Haiku, thinking off	1×6	200	0	0	0
Study 1b	Haiku, thinking off	6×1	200	0	0	0
Study 1b	Haiku-low	1×6	200	0	0	0
Study 1b	Haiku-low	6×1	200	0	0	0
Study 1b	Haiku-medium	1×6	200	0	0	0
Study 1b	Haiku-medium	6×1	200	0	0	0
Study 1b	Haiku-high	1×6	200	0	0	0
Study 1b	Haiku-high	6×1	200	0	0	0
Study 1b	Haiku-xhigh	1×6	200	0	0	0
Study 1b	Haiku-xhigh	6×1	200	0	0	0
Study 1b	Haiku-max	1×6	400	0	0	0
Study 1b	Haiku-max	6×1	400	0	0	0
Study 1b	Sonnet-low	1×6	40	0	0	0
Study 1b	Sonnet-low	6×1	40	0	0	0
Study 1b	Sonnet-medium	1×6	40	0	0	0
Study 1b	Sonnet-medium	6×1	40	0	0	0
Study 1b	Sonnet-high	1×6	40	0	0	0
Study 1b	Sonnet-high	6×1	40	0	0	0
Study 1b	Sonnet-xhigh	1×6	40	0	0	0
Study 1b	Sonnet-xhigh	6×1	40	0	0	0
Study 1b	Sonnet-max	1×6	40	0	0	0
Study 1b	Sonnet-max	6×1	40	0	0	0
Study 1b	Opus-low	1×6	40	0	0	0
Study 1b	Opus-low	6×1	40	0	0	0

Table S6: (continued)

Run	Setting	Shape	Designed calls (calls made where marked)	Missing	Truncations	Refusals
Study 1b	Opus-medium	1×6	80	0	0	0
Study 1b	Opus-medium	6×1	80	0	0	0
Study 1b	Opus-high	1×6	40	0	0	0
Study 1b	Opus-high	6×1	40	0	0	0
Study 1b	Opus-xhigh	1×6	40	0	0	0
Study 1b	Opus-xhigh	6×1	40	0	0	0
Study 1b	Opus-max	1×6	40	0	0	0
Study 1b	Opus-max	6×1	40	0	0	0
Study 1b	Haiku-max	1×12	800	0	0	0
Study 1b	Haiku-max	12×1	800	0	0	0
Study 1b	Opus-medium	1×12	80	0	0	0
Study 1b	Opus-medium	12×1	80	0	0	0
Study 1b	Haiku, thinking off	1×16	400	0	0	0
Study 1b	Haiku, thinking off	16×1	400	0	0	0
Study 1b	Haiku-low	1×16	400	0	0	0
Study 1b	Haiku-low	16×1	400	0	0	0
Study 1b	Haiku-medium	1×16	400	0	1	0
Study 1b	Haiku-medium	16×1	400	0	0	0
Study 1b	Haiku-high	1×16	400	0	0	0
Study 1b	Haiku-high	16×1	400	0	0	0
Study 1b	Haiku-xhigh	1×16	400	0	0	0
Study 1b	Haiku-xhigh	16×1	400	0	0	0
Study 1b	Haiku-max	1×16	800	0	0	0
Study 1b	Haiku-max	16×1	800	0	0	0
Study 1b	Opus-medium	1×16	160	0	0	0
Study 1b	Opus-medium	16×1	160	0	0	0
Study 1b	Trial round (all its settings; answers never scored)	—	280	—	—	—
Top-up	Haiku team, planner	1×6	80	0	0	0
Top-up	Haiku team, planner	6×1	80	0	0	0
Top-up	Haiku team, helper draws*	1×6	1,035	0	0	0
Top-up	Haiku team, helper draws*	6×1	1,044	0	0	0
Top-up	Haiku team, aggregator	1×6	80	0	0	0
Top-up	Haiku team, aggregator	6×1	80	0	0	0
Top-up	Haiku picker	1×6	80	0	0	0
Top-up	Haiku picker	6×1	80	0	0	0
Top-up	Step voting on pieces*	6×1	1,061	0	0	0
Top-up	Drift check	1×6	100	0	0	0
Top-up	Drift check	6×1	100	0	0	0

A dash means the run's row in our cost records does not state that figure; an empty cell means the figure does not apply to the total. Dollar columns are rounded to whole cents, as the Console reports them; the notes give more decimals where a check needs them. Raw spend is the figure from our per-call cost records: every call priced from its logged usage at the price paid (Batch calls at half of list price), refusals included. The billing rule bills a refusal that stopped before any output at zero, so its total is a little lower where such refusals exist. Totals to date and the Console's after-run reading are cumulative over the runs up to and including the row. The before-run reading covers the earlier runs only, so it checks the spend of all earlier runs. Study 1b's totals are read from the note on its post-run Console reading, kept with our ruling on Study 1b's data stamp; the other rows' are from their own rows in our cost records. The Console reports whole cents. Console readings are the 'used' figures recorded with each run. Before the main run, that figure (20.72) is the credit of 200 less the 179.28 left; the same dashboard reading's 'Spend this month' read 20.73 (quoted with our rulings before the main run). Within one cent: each Console reading after a run agrees with the billing-rule total to date within one cent (billing-rule total against the Console's used figure, as each record states it; a figure recorded to more than four decimals is rounded to four): smoke-2 0.0439 against 0.04; main-l14n6-1 41.70 against 41.70; study1b-1 103.26 against 103.26; topup-1 104.1353 against 104.14. The

raw total to date after topup-1, 104.1621, is more than one cent above that reading: it also prices the two zero-output setup-check refusals that the vendor does not bill (row smoke-2 (markers)). Counts are over the designed calls. *: step voting (in the main run, and on pieces in the top-up) and the Haiku team's helper draws are adaptive and have no designed count, so their figure is the calls actually made (each analysis file's label says so). Step voting's missing count in the main run is over its chains. Study 1b's trial round is outside item 16: the analysis file stamps its calls, tokens and costs only (dashes), and its records leave out 280 trial-round calls. With the trial round, each run's rows add up to the calls in its row of the main panel. Refusals are the sum of the analysis file's category counts; an empty list sums to zero.

S3.11 Further analyses on the collected data

After the pre-registered results were open, we added twelve analyses on data already collected, with no new model calls. Each output was written by one run from committed code, which checked its inputs against the stamped files (Table S7 lists the fields the paper does not repeat as written) and wrote its own labelled output file, never overwriting an earlier one; each is reported as post hoc, never as a test, and none can change a pre-registered reading. A separate checking re-run reproduced every output byte for byte, and checked that every interval had width (unless its data were all right or all wrong) and contained its point (see the correction above). For each we give its purpose, its label, its output file with that file's SHA-256, and the commit of the code that wrote it. Each was specified, and its code committed, before its output was computed; the setting accuracies it combines were already known. The numbering skips one planned analysis, a cost for a target accuracy, which became Table S33.

- **Free analysis 1**, post hoc exploratory (no Holm, no verdict; each row within one study): at matched cost on the same puzzles, one Haiku-max sample against a vote of lower-effort Haiku samples (medium in Study 1; medium, high and xhigh in Study 1b), paired over puzzles. Output `data/main/free-analyses/fa1.json`, SHA-256 a9dd18fed80c2d693824e742e60d5c07e4c8314fb5c1a39d6d23f4f993e4c5bb; code `harness/analysis/free_analyses.py` at commit 048020c.
- **Free analysis 2**, post hoc descriptive: the Haiku-max vote curves at 6 and 12 steps in Study 1b, with the cost-matched Haiku-max vote against Opus-medium at those lengths, paired. This is revision 2 of its output; the first output, whose bands had collapsed, is kept as the record (see the correction above). Output `data/main/free-analyses/fa2r2.json`, SHA-256 92aa3cfe2fb26c00b34d11edbf54b627b938a4f2db06d1306830a2f9b8e94838; code `harness/analysis/free_analyses.py` at commit cb51b58.
- **Free analysis 3**, post hoc descriptive sensitivity (spec §4: 'k-sensitivity: none'): B2's vote size at 16 steps: every vote size against Opus-medium, paired, with its cost, and each shape's own cost-matched size. Output `data/main/free-analyses/fa3.json`, SHA-256 265701337861586facf1e86cd2553819f8277eeab3bcb5716f0a38df6d18ab4e; code `harness/analysis/free_analyses.py` at commit 048020c.
- **Free analysis 4**, post hoc presentation: exact two-sided intervals (Clopper–Pearson, over puzzles), each at the level of the interval it replaces, for every zero-width interval in the result files it reads, whatever its field is called: 117 cells in all. Two kinds are left out: the per-piece rate pooled over all lengths and sets (q), and the compounded reference built from it, where no single count of puzzles carries the rate; and the cells of item 20's lookup chart below 40 samples, which no table or figure prints. This is revision 3 of its output. It keeps the second revision's 105 cells unchanged and adds item 9e's Opus-medium accuracy on the even puzzles (3 copies of one interval), which Table S19 prints. It also computes exact 90% intervals for the fairness check's 9 misapplied-rule rates; these are recorded in the output but not reported. The earlier outputs, the second revision and the first version (13 cells), are kept as the record. Output `data/main/free-analyses/fa4r3.json`, SHA-256 61ecbaf4ae31a1be4ba280323af0224be375d496a1a022d581b3f9a96e6b2f91; code `harness/analysis/free_analyses.py` at commit c224d69.

Table S7: Data README: the stamped fields of the result files that the paper does not repeat as written, each beside the wording or key the paper uses instead (disclosure).

Stamped field	Stamped value (or [a description])	In the paper
main: confirmatory.headline_wording.t5	matches	Test 5's stamped headline word, the pre-registration's own word for the no-worse-than reading within the margin. The paper writes: shown no worse than the comparator by more than the margin (Holm-adjusted). The values for Tests 1–4 read 'does not match as deployed'; the paper uses the pre-registered decision rule.
main: original_repeat_rule_appendix.format_compliance	[repeats the official-rule values, e.g. Sonnet-medium chain: 97.50%]	Original-rule format compliance is read from secondary.rates (and item 10 for the other settings), e.g. Sonnet-medium chain: 65.00% (Table S16, Panel 1).
groupc: label	pre-specified descriptive (definition fixed and timestamped before the data, [the 12 October analysis] stamp; computed after the confirmatory results were opened, because it was not in the stamped driver)	The keep-list definitions were fixed before any result was opened, inside the public analysis stamp; they were computed after the confirmatory results were opened. Table S18, which holds these analyses, says so in its caption: "The planned descriptive analyses, computed after the confirmatory results were opened". That caption uses this wording, not the stamped label's.
groupc: inputs.k	[a scratch path on the build machine]	The k.json used is the committed Study 1 k.json: its SHA-256 equals the pinned file's (both in Panel 1).

Table S7: (continued)

Stamped field	Stamped value (or [a description])	In the paper
topup: costs.price_table_date	2026-10-12	The analysis code's own price-table date; the run's price date is 2026-10-13, as our ruling of 2026-10-13 records. The prices it applies are the same list prices the run was cleared on, re-checked unchanged before the run; the paper states list prices once, with their check date.
topup: inputs.plan_sha256	[a SHA-256 hash, in full in Panel 1]	The driver was given the run folder's own copy of the plan (topup_plan.json), so its plan check compares that copy with itself; the value is that copy's file hash, as in the run's row of our cost records. Before the driver ran, the cleared plan's hash (e038a46e…, as in that row) and all 13 run-file hashes (the top-up's and the main run's, as in their rows of our cost records) were checked (our work log's entry for the driver run).
main: "reporting.group_b.item_9b.arms.rt-haiku-med:1x6.batch_mean"	[equals its list mean; its batch_label reads 'real time (list)']	The six real-time settings were billed at list price, so their 'batch_mean' holds the list cost; the paper reports them at list.

Table S7: (continued)

Stamped field	Stamped value (or [a description])	In the paper
topup: comparisons.topup:e2-e3:6x1.comparator	c13	The comparison's key reads as the Haiku team against the main run's step voting, but on pieces its comparator is the top-up's step voting on pieces, as this field says; in the main run, step voting ran on chains only. The paper follows this field and calls the comparator step voting on pieces.
topup: test4_pieces.label	unadjusted (exploratory, outside Holm)	The key's name reads as Test 4 at pieces, the top-up spec's convenience label. The paper never names it a test (only Tests 1–5 and B1–B3 are tests). It writes: step voting on pieces against one Opus-medium answer (the pieces counterpart of Test 4), unadjusted, exploratory, outside the pre-registered family; the drift check between runs: not fired (Table S1).
topup: test4_pieces.methods_note	"this arm ran in a different window from its comparator (the top-up run's own window, after the main run's; spec section 1, [review note])"	The top-up was a run of its own, which started within 48 hours of the main run's start by a disclosed change; comparisons across the two runs carry the status of the drift check between runs (Table S15). In the stamped value, the words 'arm' and 'window' read, in the paper, design and run.
main: fairness.flag_k.reason	"does not fire (TI_house not assessable at pilot scale: opus+sonnet short (0 exposed slips < 3; counts {'exposed_slips': 0, 'exposed_steps': 457, 'G_slips': 11, 'G_steps': 1249}))"	Flag K did not fire because the house-rule targeting index could not be assessed: Opus and Sonnet together had 0 house-rule slips, fewer than the 3 the index needs. The main run is not a pilot: the paper gives the counts and does not repeat 'pilot scale' (Table S30).

Panel 1: SHA-256 values named in the rows

Field	SHA-256
groupc: inputs.k_json_sha256	`b0f8b05f729b5e6cee6deaa2384c986646217adf72a96772d5096bd2d6e3234b`
The pinned Study 1 k.json (Table S34)	`b0f8b05f729b5e6cee6deaa2384c986646217adf72a96772d5096bd2d6e3234b`
topup: inputs.plan_sha256	`c24960360b2ff0fdf1e69b599120bf909197d404baae02b8359b23b237e3b3dc`

Each row is a stamped field that the paper does not repeat as written, beside the wording or key the paper uses instead. The first column gives the file (main: the Study 1 analysis; groupc: the Study 1 keep-list analyses; topup: the top-up analysis; Table S34 gives their paths) and the field's key path in it, its keys joined by dots. The key paths and the values in the 'Stamped value' column are quoted verbatim from the stamped files, as is every stamped word in single quotes in the 'In the paper' column; some are words the paper's own text does not use, and a key path or value that holds one is set in double quotation marks. Square brackets mark our changes: a value wholly in square brackets describes the stamped value rather than quoting it, and a bracketed phrase inside a quoted value replaces the words the next notes name. Panel 1 gives the SHA-256 values these rows name, in full. The released data files' labels cite internal decision numbers; the decision register is not released, so each is given here in plain words. In stamped text, a reviewing session's name and review round are replaced by [review note].

- **Free analysis 5**, post hoc sensitivity: Study 1's Tests 1–4 under a multiple-testing correction (Holm in the inferiority direction, from the locked code and seeds), chosen after the main results were open, with a stricter 99% interval beside each test. Output `data/main/free-analyses/fa5.json`, SHA-256 13fe74329bebcb83e1a4cc15ba90b393e98a625781ba2233d2bf349ae26ec21e; code `harness/analysis/free_analyses.py` at commit 048020c.
- **Free analysis 6**, post hoc exploratory (no causal claim): the wrong-answer repeat share at each Haiku effort and length, and the Haiku-max vote at 16 steps grouped by Haiku-medium's single-sample accuracy; the groups' sizes (output `data/main/free-analyses/fa6a.json`, SHA-256 6af8ea0b47f938fcb3855ed3e58caf417393f45f9925a64af86a990018874258, code 048020c) were written before any Haiku-max value was grouped. Output `data/main/free-analyses/fa6b.json`, SHA-256 76d0baaf276c045dc63fa5d38a83d045a9f1fcdc9f342aa86cb9a296e8486c45; code `harness/analysis/free_analyses.py` at commit 1a7649a.
- **Free analysis 7**, exploratory (labelled post hoc script): Study 1: one Haiku-max sample against one Opus-medium answer per shape, paired over puzzles. Output `data/main/free-analyses/fa7.json`, SHA-256 9f71dc7893cf35c1b1e96fbeee9dd189c678731e88b47ae86949e063b387db26; code `harness/analysis/free_analyses.py` at commit 048020c.
- **Free analysis 8**, post hoc descriptive: Haiku, thinking off: replies that did not keep the format separated from wrong answers, with the share right among replies that kept it. Output `data/main/free-analyses/fa8.json`, SHA-256 33f5ed5fad5bb936cd7aa96d9b6718be6900547a4044fb9c1ba23a63ac912f0c; code `harness/analysis/free_analyses.py` at commit 048020c.
- **Free analysis 9**, post hoc exploratory; against Test 5 a cross-study reading: Sonnet-high, Sonnet's default effort, on Study 1b's 6-step set, paired against Opus-low, Opus-medium, one Haiku-max sample and Sonnet-medium, with its cost against each. Output `data/main/free-analyses/fa9.json`, SHA-256 88db8da4553ea4b6359bdbbc8e77ad070682c1441a85dba12ead6174b5a94f44; code `harness/analysis/free_analyses.py` at commit e4321f9.
- **Free analysis 10**, pre-specified descriptive (items 9g, 19) plus post hoc descriptive: the API's own thinking-token counts, and token and cost structure by setting and length, including Haiku-max's output-token and cost ratios to Opus-medium. Output `data/main/free-analyses/fa10.json`, SHA-256 74bdca2c18c23b6542af2e32555c1e62fd1f8e96f7789ac1f807c1711154df54; code `harness/analysis/free_analyses.py` at commit 048020c.
- **Free analysis 12**, post hoc sensitivity: the cost-matched vote size and the vote's paired gap for Tests 1–3 and B2 re-priced at comparator-to-Haiku price ratios of 10, 20, 40 and 80 times, and the break-even price ratio for Haiku-max at 6, 12 and 16 steps. Output `data/main/free-analyses/fa12.json`, SHA-256 63d009747ae20318f3332a61fd6f088c4eead9694a754e122af5146f68c34a81; code `harness/analysis/free_analyses.py` at commit e4321f9.
- **Free analysis 13**, post hoc: the tie view for Test B2, from the locked data: how often a tie at the top decided a B2 vote, and B2's paired difference with a vote decided by a tie counted wrong. Output `data/main/free-analyses/fa13.json`, SHA-256 ca4d0805ded485d6787fc3b099064101d47c4e3f8b9f70c27937ce45d2f92a32; code `harness/analysis/free_analyses.py` at commit 3590871.

Table S8 gives the exact intervals where a bootstrap interval collapsed to zero width (free analysis 4).

Table S8: Exact intervals where a bootstrap interval collapsed to zero width (post hoc† presentation). Each row replaces the stamped, collapsed interval with an exact two-sided 95% Clopper–Pearson interval, counting each puzzle once (the conservative unit), within its own study, puzzle set and length, with both shapes together only where the row's key says pooled; no row pools studies or lengths. The rows come from Study 1, Study 1b, the top-up run and the length sweep (with the Pilot 1 markers the length-sweep files carry), and a panel gives the collapsed bars of Figure S10.

What it is	Stamped key (of the collapsed interval; its field name follows where it is not ci95)	Study or run	Label of the stamped interval (from its source; see the labels note)
Study 1b Opus-medium, 6 steps, pieces	"descriptive.arms.om@6.6x1"	Study 1b	descriptive
Top-up comparator Opus-medium (main run), pieces	accuracy.main-om.6x1	Top-up run	exploratory (no Holm, no verdict)
C10 Opus-low, pieces (main run)	item_C10.pairs.opus-low-vs-opus-med:6x1	Study 1, keep-list analyses	exploratory: no Holm, no verdict
C10 Opus-medium, pieces (main run)	item_C10.pairs.opus-low-vs-opus-med:6x1	Study 1, keep-list analyses	exploratory: no Holm, no verdict
Item 7 Opus-medium per-piece accuracy	"reporting.group_b.item_7.per_piece_accuracy.arms.opus-med"	Study 1, main run	secondary, descriptive (no test)
Item 7 Opus-low per-piece accuracy	"reporting.group_b.item_7.per_piece_accuracy.arms.opus-low"	Study 1, main run	secondary, descriptive (no test)
Length sweep Opus-medium per-step, 6 steps, pooled	models.om.by_n.6.per_step.pooled	Length sweep, scored under Deviation 2	descriptive
Length sweep Opus-medium per-step, 6 steps, pieces (q's input)	models.om.by_n.6.per_step.nx1	Length sweep, scored under Deviation 2	descriptive
Length sweep Opus-medium per-step, 16 steps, pooled	models.om.by_n.16.per_step.pooled	Length sweep, scored under Deviation 2	descriptive
Length sweep Opus-medium per-step, 16 steps, pieces (q's input)	models.om.by_n.16.per_step.nx1	Length sweep, scored under Deviation 2	descriptive
Study 1 Haiku-max 3-sample vote, chains	"reporting.group_b.item_10.arms.haiku-max 1x6.vote_curve.points.2"	Study 1, main run	exploratory (no Holm, no verdict)
Study 1b Haiku-max 10-sample vote, 16 steps, pieces	descriptive.vote_curve_n16.points.9.16x1	Study 1b	descriptive
Study 1b Haiku, thinking off, 16 steps, chains	"descriptive.arms.ho@16.1x16"	Study 1b	descriptive

Table S8: (continued)

What it is	Stamped key (of the collapsed interval; its field name follows where it is not ci95)	Stamped bootstrap 95% accuracy interval (%)	Stamped bootstrap 95% interval of a paired difference (points)	Samples per puzzle (a vote: its k)	Puzzles right (of the cell's puzzles; post hoc)	Exact 95% accuracy interval, Clopper–Pearson (%, post hoc)
Study 1b Opus-medium, 6 steps, pieces	"descriptive.arms.om@6.6x1"	[100.00, 100.00]	—	2	40 of 40	[91.19, 100.00]
Top-up comparator Opus-medium (main run), pieces	accuracy.main-om.6x1	[100.00, 100.00]	—	1	80 of 80	[95.49, 100.00]
C10 Opus-low, pieces (main run)	item_C10.pairs.opus-low-vs-opus-med:6x1	—	[0.00, 0.00]	1	80 of 80	[95.49, 100.00]
C10 Opus-medium, pieces (main run)	item_C10.pairs.opus-low-vs-opus-med:6x1	—	[0.00, 0.00]	1	80 of 80	[95.49, 100.00]
Item 7 Opus-medium per-piece accuracy	"reporting.group_b.item_7.per_piece_accuracy.arms.opus-med"	[100.00, 100.00]	—	1	80 of 80	[95.49, 100.00]
Item 7 Opus-low per-piece accuracy	"reporting.group_b.item_7.per_piece_accuracy.arms.opus-low"	[100.00, 100.00]	—	1	80 of 80	[95.49, 100.00]
Length sweep Opus-medium per-step, 6 steps, pooled	models.om.by_n.6.per_step.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep Opus-medium per-step, 6 steps, pieces (q's input)	models.om.by_n.6.per_step.nx1	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep Opus-medium per-step, 16 steps, pooled	models.om.by_n.16.per_step.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep Opus-medium per-step, 16 steps, pieces (q's input)	models.om.by_n.16.per_step.nx1	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Study 1 Haiku-max 3-sample vote, chains	"reporting.group_b.item_10.arms.haiku-max 1x6.vote_curve.points.2"	[100.00, 100.00]	—	3	80 of 80	[95.49, 100.00]
Study 1b Haiku-max 10-sample vote, 16 steps, pieces	descriptive.vote_curve_n16.points.9.16x1	[100.00, 100.00]	—	10	80 of 80	[95.49, 100.00]
Study 1b Haiku, thinking off, 16 steps, chains	"descriptive.arms.ho@16.1x16"	[0.00, 0.00]	—	5	0 of 80	[0.00, 4.51]

Table S8: (continued)

What it is	Stamped key (of the collapsed interval; its field name follows where it is not ci95)	Study or run	Label of the stamped interval (from its source; see the labels note)
Item 24: a vote at its own cost-matched k	reporting.group_b.item_24.table.haiku-max 1x6.vote_at_own_k.own k vs opus-med	Study 1, main run	exploratory (no Holm, no verdict)
Item 24: a vote at its own cost-matched k	reporting.group_b.item_24.table.haiku-max 1x6.vote_at_own_k.own k vs sonnet-med	Study 1, main run	exploratory (no Holm, no verdict)
Study 1b: a figure's copy of a setting	descriptive.effort_figure.N=16.haiku.0.1x16	Study 1b	descriptive
Study 1b: a figure's copy of a setting	descriptive.effort_figure.N=6.opus.1.6x1	Study 1b	descriptive
Study 1b: a figure's copy of a setting	descriptive.length_curve.opus-medium.6.6x1	Study 1b	descriptive
Study 1b: the length sweep's points as copied	descriptive.length_curve.ladder.opus-medium.6.pooled	Study 1b	descriptive
Study 1b: the length sweep's points as copied	descriptive.length_curve.ladder.opus-medium.16.pooled	Study 1b	descriptive
Study 1b: a test's setting per shape	per_form.B3.6x1.o	Study 1b	descriptive per-shape breakdown
Length sweep (original rule), Opus-medium per-step, 4 steps, chains	models.om.by_n.4.per_step.1xn	Length sweep, original marking rule	descriptive
Length sweep (original rule), Opus-medium per-step, 4 steps, pieces	models.om.by_n.4.per_step.nx1	Length sweep, original marking rule	descriptive
Length sweep (original rule), Opus-medium per-step, 4 steps, pooled	models.om.by_n.4.per_step.pooled	Length sweep, original marking rule	descriptive
Length sweep (original rule), Opus-medium per-step, 6 steps, chains	models.om.by_n.6.per_step.1xn	Length sweep, original marking rule	descriptive
Length sweep (original rule), Opus-medium per-step, 6 steps, pieces	models.om.by_n.6.per_step.nx1	Length sweep, original marking rule	descriptive
Length sweep (original rule), Opus-medium per-step, 6 steps, pooled	models.om.by_n.6.per_step.pooled	Length sweep, original marking rule	descriptive

Table S8: (continued)

What it is	Stamped key (of the collapsed interval; its field name follows where it is not ci95)	Stamped bootstrap 95% accuracy interval (%)	Stamped bootstrap 95% interval of a paired difference (points)	Samples per puzzle (a vote: its k)	Puzzles right (of the cell's puzzles; post hoc)	Exact 95% accuracy interval, Clopper–Pearson (%, post hoc)
Item 24: a vote at its own cost-matched k	reporting.group_b.item_24.table.haiku-max 1x6.vote_at_own_k.own k vs opus-med	[100.00, 100.00]	—	3	80 of 80	[95.49, 100.00]
Item 24: a vote at its own cost-matched k	reporting.group_b.item_24.table.haiku-max 1x6.vote_at_own_k.own k vs sonnet-med	[100.00, 100.00]	—	3	80 of 80	[95.49, 100.00]
Study 1b: a figure's copy of a setting	descriptive.effort_figure.N=16.haiku.0.1x16	[0.00, 0.00]	—	5	0 of 80	[0.00, 4.51]
Study 1b: a figure's copy of a setting	descriptive.effort_figure.N=6.opus.1.6x1	[100.00, 100.00]	—	2	40 of 40	[91.19, 100.00]
Study 1b: a figure's copy of a setting	descriptive.length_curve.opus-medium.6.6x1	[100.00, 100.00]	—	2	40 of 40	[91.19, 100.00]
Study 1b: the length sweep's points as copied	descriptive.length_curve.ladder.opus-medium.6.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Study 1b: the length sweep's points as copied	descriptive.length_curve.ladder.opus-medium.16.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Study 1b: a test's setting per shape	per_form.B3.6x1.o	[100.00, 100.00]	—	2	40 of 40	[91.19, 100.00]
Length sweep (original rule), Opus-medium per-step, 4 steps, chains	models.om.by_n.4.per_step.1xn	[100.00, 100.00]	—	1	19 of 19	[82.35, 100.00]
Length sweep (original rule), Opus-medium per-step, 4 steps, pieces	models.om.by_n.4.per_step.nx1	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (original rule), Opus-medium per-step, 4 steps, pooled	models.om.by_n.4.per_step.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (original rule), Opus-medium per-step, 6 steps, chains	models.om.by_n.6.per_step.1xn	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (original rule), Opus-medium per-step, 6 steps, pieces	models.om.by_n.6.per_step.nx1	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (original rule), Opus-medium per-step, 6 steps, pooled	models.om.by_n.6.per_step.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]

Table S8: (continued)

What it is	Stamped key (of the collapsed interval; its field name follows where it is not ci95)	Study or run	Label of the stamped interval (from its source; see the labels note)
Length sweep (original rule), Opus-medium whole-answer, 6 steps, pooled	models.om.by_n.6.whole_answer.pooled	Length sweep, original marking rule	descriptive
Length sweep (original rule), Opus-medium per-step, 8 steps, pieces	models.om.by_n.8.per_step.nx1	Length sweep, original marking rule	descriptive
Length sweep (original rule), Opus-medium per-step, 12 steps, chains	models.om.by_n.12.per_step.1xn	Length sweep, original marking rule	descriptive
Length sweep (original rule), Opus-medium per-step, 12 steps, pieces	models.om.by_n.12.per_step.nx1	Length sweep, original marking rule	descriptive
Length sweep (original rule), Opus-medium per-step, 12 steps, pooled	models.om.by_n.12.per_step.pooled	Length sweep, original marking rule	descriptive
Length sweep (original rule), Opus-medium per-step, 16 steps, chains	models.om.by_n.16.per_step.1xn	Length sweep, original marking rule	descriptive
Length sweep (original rule), Opus-medium per-step, 16 steps, pieces	models.om.by_n.16.per_step.nx1	Length sweep, original marking rule	descriptive
Length sweep (original rule), Opus-medium per-step, 16 steps, pooled	models.om.by_n.16.per_step.pooled	Length sweep, original marking rule	descriptive
Length sweep (original rule), Opus-medium whole-answer, 16 steps, pooled	models.om.by_n.16.whole_answer.pooled	Length sweep, original marking rule	descriptive
Length sweep (original rule): a Pilot 1 marker	pilot1.om.l14_only	Pilot 1, carried beside the length sweep (in its original-rule file)	exploratory
Length sweep (Deviation 2 rule), Opus-medium per-step, 4 steps, chains	models.om.by_n.4.per_step.1xn	Length sweep, scored under Deviation 2	descriptive
Length sweep (Deviation 2 rule), Opus-medium per-step, 4 steps, pieces	models.om.by_n.4.per_step.nx1	Length sweep, scored under Deviation 2	descriptive
Length sweep (Deviation 2 rule), Opus-medium per-step, 4 steps, pooled	models.om.by_n.4.per_step.pooled	Length sweep, scored under Deviation 2	descriptive
Length sweep (Deviation 2 rule), Opus-medium per-step, 6 steps, chains	models.om.by_n.6.per_step.1xn	Length sweep, scored under Deviation 2	descriptive

Table S8: (continued)

What it is	Stamped key (of the collapsed interval; its field name follows where it is not ci95)	Stamped bootstrap 95% accuracy interval (%)	Stamped bootstrap 95% interval of a paired difference (points)	Samples per puzzle (a vote: its k)	Puzzles right (of the cell's puzzles; post hoc)	Exact 95% accuracy interval, Clopper–Pearson (%, post hoc)
Length sweep (original rule), Opus-medium whole-answer, 6 steps, pooled	models.om.by_n.6.whole_answer.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (original rule), Opus-medium per-step, 8 steps, pieces	models.om.by_n.8.per_step.nx1	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (original rule), Opus-medium per-step, 12 steps, chains	models.om.by_n.12.per_step.1xn	[100.00, 100.00]	—	1	19 of 19	[82.35, 100.00]
Length sweep (original rule), Opus-medium per-step, 12 steps, pieces	models.om.by_n.12.per_step.nx1	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (original rule), Opus-medium per-step, 12 steps, pooled	models.om.by_n.12.per_step.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (original rule), Opus-medium per-step, 16 steps, chains	models.om.by_n.16.per_step.1xn	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (original rule), Opus-medium per-step, 16 steps, pieces	models.om.by_n.16.per_step.nx1	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (original rule), Opus-medium per-step, 16 steps, pooled	models.om.by_n.16.per_step.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (original rule), Opus-medium whole-answer, 16 steps, pooled	models.om.by_n.16.whole_answer.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (original rule): a Pilot 1 marker	pilot1.om.l14_only	[100.00, 100.00]	—	not stamped	5 of 5	[47.82, 100.00]
Length sweep (Deviation 2 rule), Opus-medium per-step, 4 steps, chains	models.om.by_n.4.per_step.1xn	[100.00, 100.00]	—	1	19 of 19	[82.35, 100.00]
Length sweep (Deviation 2 rule), Opus-medium per-step, 4 steps, pieces	models.om.by_n.4.per_step.nx1	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (Deviation 2 rule), Opus-medium per-step, 4 steps, pooled	models.om.by_n.4.per_step.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (Deviation 2 rule), Opus-medium per-step, 6 steps, chains	models.om.by_n.6.per_step.1xn	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]

Table S8: (continued)

What it is	Stamped key (of the collapsed interval; its field name follows where it is not ci95)	Study or run	Label of the stamped interval (from its source; see the labels note)
Length sweep (Deviation 2 rule), Opus-medium whole-answer, 6 steps, pooled	models.om.by_n.6.whole_answer.pooled	Length sweep, scored under Deviation 2	descriptive
Length sweep (Deviation 2 rule), Opus-medium per-step, 8 steps, pieces	models.om.by_n.8.per_step.nx1	Length sweep, scored under Deviation 2	descriptive
Length sweep (Deviation 2 rule), Opus-medium per-step, 12 steps, chains	models.om.by_n.12.per_step.1xn	Length sweep, scored under Deviation 2	descriptive
Length sweep (Deviation 2 rule), Opus-medium per-step, 12 steps, pieces	models.om.by_n.12.per_step.nx1	Length sweep, scored under Deviation 2	descriptive
Length sweep (Deviation 2 rule), Opus-medium per-step, 12 steps, pooled	models.om.by_n.12.per_step.pooled	Length sweep, scored under Deviation 2	descriptive
Length sweep (Deviation 2 rule), Opus-medium per-step, 16 steps, chains	models.om.by_n.16.per_step.1xn	Length sweep, scored under Deviation 2	descriptive
Length sweep (Deviation 2 rule), Opus-medium whole-answer, 16 steps, pooled	models.om.by_n.16.whole_answer.pooled	Length sweep, scored under Deviation 2	descriptive
Length sweep (Deviation 2 rule): a Pilot 1 marker	pilot1.om.l14_only	Pilot 1, carried beside the length sweep (in its Deviation 2 file)	exploratory
Free analysis 2: a Haiku-max vote-curve point	result.N=6.curve.8.1x6	Study 1b's samples, free analysis 2 (post hoc)	post hoc descriptive
Free analysis 2: a Haiku-max vote-curve point	result.N=6.curve.9.1x6	Study 1b's samples, free analysis 2 (post hoc)	post hoc descriptive
Item 9e: the cascade's Opus-medium accuracy on its half of the puzzles	reporting.group_b.item_9e.shapes.6x1.5.even (field opus_accuracy_ci95)	Study 1, main run	escalate-to-Opus cascade (offline, exploratory)
Item 9e: the cascade's Opus-medium accuracy on its half of the puzzles	reporting.group_b.item_9e.shapes.6x1.10.even (field opus_accuracy_ci95)	Study 1, main run	escalate-to-Opus cascade (offline, exploratory)
Item 9e: the cascade's Opus-medium accuracy on its half of the puzzles	reporting.group_b.item_9e.shapes.6x1.20.even (field opus_accuracy_ci95)	Study 1, main run	escalate-to-Opus cascade (offline, exploratory)

Table S8: (continued)

What it is	Stamped key (of the collapsed interval; its field name follows where it is not ci95)	Stamped bootstrap 95% accuracy interval (%)	Stamped bootstrap 95% interval of a paired difference (points)	Samples per puzzle (a vote: its k)	Puzzles right (of the cell's puzzles; post hoc)	Exact 95% accuracy interval, Clopper–Pearson (%, post hoc)
Length sweep (Deviation 2 rule), Opus-medium whole-answer, 6 steps, pooled	models.om.by_n.6.whole_answer.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (Deviation 2 rule), Opus-medium per-step, 8 steps, pieces	models.om.by_n.8.per_step.nx1	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (Deviation 2 rule), Opus-medium per-step, 12 steps, chains	models.om.by_n.12.per_step.1xn	[100.00, 100.00]	—	1	19 of 19	[82.35, 100.00]
Length sweep (Deviation 2 rule), Opus-medium per-step, 12 steps, pieces	models.om.by_n.12.per_step.nx1	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (Deviation 2 rule), Opus-medium per-step, 12 steps, pooled	models.om.by_n.12.per_step.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (Deviation 2 rule), Opus-medium per-step, 16 steps, chains	models.om.by_n.16.per_step.1xn	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (Deviation 2 rule), Opus-medium whole-answer, 16 steps, pooled	models.om.by_n.16.whole_answer.pooled	[100.00, 100.00]	—	1	20 of 20	[83.16, 100.00]
Length sweep (Deviation 2 rule): a Pilot 1 marker	pilot1.om.l14_only	[100.00, 100.00]	—	not stamped	5 of 5	[47.82, 100.00]
Free analysis 2: a Haiku-max vote-curve point	result.N=6.curve.8.1x6	[100.00, 100.00]	—	9	40 of 40	[91.19, 100.00]
Free analysis 2: a Haiku-max vote-curve point	result.N=6.curve.9.1x6	[100.00, 100.00]	—	10	40 of 40	[91.19, 100.00]
Item 9e: the cascade's Opus-medium accuracy on its half of the puzzles	reporting.group_b.item_9e.shapes.6x1.5.even (field opus_accuracy_ci95)	[100.00, 100.00]	—	1	40 of 40	[91.19, 100.00]
Item 9e: the cascade's Opus-medium accuracy on its half of the puzzles	reporting.group_b.item_9e.shapes.6x1.10.even (field opus_accuracy_ci95)	[100.00, 100.00]	—	1	40 of 40	[91.19, 100.00]
Item 9e: the cascade's Opus-medium accuracy on its half of the puzzles	reporting.group_b.item_9e.shapes.6x1.20.even (field opus_accuracy_ci95)	[100.00, 100.00]	—	1	40 of 40	[91.19, 100.00]

Table S8: (continued)

Panel 1: Item 20: the Figure S10 bars whose bootstrap interval collapsed (the Haiku-medium vote at its largest k, by single-sample accuracy bin and agreement third)

Shape	Haiku-medium single-sample accuracy bin (%)	Agreement third	Puzzles right (of the cell's puzzles; post hoc)	Exact 95% accuracy interval, Clopper–Pearson (%, post hoc)
1×6	[0,10)	lowest third	0 of 6	[0.00, 45.93]
1×6	[0,10)	middle third	0 of 5	[0.00, 52.18]
1×6	[0,10)	highest third	0 of 5	[0.00, 52.18]
1×6	[10,20)	middle third	0 of 3	[0.00, 70.76]
1×6	[10,20)	highest third	0 of 2	[0.00, 84.19]
1×6	[20,30)	lowest third	2 of 2	[15.81, 100.00]
1×6	[20,30)	middle third	0 of 2	[0.00, 84.19]
1×6	[20,30)	highest third	0 of 2	[0.00, 84.19]
1×6	[30,40)	lowest third	3 of 3	[29.24, 100.00]
1×6	[30,40)	middle third	0 of 3	[0.00, 70.76]
1×6	[30,40)	highest third	0 of 2	[0.00, 84.19]
1×6	[40,50)	lowest third	3 of 3	[29.24, 100.00]
1×6	[50,60)	lowest third	3 of 3	[29.24, 100.00]
1×6	[50,60)	middle third	3 of 3	[29.24, 100.00]
1×6	[50,60)	highest third	2 of 2	[15.81, 100.00]
1×6	[60,70)	lowest third	1 of 1	[2.50, 100.00]
1×6	[60,70)	middle third	1 of 1	[2.50, 100.00]
1×6	[60,70)	highest third	1 of 1	[2.50, 100.00]
1×6	[70,80)	lowest third	3 of 3	[29.24, 100.00]
1×6	[70,80)	middle third	2 of 2	[15.81, 100.00]
1×6	[70,80)	highest third	2 of 2	[15.81, 100.00]
1×6	[80,90)	lowest third	2 of 2	[15.81, 100.00]
1×6	[80,90)	middle third	2 of 2	[15.81, 100.00]
1×6	[90,100]	lowest third	3 of 3	[29.24, 100.00]
1×6	[90,100]	middle third	1 of 1	[2.50, 100.00]
1×6	[90,100]	highest third	1 of 1	[2.50, 100.00]
1×6	[90,100]	too few wrong samples	6 of 6	[54.07, 100.00]
6×1	[0,10)	lowest third	0 of 3	[0.00, 70.76]
6×1	[0,10)	middle third	0 of 3	[0.00, 70.76]
6×1	[0,10)	highest third	0 of 2	[0.00, 84.19]
6×1	[10,20)	lowest third	0 of 2	[0.00, 84.19]
6×1	[10,20)	middle third	0 of 2	[0.00, 84.19]
6×1	[10,20)	highest third	0 of 2	[0.00, 84.19]
6×1	[20,30)	middle third	0 of 3	[0.00, 70.76]
6×1	[20,30)	highest third	0 of 3	[0.00, 70.76]
6×1	[30,40)	lowest third	0 of 1	[0.00, 97.50]
6×1	[30,40)	middle third	0 of 1	[0.00, 97.50]
6×1	[30,40)	highest third	0 of 1	[0.00, 97.50]
6×1	[40,50)	lowest third	4 of 4	[39.76, 100.00]
6×1	[40,50)	middle third	4 of 4	[39.76, 100.00]
6×1	[50,60)	lowest third	3 of 3	[29.24, 100.00]
6×1	[50,60)	middle third	3 of 3	[29.24, 100.00]
6×1	[50,60)	highest third	2 of 2	[15.81, 100.00]

Table S8: (continued)

Shape	Haiku-medium single-sample accuracy bin (%)	Agreement third	Puzzles right (of the cell's puzzles; post hoc)	Exact 95% accuracy interval, Clopper–Pearson (%, post hoc)
6×1	[60,70)	lowest third	3 of 3	[29.24, 100.00]
6×1	[60,70)	middle third	2 of 2	[15.81, 100.00]
6×1	[60,70)	highest third	2 of 2	[15.81, 100.00]
6×1	[70,80)	lowest third	3 of 3	[29.24, 100.00]
6×1	[70,80)	middle third	3 of 3	[29.24, 100.00]
6×1	[70,80)	highest third	3 of 3	[29.24, 100.00]
6×1	[80,90)	lowest third	3 of 3	[29.24, 100.00]
6×1	[80,90)	middle third	3 of 3	[29.24, 100.00]
6×1	[80,90)	highest third	3 of 3	[29.24, 100.00]
6×1	[90,100]	lowest third	4 of 4	[39.76, 100.00]
6×1	[90,100]	too few wrong samples	5 of 5	[47.82, 100.00]

Each stamped bootstrap interval has zero width because every puzzle in the cell was right, or every one wrong; the exact interval replaces it. Nothing here is a test, and nothing here changes a pre-registered reading. Exact interval: the two-sided 95% Clopper–Pearson interval for the puzzles right out of the cell's puzzles, with the puzzle as the unit (the conservative unit: a puzzle with several samples, or a vote, counts once). It counts puzzles, not samples. Scope (free analysis 4, revision 3): the rows of this table and of its item 20 panel are every zero-width interval (a stamped interval whose two ends are equal) that free analysis 4 covers, under any interval field name, in the result files its scan reads (Study 1, main run; Study 1, keep-list analyses; Study 1b; Study 1b's samples, free analysis 2 (post hoc); Top-up run; Length sweep, original marking rule; Length sweep, scored under Deviation 2), except the 9 that are not reported (below). Cells: the previous revision's 105 cells, unchanged (the first version's 13, and 92 added by rule, all under the key ci95), and the 12 that this revision added by scanning every interval field by name: 3 are rows here (item 9e's Opus-medium accuracy on the even-indexed puzzles, stamped under opus_accuracy_ci95) and 9 are not reported. Not reported: this revision also computed 9 exact 90% intervals (Clopper–Pearson, at the stamped interval's own level) for the fairness check's misapplied-rule rates, stamped under ci90; they are not reported (beyond the approved scope), and no table or figure prints them. Left out (each keeps its zero-width stamped interval and has no exact interval), with free analysis 4's counts and reasons: 1,061 zero-width intervals in Study 1, main run ("item 20 at k below 40: Figure S10 draws the vote at $k = 40$ only, so these cells are printed nowhere"); 1 zero-width interval in Length sweep, original marking rule ("q pools every piece-step over all lengths and sets (clusters stratified by set), so no single count of base puzzles carries it, and its q^N bands are powers of its interval; no exact interval"); 6 zero-width intervals in Length sweep, original marking rule ("q's q^N bands: powers of q's interval; q has no exact interval"); 1 zero-width interval in Length sweep, scored under Deviation 2 ("as for the original rule's q"); 6 zero-width intervals in Length sweep, scored under Deviation 2 ("q's q^N bands: powers of q's interval; q has no exact interval"). Recount: the scan finds 1,191 zero-width intervals in these files, under any field name; 107 are behind the rows and the item 20 panel (a paired difference's one interval has a row for each of its two settings), 1,075 are left out and 9 are not reported. Scan: free analysis 4 reads every two-number field in these files; it counts 4,819 of them as intervals, those whose field name is one of: ci, cp or wilson followed by two digits (for example ci95), with or without a prefix that ends in an underscore; exactly 'band'; or any name ending in _bars, and the only one it does not count as an interval is quartiles (3 fields: the first and third quartiles of a spread, not an interval for an estimate). C10 rows: "the stamped interval is the paired difference's [0, 0]; each arm's exact accuracy interval is given instead" (free analysis 4's note, where its "arm" is a setting). The difference is Opus-low minus Opus-medium on pieces, in points, and is shown in its own column; a difference has no exact counterpart. Length sweep: q is a model's per-step accuracy, the share of all its pieces right pooled over the sweep's lengths (4, 6, 8, 12 and 16 steps), and q to the power N is the reference for a chain of N steps. The length sweep's by-length rows here are all Opus-medium's, and the inputs of q are its pieces per-step rows (keys ending per_step.nx1), here at 4, 6, 8, 12 and 16 steps under both marking rules, each over that length's puzzles; the mark (q's input) in a row's name is the first version's own wording, so only its rows carry it (the rows added by rule are named by their family, model, measure, length and shape, and their keys show which are inputs of q); Opus-medium's q itself, pooled over every length, has a zero-width stamped interval under each rule and no exact interval (the scope note gives free analysis 4's reason), while the other models' q intervals have width. Labels: the counts and the exact intervals are post hoc presentation; each stamped interval keeps its source's label, given in its own column: the source's stamped label (for Study 1b's descriptive blocks, the block's name; for the length sweep, its stamped plan line; for the Pilot 1 markers the length-sweep files carry, exploratory, the label the plan gives Pilot 1's 24-step markers (they ran before Lock 2, with a different rule mix, sets and timing); for free analysis 2, its stamped label), or, where the source stamps none, the plan's label for the output that reports it (Figure S10 for item 20; Table S22 for Study 1b's per-test cells). Samples per puzzle: a vote row gives its k, the samples it votes over; a row with several single samples averages over them; either way the puzzle counts once in the count of puzzles right; a Pilot 1 marker's samples are not stamped in the length-sweep files. Item 24: its two rows are item 10's own-k votes (checked equal in k, accuracy and cost), and the main analysis stamps no label for item 24, so they carry item 10's stamped label, "exploratory (no Holm, no verdict)", as Table S17 prints them. Names: Item 7 is a Study 1 reporting item, per-piece accuracy: the share of a sample's pieces that are right, averaged over the puzzle's designed samples and then over puzzles; Item 9e is a Study 1 reporting item, the offline escalate-to-Opus cascade: its key ends in the number of

Haiku-medium samples in the cascade's vote (k_c) and then even, the even-indexed puzzles, where the cascade is scored; Opus-medium's accuracy there is one estimate, the same for every k_c, so these rows repeat it; C10 is the keep-list analysis of the paired difference in single-sample accuracy between two effort levels of one model; a vote of k samples scores one answer per puzzle from those samples; a vote-curve point is the vote at one k; 'pooled' in a row name or key means both shapes together, each puzzle counted once, not pooled over lengths; 'chains' is the chain shape. Keys: a shape is written with the letter x, so 6x1 is the shape 6×1; 1xn and nx1 are the chain and the pieces at the row's length; the number after 'by_n.' or after 'N=' in a key is the row's length in steps; per_step is per-step accuracy and whole_answer whole-answer accuracy; a model is written as a tag (om: Opus-medium; ho: Haiku, thinking off), and tag@N is that model at N steps; points.j and curve.j are a point of a vote curve by its place in the list (its k is in the samples column); Study 1b's descriptive.effort_figure and descriptive.length_curve keys are the copies its figures draw of a setting, or, under length_curve.ladder, of a length-sweep block, each checked equal to what it copies; 'own k vs opus-med' (item 24) is the vote at the row's own cost-matched k against Opus-medium; 'own k vs sonnet-med' (item 24) is the vote at the row's own cost-matched k against Sonnet-medium; per_form.B3.6x1.o is Test B3's cell on the pieces for its Opus-medium setting (the test's per-shape breakdown); pilot1.om.l14_only is a Pilot 1 result that the length-sweep files carry beside the sweep, stamped "pilot-1, presets l14, before lock-2 (different rule mix, sets and timing)". Words: each row's description is free analysis 4's own, in this paper's words where it used a word or a form of a name this paper no longer uses (setting for "arm", sample for "try", puzzle for "base puzzle", cost-matched for "equal-money", "Haiku, thinking off" for "Haiku thinking-off", "-sample vote" after a vote's k for "vote@k"), each name starting with a capital, and a length-sweep row added by rule named in full from its key (its family, then the model, measure, length and shape); a stamped key that holds such a word is printed as stamped, in quotation marks. Values stamped as shares are printed in %. Item 20 (Study 1, main run): the Figure S10 bars whose stamped bootstrap interval collapsed, 54 of Figure S10's 59; each is the Haiku-medium vote at $k = 40$ (its largest k: all of a puzzle's samples) over the puzzles of one single-sample accuracy bin and one agreement third (or the bin's "too few wrong samples" group); Figure S10's other bars keep stamped intervals with width and are not here, and free analysis 4 names these cells "item 20: the vote at $k = 40$ within a bin and third". Each cell's stamped bootstrap interval has zero width, because every puzzle in it was right, or every one wrong (the count right out of the cell's puzzles says which); the exact interval replaces it. Agreement thirds split a bin's puzzles by how often their wrong samples repeat each other, lowest first; "too few wrong samples" holds the puzzles with too few wrong samples to be placed in a third. Labels: each stamped interval keeps the plan's label for Figure S10, which draws it (the main analysis stamps no label for item 20): pre-specified descriptive; no test, no cause; the counts and the exact intervals are post hoc presentation. Samples per puzzle: 40 (the vote's k); a puzzle counts once in the count of puzzles right. Item 20's cells at smaller k are left out of free analysis 4 (the scope note gives its count and reason).

S3.12 Changes to the locked plans (Table S9)

Table S9 lists every change to a locked plan, with its date, its trigger and the data that existed when it was made. Deviation 1 (Study 1 ran at 6 steps, not at the planned 24 steps) and Deviation 2 (repeated step lines) are described in the Method. The other changes are smaller, and none touched a test.

- **The main run's rules.** Before the main run, a fixed hard spending cap replaced the pre-registered cap, a multiple of the projected cost, so that the run would not need many small waves; the pause on truncated replies was kept, and the run was allowed to start before Lock 3's Zenodo record was registered (its gist and Wayback snapshot came first). Actual spend was unaffected. No main-run data existed.
- **Amendment 1's erratum.** Amendment 1 filled in only the values the locked rules chose after the length sweep, with its disclosures; it added no test and changed no rule. One of its lines says that Haiku's and Opus's numbers are the same under both marking rules. That understates what moved under Deviation 2's rule: Pilot 1's Opus-medium marker at 24 steps also changed, and so, slightly, did Haiku-medium's per-step rate on chains at 8 and 12 steps and its count of chain samples that could not be placed at 8, 12 and 16 steps (Table S24 and its note). We correct the line here as a disclosed erratum.
- **Study 1b's 12-step point.** It was sent once more after earlier submits had failed to upload; those created no batch and cost nothing. This is a disclosed deviation from the resubmit limit, and the 12-step point is descriptive only. When it was sent, only the other waves' counts, stop reasons and costs had been seen, no answer or score.
- **Lock 4's record merge.** Merging the Lock 4 tag into the main branch gave the tag a second child commit. The run commit is the tag's only first-parent child, and the merge holds records only, no run code. It came before any Study 1b or top-up data.

Table S9: Deviations and disclosed changes in Study 1, Study 1b and the top-up run (disclosure).

Item	Ruling or record	Date	Change	Trigger	Data existing at the time	Where reported
Deviation 1	Our ruling on shorter puzzles (a disclosed deviation); spec Amendments, heading Deviation 1	2026-10-12	Puzzle length moved from 24 steps to 6 steps; the new length was picked by a Haiku-only rule fixed before the length sweep ran	Pilot 1 scored Haiku-medium at about zero at every preset, so the pre-registered preset rule returned the decision to us	Pilot 1 had run; its Haiku results were read and its Opus and Sonnet answers and scores were still sealed; the Haiku-only effort diagnostic had run and was read; no length-sweep, main-run or later data	Section 3.3 of the paper; Section 4.1 of the paper, where Tests 1–5 are first reported; Appendix B of the paper
Deviation 2	Our ruling on the marking rule (a disclosed deviation); spec Amendments, heading Deviation 2	2026-10-12	A step line written more than once with identical balances is accepted; a repeat with different balances still fails; the same rule for every model	Sonnet-medium's length-sweep replies often repeated every step line, which the old rule marked wrong even when every value was right (at the chosen 6 steps, 3 of its 20 chain answers)	The length sweep had run and its Sonnet and Opus replies were read after the Haiku-only pick was committed; no main-run data	Section 3.4 of the paper; Appendix E of the paper (Deviation 2 and the length curve); Table S16
Amendment 1	spec Amendments, heading Amendment 1	2026-10-12	Filled in only the values the locked rules chose after the length sweep, plus four disclosures; no test added and no rule changed. Its line 'Haiku's and Opus's numbers are the same, apart from . . . ' understates what moved under Deviation 2's rule (among them Pilot 1's Opus-medium 24-step marker, exploratory, with 95% intervals: 92.00% [84.00, 98.00] under the original rule, 94.00% [88.00, 98.00] under Deviation 2; Table S24) and is corrected as a disclosed erratum	The length sweep: its pre-set rules chose the values	Length-sweep data; the pick was committed on Haiku data before any Opus or Sonnet length-sweep answer was opened; no main-run data	Appendix B of the paper (Changes to the locked plans); Table S24

Table S9: (continued)

Item	Ruling or record	Date	Change	Trigger	Data existing at the time	Where reported
Main-run rules	Our rulings before the main run	2026-10-12	A hard cap of $100 replaced the pre-registered multiple of the projected cost; the truncation pause was kept; the run was allowed to start before Zenodo v3	Speed: the old formula would have forced many small waves (up to about 17) and could have run past the main-run clearance's expiry (the end of 2026-10-13 UTC); actual spend was unaffected	Pilot 1, the length sweep and Lock 3 existed; no main-run data	Appendix B of the paper (Changes to the locked plans)
Reporting definitions after Lock 3	Our rulings before the main run and our later acceptance of Addendum C	2026-10-12	Reporting items 1–24 were approved by name in our rulings before the main run, after Lock 3 and before launch. Addendum C (committed before launch; it rewords item 7) and Addendum C4 (committed about 12 minutes after launch) were checked, but neither was ruled on by name at the time; we accepted Addendum C, C4 included, on 2026-10-15, after the results had been opened	Pre-registered descriptive outputs lacked locked definitions; the definitions were written and reviewed before the main run's public analysis stamp	No main-run result had been opened when the definitions and Addendum C4 were committed; the definitions are inside the main run's public analysis stamp; the first main-run calls were already sent when Addendum C4 was committed	Appendix B of the paper (Locks and public fingerprints; Changes to the locked plans)
12-step retry	Our ruling on one final 12-step attempt	2026-10-13	A final attempt to send Study 1b's point at 12 steps after three failed submits; a disclosed deviation from the resubmit limit; descriptive only	The earlier submits failed to upload; no batch was created and nothing was charged	Study 1b's other waves had returned; only counts, stop reasons and costs had been seen, no answer or score	Appendix B of the paper (Changes to the locked plans)
Lock 4 lineage clarification	record merge 2080573	2026-10-13	Merging the Lock 4 tag commit into main made a second child of the tag; ruled a disclosed clarification: the run commit is the tag's only first-parent child and the merge is a record merge with no run code	The record merge of the tag commit into main, after the tag was cut	No Study 1b or top-up data: the merge (07:36:37 UTC) came after Lock 4's public gist (07:34:10 UTC) and before Study 1b's first launch (08:33:00 UTC)	Appendix B of the paper (Changes to the locked plans)

Table S9: (continued)

Item	Ruling or record	Date	Change	Trigger	Data existing at the time	Where reported
Top-up start	Our rulings on the top-up's go, timing and run commit	2026-10-13	The top-up started within 48 hours of the main run's start by a disclosed change to the locked plan (its earliest start was brought forward); the live run started at 11:50:23 UTC	We asked to start it before 48 hours had passed since the main run's start. It was paused at 11:52:36 UTC with its first batch of requests in flight, before any reply had been collected, and resumed at 13:26:53 UTC, collecting that batch without sending it again; each call was sent once (Table S10). No request, setting or analysis changed	Study 1's results were open; Study 1b's were cleared to read at 11:35:43 UTC, after the disclosed change was committed and before the run's clearance and launch; no top-up data; the top-up's definitions were written before any top-up data	Appendix B of the paper (Changes to the locked plans)
Post-lock code	Code used after the locks (our list); commits 5e2b860, 0b45978, d364c24 → 7c8bb4b → b4ff738, aa228e1 / 20e093e, 89ffd02, c273eb9	—	Code used in paid runs or the analysis after the locks: after Lock 1 (the pilot fix, the effort-diagnostic runner) and after Lock 3 (the main runner with operational changes to the batch and transport code, the step-voting design and the rewording-check setting, and the converter and driver)	After Lock 1, a runner fix so that the pilot's real-time check could run on a ledger set, and a runner for the separate Haiku-only effort diagnostic (a run with its own rulings); after Lock 3, operational needs (crash resume and read retries) and new settings; the converter and driver were written after Lock 3	Each runner was committed and reviewed before its own run; the converter and driver are in the main run's public analysis stamp, made after collection ended and before any result was opened	Appendix B of the paper (Changes to the locked plans); Table S34

Dates are the dates of our rulings (local calendar dates, not UTC), the spec heading's date or our log entry's date (also a local calendar date); a dash means no single date is stamped. In the commit list of the post-lock code row, "→" leads from one commit of the main runner to the next, the last being the commit the main run ran from, and "/" separates the two commits of the step-voting design. The change, trigger and data cells are short summaries of the records each cell cites in this table's source file, released with the paper outputs in data/paper-outputs/; every number printed in digits (dates, times, counts, measured values and commit IDs) is read from those records, except digits that are part of a name, such as Lock 1, Tests 1–5, the 24-step marker, 95% intervals or a table's number.

- **Code written after the locks.** The paid runs and the analysis also used code written after the locks: after Lock 1, a fix to the pilot runner and the effort-diagnostic runner; after Lock 3, the main runner with operational changes to the batch and transport code (resuming after a crash, retrying reads), the step-voting and rewording-check settings, and the converter and driver of Study 1's analysis. Each runner was committed before its own run; the converter and driver were stamped publicly after collection ended and before any result was opened. Table S34 gives the commits.
- **Addendum C.** It was committed with the reporting definitions, its last part about 12 minutes after the main run began and before any result was opened (above). We accepted it, C4 included, on 2026-10-15, after the results had been opened.
- **The top-up run's start.** Lock 4 fixed the top-up run's earliest start at 2026-10-14 13:01 UTC, when the 48-hour period set aside for the main run ended. The top-up run instead started on 2026-10-13 at 11:50 UTC, within that 48-hour period but after the main run itself had finished. This was decided on 2026-10-13, before any top-up data existed. The run used a plan that differs from the locked one only in that start time, and no setting, prompt, puzzle, limit or analysis changed. The change brings the three new designs closer in time to the main-run settings they are compared with; the comparisons across the two runs and the drift check between runs are reported as specified. We paused it at 11:52, after its first batch of requests had been submitted and before any reply had been collected. It resumed at 13:26 from the same frozen plan, collected the batch already submitted without sending it again, and finished at 14:09 with all 3,820 calls complete, each sent once. Table S10 gives the pre-registration timeline, with the time span of every run that sent an API call.

Table S10: The pre-registration timeline (disclosure): a dated table of events, then three panels: each lock's fingerprints and public proofs; the time span of every run that sent an API call, from the setup checks to the top-up run; and the pre-registered check of Study 1b without its exposed puzzles, with the puzzles per test. Figure S9 draws the same sequence.

Date	Time (UTC)	Event	Commit or record	Seal or note
2026-10-10	—	Setup checks: smoke-1 (rejected by the API: HTTP 401; unbilled) and smoke-2; the first API calls, before any lock	smoke-2 run commit 568e8a4	—
2026-10-11	10:16–10:37	Setup check check-ledger-1 (the refusal check on Opus and Sonnet; before any lock)	run commit e1d6d60	Sealed: the answers are not read or graded until Lock 1's tag
	19:09–19:18	Setup check s3-ledger-1 (the bill check: one Batch of 5 requests on Opus and Sonnet; before any lock)	run commit 582d8d1	Sealed for good: the answers are never read or graded; the items are excluded from every analysis
	19:32:37	Lock 1 cut (tag prereg-lock-1)	commit b4b6b84	—
	19:35:19	Lock 1 public proof: gist with a Wayback snapshot	gist f8da61f9…; Wayback 20261011193547	—
	20:20:40	Lock 1 public proof: Zenodo v1 registered (DataCite)	DOI 10.5281/zenodo.23304164	—

Table S10: (continued)

Date	Time (UTC)	Event	Commit or record	Seal or note
	20:48–21:27; first paid call 21:06	Pilot 1 (pilot-ledger-1) under Lock 1	run commit 1b058a5 (Lock 1 plus a reviewed real-time fix, 5e2b860)	Preset pick (Haiku only, committed 21fa9f0 before any Opus or Sonnet answer or score was opened): none (Haiku-medium's pilot medians were 0.0% at every preset; this led to Deviation 1). Opus and Sonnet answers and scores stayed sealed until after Lock 2; then Pilot 1's Opus and Sonnet results were opened as exploratory
	21:57–22:55; paid calls 22:12–22:55	Effort diagnostic (diag-haiku-effort-1), under no lock; disclosed in Zenodo v3	run commit 0b45978	—
2026-10-12 (local date of the ruling)	—	Deviation 1 ruled: shorter puzzles, the length to be picked by a Haiku-only rule on the length sweep	spec heading Deviation 1	At the time of the ruling, Opus and Sonnet pilot answers and scores stayed unopened; their counts, tokens and costs were seen, for pricing
2026-10-12	06:25:10	Puzzle sets frozen at every candidate length for Lock 2, among them Study 1b's later test sets (unused by Study 1)	commit aae04fe (a git commit time is self-reported); public proof: Lock 2's Zenodo record (07:20:23), before any Haiku-max API answer existed	Checked before Lock 4: all three sets never sent and in no other file (the check covers only API logs and repo files); but see the exposure-scan row and the sensitivity panel
	07:11:26	Lock 2 cut (tag prereg-lock-2)	commit 95fb5f8	—
	07:20:23	Lock 2 public proof: Zenodo v2 registered (DataCite) only (no gist or Wayback)	DOI 10.5281/zenodo.23311831	—
	07:28–07:53; first call 07:29:27	Length sweep (ladder-l14n-1) under prereg-lock-2	run commit 95fb5f8	Haiku-only pick of the length committed at 07:54:40, before any Opus or Sonnet length-sweep answer was opened

Table S10: (continued)

Date	Time (UTC)	Event	Commit or record	Seal or note
	—	Deviation 2 ruled (identical repeated step lines accepted, for every model); Amendment 1 written (only the values the pre-set rules chose after the length sweep)	spec headings Deviation 2 and Amendment 1	Ruled before any main-run data
	09:11:59	Lock 3 cut (tag prereg-lock-3): Amendment 1 and Deviation 2	commit 086fbfd	—
	10:25:49	Lock 3 public proof: gist with a Wayback snapshot; Zenodo was unavailable	gist eb3ec6a6…; Wayback 20261012102601	—
	13:01:21–13:54:17; first call 13:01:25	Main run (main-l14n6-1) under prereg-lock-3; ruled before the run: the run may start before Zenodo v3 is published	run commit b4ff738 (Lock 3 plus reviewed runner code)	Sealed: no answer, score or accuracy opened; counts, stop reasons, kinds and costs only
	13:12:58	Addendum C4 (four edge-case pins to the reporting definitions) committed about 12 minutes after launch; Addendum C, C4 included, was not ruled on by name at the time; we accepted it on 2026-10-15	commit 93334b8	Addendum C4 came before any result was opened and is inside the stamp
	13:28:29	Lock 3 public proof: Zenodo v3 registered (DataCite); deposited during the run, as the record says	DOI 10.5281/zenodo.23313537	—
	14:18:44	Analysis stamp for Study 1: driver c273eb9 and reporting definitions 93334b8 with the main run's raw-file fingerprints	gist a604ef9b…; Wayback 20261012141856	No main-run result has been opened
	14:21:29–14:23:36	k fixed from costs only, before any score was seen (k.json committed at 14:21:14); then the stamped driver's one run	driver at c273eb9; output SHA-256 43d18eed…	Results may now be read, after a byte-identical re-run

Table S10: (continued)

Date	Time (UTC)	Event	Commit or record	Seal or note
2026-10-13	07:21:34	Final exposure scan before Lock 4: no new exposure; Study 1b's three partly displayed test puzzles had parts of their answers shown inside Claude Code during our own work sessions; kept, disclosed, and given a pre-registered sensitivity without them (sensitivity panel)	scan file SHA-256 a78fb109…	Those puzzles: they were never shown to a model under test, and nothing was tuned on them
	07:33:17	Lock 4 cut (tag prereg-lock-4): Study 1b and the top-up run, before either sent a call	commit d1c565f	—
	07:34:10	Lock 4 public proof: gist with a Wayback snapshot	gist f0ec2ea8…; Wayback 20261013073439	—
	07:36:37	Record merge of the tag into main: it made a second child of the tag; ruled a disclosed clarification	commit 2080573	2080573 is disclosed as a record merge with no run code
	07:38:42	Price-date commit, the tag's only first-parent child; it changed only the price table's date (with its re-check note and the test that checks it); no price changed	commit 521f2a1	—
	07:43:34	Lock 4 public proof: Zenodo v4 record created	DOI 10.5281/zenodo.23335207	—
	08:33:00–11:19:29	Study 1b (study1b-1) under prereg-lock-4	run commit 521f2a1	Sealed: counts, stop reasons and costs only; no answer or score opened
	—	Study 1b's k fixed from costs only ($k = 6$)	commit 534e415; k.json SHA-256 4204d537…	No answer or score opened
	11:26:06	Study 1b data stamp: run-file and k.json fingerprints	gist 52dd104c…; Wayback 20261013112716	It holds no data and no results
	11:28:10–11:31:26	Study 1b driver run once at the Lock 4 commit	driver at d1c565f; output SHA-256 239173fc…	The results may now be read, after a byte-identical re-run (11:32:03–11:35:43)

Table S10: (continued)

Date	Time (UTC)	Event	Commit or record	Seal or note
	11:50:13–14:09:51	Top-up run (topup-1) under prereg-lock-4 with the disclosed start change; paused 11:52:36 with its first batch of requests in flight, resumed 13:26:53; every call at attempt 1 (no re-send, no crash re-run, no failed submit)	run commit 056dd26	Sealed: counts, stop reasons and costs only; no answer or score opened
	14:15:17	Top-up data stamp: run-file fingerprints	gist c9870cf2…; Wayback 20261013141633	It holds no data and no results
	14:18:45–14:19:00	Top-up driver run once at the Lock 4 commit	driver at d1c565f; output SHA-256 c76478bf…	The top-up results may now be read, after a byte-identical re-run (14:19:43–14:19:59)

Table S10: (continued)

Panel 1: Locks (fingerprints and public proofs)

Lock	Tag	Commit	Tar SHA-256	Tag cut (UTC, month-day)	Gist created (UTC, month-day)	Wayback snapshot	Zenodo DOI	Zenodo time (UTC, month-day)	Next paid data (UTC)
Lock 1	prereg-lock-1	`b4b6b847 4f395380 55b3e358 b48243d5 1aca5e3b`	`37454153 435ae927 ccd5aa3a 5f61e4f7 033e53a3 5875b830 a3f9ccfd 2c610adf`	10-11 19:32:37	10-11 19:35:19	20261011193547	10.5281/zenodo.23304164	10-11 20:20:40 (DataCite)	Pilot 1 first call 21:06
Lock 2	prereg-lock-2	`95fb5f81 b9a3a131 da77f5a7 76af200c 8e60477d`	`93a23815 71d0eb12 ba27bd90 84701ecc 4ca1e725 36bc0300 e509ac58 3e6c615a`	10-12 07:11:26	no gist or Wayback	—	10.5281/zenodo.23311831	10-12 07:20:23 (DataCite)	length sweep 07:29:27
Lock 3	prereg-lock-3	`086fbfd0 4063a10e 2c64f2a7 37e5771b 0aa70dc8`	`1221e839 4529d1f5 e81bdfeb c7827366 b32e3fb3 aa24cfa3 9bfe71e9 8144deb5`	10-12 09:11:59	10-12 10:25:49	20261012102601	10.5281/zenodo.23313537	10-12 13:28:29 (DataCite; during the main run)	main run 13:01:25–13:54:17
Lock 4	prereg-lock-4	`d1c565fc 9188d56e 970bbb15 ce908271 627ccf0a`	`47e40b35 97b10808 89616022 67001503 7d356914 d0dc379d 0ee669ac b1b595ad`	10-13 07:33:17	10-13 07:34:10	20261013073439	10.5281/zenodo.23335207	10-13 07:43:34 (record created)	Study 1b first invocation 08:33:00

Table S10: (continued)

Panel 2: Runs (every run that sent an API call)

Run	Date	Lock in force	Run commit	Logged span (UTC)	Invocations (UTC)	Run started (UTC)	Pauses	First call (UTC)	Last commit or end (UTC)
smoke-1 (rejected by the API: HTTP 401; unbilled)	2026-10-10	before any lock	—	—	—	—	—	—	—
smoke-2	2026-10-10	before any lock	568e8a4	—	—	—	—	—	—
check-ledger-1	2026-10-11	before any lock	e1d6d60	10:16–10:37	—	—	—	—	—
s3-ledger-1	2026-10-11	before any lock	582d8d1	19:09–19:18	—	—	—	—	—
pilot-ledger-1	2026-10-11	Lock 1 plus a reviewed real-time fix, 5e2b860	1b058a5	20:48–21:27	—	—	no halts	21:06	—
diag-haiku-effort-1	2026-10-11	under no lock	0b45978	21:57–22:55	—	—	—	22:12	22:55
ladder-l14n-1	2026-10-12	prereg-lock-2	95fb5f8	07:28–07:53	—	—	no halts	07:29:27	07:53:17
main-l14n6-1	2026-10-12	prereg-lock-3 (runner: Lock 3 plus reviewed runner code)	b4ff738	—	13:01:21 (one invocation)	—	no halts	13:01:25	13:54:17
study1b-1	2026-10-13	prereg-lock-4	521f2a1	—	08:33:00, 10:02:40, 10:24:12, 11:07:47	08:35:49	truncation pause: none (every truncation check under 2%); the run stopped after each failed upload of the 12-step point and was invoked again (see the notes)	—	11:19:29
topup-1	2026-10-13	prereg-lock-4 with the disclosed start change	056dd26	11:50:13–14:09:51	11:50:13, 13:26:53	11:50:23	paused 11:52:36 with its first batch of requests in flight; resumed 13:26:53; every call at attempt 1 (no re-send, no crash re-run, no failed submit)	—	14:09:51 (run end)

Table S10: (continued)

Panel 3: Exposed-puzzle sensitivity (Study 1b, pre-registered)

Test	Test set	Puzzles	Puzzles without the exposed ones	Exposed puzzles left out	Reading, 5-point margin	Reading without them, 5-point margin	Shown no worse (Holm-adjusted), as deployed, 5-point margin	Shown no worse (Holm-adjusted) without them, as deployed, 5-point margin
Test B1	main-ledger-l14n16-v1	80	79	main-ledger-l14n16-v1-000	"inconclusive (could not show)"	"inconclusive (could not show)"	no	no
Test B2	main-ledger-l14n16-v1	80	79	main-ledger-l14n16-v1-000	"holds (no worse in all three views)"	"holds (no worse in all three views)"	yes	yes
Test B3	ext40-ledger-l14n6-v1	40	38	ext40-ledger-l14n6-v1-000, ext40-ledger-l14n6-v1-003	"inconclusive (could not show)"	"inconclusive (could not show)"	no	no

All times are UTC. The lock tag times, and the commit times of the puzzle sets, Addendum C4, the record merge and the price-date commit, are git's tagger and committer times, as re-read from git in our final fact-check; the commit times of the length pick and of Study 1's k.json are from our own process record; run times are from our records of each run (a logged span starts when the run's command started, before its first call); the exposure scan's time is the Lock 4 manifest's; gist, Wayback and Zenodo times are the public records' own. A date is printed on the first row of each UTC day; the rows below it fall on the same day. A row with no time is a ruling, or a run or step whose record gives only its date; it sits in order between the rows around it. Deviation 1's ruling has no recorded time: our decision records give only its date, with no time or time zone. That date, printed beside it, is a local calendar date, not a UTC one. It was ruled after the effort diagnostic and before the puzzle sets were committed. Run rows give each run's span from its records; the second panel splits it into invocations, the run's start, pauses, first call and last commit or end. "Run started" is the time our cost records log as the run's "window started" (after its first invocation). The top-up's "First call" is a dash: its work log records "Window started" and then its first batch of requests as sent, with no time for that send. Commits, gists and hashes are shortened in the timeline; the lock panel prints each lock's full commit ID and tar SHA-256, and Table S34 the analysis files' hashes. Study 1b's later invocations each re-submitted the 12-step point after its upload failed: waves w6–w8 (the 12-step point) failed to upload, and no batch was created; they were closed at $0, and the 12-step point went through on its 4th attempt, a disclosed deviation (Table S9). Lock 3's Zenodo v3: our project log records the DataCite registration at 13:28:29 UTC and the publication time read from the Zenodo API at 13:28:28 UTC; both fall inside the main run. Our lock procedure set a Zenodo record plus a gist with a Wayback snapshot as each lock's public proof; Lock 1 has all three. Lock 2 has only its Zenodo record. Lock 3 was first public as a gist, with Zenodo v3 deposited during the main run, as its record states. Study 1b ran at the tag's only first-parent child; the top-up at a later commit that also moved the plan's earliest start time earlier, as a disclosed change (Table S9). Exposed-puzzle sensitivity (third panel): a pre-registered sensitivity analysis. It is reported outside the main text (the paper's Appendix D; here, Table S21's sensitivity panel, which gives its d, 95% interval, p and error-rate ratio) and never replaces the primary result. Readings in quotation marks are the Study 1b file's own template words; the ruled reading, shown no worse (Holm-adjusted), is in the columns headed "Shown no worse (Holm-adjusted)". A dash: the records give nothing for that cell; in the lock panel, Lock 2's dash means it has no Wayback snapshot. Lock panel, last column: the next paid data after each lock, as our final fact-check lists it: the first call of Pilot 1, of the length sweep and of the main run (the main run's entry runs on to its last commit), and Study 1b's first invocation. The timeline's run rows instead start each run's span when its command started.

S4 Study 1, further results

S4.1 Study 1's tests and checks in detail

These paragraphs give the detail behind the main text's account of Study 1, in the order the main text uses. Table S12 gives the eight confirmatory tests at a glance.

Holm's correction, used because five tests were run together, passed only Test 5 (Table S11 applies it to Tests 1–4 in the inferiority direction, a post hoc sensitivity view). It took the tests in the order 5, 3, 4, 2, 1 and stopped at Test 3 ($p = 0.98913$, above its threshold of 0.00625). The "worse by more than the margin" readings for Tests 1–4 come from the pre-registered decision rule on the unadjusted 95% intervals, not from Holm's correction. Their upper ends are −23.75 (Test 1), −26.96 (Test 2), −12.20 (Test 3) and −30.00 (Test 4) against the −10 margin: 13.75 to 20.00 points beyond it for Tests 1, 2 and 4, and 2.20 points for Test 3, which also depends on the marking rule. Table S13 gives every test in full.

Tests 1 and 2 carry the pre-registered near-ceiling label, a warning fixed in advance for when the comparator scores near a perfect score. It was set from the length sweep's Opus-medium score of 40 of 40 at 6 steps, before the main-run data, and was not re-decided afterwards; the main run's Opus-medium pooled score, 98.13% (95% interval 95.63 to 100.00), is shown for information only. At 40 samples the Test 1 vote uses each puzzle's one full set of samples, so unlike Test 2's vote it involves no averaging over subsets and so depends more on the particular samples drawn; its reading does not change across the sensitivity rows.

Before the main run, the power model fixed in the plan, fed the length sweep's measured 6-step chain inputs (Haiku-medium right on 92 of 200 samples; Opus-medium 20 of 20, entered as 0.99), forecast that Test 2 would read "worse by more than the margin" with probability 1.00 at 80 puzzles. The plan itself calls this "the pinned power model's forecast under the measured inputs, not a result", and the forecast covers Test 2 only. What the main run adds is the measured size of the gap, the error-rate ratios, the vote curves and the costs (our reading, not a test).

Test 3 and Test 5 under other views. Test 3 holds on the final answer alone ($d = -25.38$, -36.52 to -13.46) and with ties counted wrong ($d = -27.43$, -38.92 to -14.84; a sensitivity check). Under the original marking rule Test 5's difference is +41.25 points (95% interval +28.75 to +52.50), and its reading is unchanged. On pieces (exploratory, outside the Holm family; not a test), Opus-low minus Sonnet-medium was +25.00 points (95% interval +16.25 to +35.00).

Step voting's draws. Step voting averaged 2.26 draws per step (1,086 draws over 480 steps). At Opus-medium's price per chain puzzle ($0.0402), it could have bought about 26 draws per step (post hoc arithmetic). The locked rule stopped as soon as one answer led by 2, so a step-voting run given Opus-medium's budget was not run and cannot be rebuilt offline from the logged draws.

Format and cost per correct answer. Under the marking rule as run, the confirmatory single-call settings kept the required format in 97.50–100.00% of replies: Haiku-medium 98.59% (3,155 of 3,200) on chains and 99.72% (3,191 of 3,200) on pieces; Opus-medium and Sonnet-medium 97.50% (78 of 80) on chains and 100.00% on pieces; Opus-low 98.75% (79 of 80) and 100.00%. No Test 1–5 reading turns on the remaining parse failures; the marking rule itself matters for Test 3. The large format failures sit in the exploratory thinking-off setting, the only Study 1 setting whose format flag fires (18.28% non-compliance on chains). At the cost-matched vote size, the Haiku-medium vote cost more per correct answer than one comparator answer: on chains $0.0676 (95% interval 0.0573 to 0.0808) against Opus-medium's $0.0418 (0.0395 to 0.0447); on pieces $0.0654 (0.0564 to 0.0778) against $0.0481 (0.0471 to 0.0491); and against Sonnet on chains $0.0259 (0.0222 to 0.0307) against $0.0184 (0.0166 to 0.0209). A single Haiku-medium sample costs far less per correct answer ($0.00252 on chains, $0.00214 on pieces), which is why cost per correct answer alone misleads. Table S14 gives every setting's accuracy, format and cost.

Table S11: Tests 1–4 under Holm's correction in the inferiority direction, applied over Tests 1–5 at a one-sided level of 0.025 (post hoc† sensitivity, chosen after the main results were open; Study 1, main run, 80 puzzles per test). The pre-registered decision rule on the unadjusted 95% interval stays the result; its 99% interval uses the same bootstrap draws (10,000 resamples).

Test	Shape	Challenger	Comparator	d (points)	95% interval (unadjusted)	Error-rate ratio (challenger's over comparator's, 95% interval)	Reading (pre-registered decision rule)
Test 1	6×1	Haiku-medium vote	Opus-medium	−33.13	[−43.75, −23.75]	undefined (comparator: 0 errors in 80; its error rate at most 4.51, exact 95%)	worse by > margin
Test 2	1×6	Haiku-medium vote	Opus-medium	−37.64	[−48.82, −26.96]	11.04 [4.75, unbounded]	worse by > margin
Test 3	1×6	Haiku-medium vote	Sonnet-medium	−24.70	[−36.12, −12.20]	2.23 [1.47, 3.92]	worse by > margin
Test 4	1×6	Step voting	Opus-medium	−42.50	[−53.75, −30.00]	12.33 [5.17, unbounded]	worse by > margin
Test 5	1×6	Opus-low	Sonnet-medium	+17.50	[8.75, 27.50]	0.13 [0.00, 0.40]	shown no worse by > margin (Holm-adjusted)

Test	Locked p (one-sided)	Inferiority p, one minus the locked p (post hoc)	Holm threshold, inferiority direction (post hoc)	Inferiority p below its Holm threshold (post hoc)	Stricter interval (post hoc)	Stricter interval wholly below the margin (post hoc)
Test 1	0.99996	≤ 0.00004	0.005	yes	[−47.50, −20.63]	yes
Test 2	0.99993	≤ 0.00007	0.00625	yes	[−52.23, −23.55]	yes
Test 3	0.98913	0.01087	0.0125	yes	[−39.28, −8.25]	no
Test 4	0.99987	≤ 0.00013	0.00833	yes	[−57.50, −26.25]	yes
Test 5	≤ 0.00004	0.99996	0.025	—	—	—

In the reading column, "> margin" stands for "by more than the margin": "worse by > margin" is worse by more than the margin, as the other tables of this supplement write it. Post hoc sensitivity, not a test: the correction was fixed before this check was computed, but "chosen after the main results were open; the pre-registered unadjusted three-way reading stays the result" (free analysis 5's own words; its "three-way reading" is the reading from the pre-registered decision rule, applied to the unadjusted interval). Holm in the inferiority direction (null hypothesis: d at or above the margin of −10 points) over all five pre-registered tests at one-sided alpha 0.025; each inferiority p is one minus the locked one-sided p, from the same bootstrap (same differences and seeds, B = 10,000). Holm order, smallest inferiority p first: Test 1, Test 2, Test 4, Test 3, Test 5. The stricter interval is the two-sided 99% interval from the same bootstrap draws; it is shown beside Holm, not as part of it. Test 3 (post hoc): its inferiority p, 0.01087, is below its Holm threshold of 0.0125, but its 99% interval [−39.28, −8.25] does not lie wholly below the margin of −10 points. Test 3 under the original marking rule (the sensitivity check Deviation 2 fixed in Lock 3, before any main-run data): d = +0.30 points, 95% interval [−13.58, 14.95]; reading: inconclusive. Test 5's confirmatory result is its Holm-adjusted result in the pre-registered family of one-sided tests. Opus-low is shown no worse than Sonnet-medium by more than 10 points (Holm-adjusted): yes ($p \leq 0.00004$, below its Holm threshold of 0.005). Sonnet ran at medium effort, one level below its default, in Test 3 and Test 5. Here Test 5 is only the fifth member of the inferiority-direction Holm family, not an inferiority claim; its full report is in Table S13. The error-rate ratio is the challenger's error rate over the comparator's, given beside d as in Table S13 (pre-specified descriptive, reported beside the test, not part of its reading); the main analysis labels Test 1 and Test 2 as near ceiling (Amendment 1), where the error-rate ratio travels with d. Exact: the comparator's error-rate bound, in %, is the upper end of its exact (Clopper–Pearson) 95% interval. ≤ marks a p at the smallest value the bootstrap can return (no resample reached the margin).

Table S12: The medium-effort Haiku tests read worse than their comparators by more than the margin (Test 3 only under the marking rule as run), while at 16 steps a small Haiku-max vote (its accuracy estimated from 10 samples) was shown no worse than one Opus-medium answer by more than 5 points (Holm-adjusted) and a single Haiku-max sample could not be shown to be within 5 points. The eight tests at a glance. Tests 1–5 (Study 1, main run, 6 steps) and Tests B1–B3 (Study 1b; B1 and B2 at 16 steps, B3 at 6) are confirmatory, with Holm's correction within each study only; the two studies are never pooled. For each test: what it compares, its margin, the vote size k and the spend share or cost ratio (not tested; labels in the note), d in points with its 95% interval (paired BCa bootstrap, 10,000 resamples), its pre-registered reading, whether it passed Holm's correction (Holm-adjusted), and its error-rate ratio (pre-specified descriptive, reported beside the test, not tested). Costs at list price.

Test	Study	Puzzles	Steps	What is compared	Margin (points)
Test 1	Study 1	80	6	Haiku-medium vote vs Opus-medium, 6×1	−10
Test 2	Study 1	80	6	Haiku-medium vote vs Opus-medium, 1×6	−10
Test 3	Study 1	80	6	Haiku-medium vote vs Sonnet-medium, 1×6	−10
Test 4	Study 1	80	6	Step voting vs Opus-medium, 1×6	−10
Test 5	Study 1	80	6	Opus-low vs Sonnet-medium, 1×6	−10
B1	Study 1b	80	16	One Haiku-max sample vs one Opus-medium answer	−5
B2	Study 1b	80	16	Haiku-max vote vs one Opus-medium answer	−5
B3	Study 1b	40	6	One Haiku-max sample vs one Opus-medium answer	−5

Test	k	Spend share or cost ratio	d (points)	95% interval	Error-rate ratio [95%]
Test 1	40	spend share 0.909	−33.13	[−43.75, −23.75]	undefined (comparator errors 0 of 80)
Test 2	36	spend share 0.984	−37.64	[−48.82, −26.96]	11.04 [4.75, unbounded]
Test 3	13	spend share 0.970	−24.70	[−36.12, −12.20]	2.23 [1.47, 3.92]
Test 4	—	cost ratio 0.088	−42.50	[−53.75, −30.00]	12.33 [5.17, unbounded]
Test 5	—	cost ratio 2.48 [2.36, 2.59]	+17.50	[8.75, 27.50]	0.13 [0.00, 0.40]
B1	—	cost ratio 0.149	−5.19	[−7.75, −2.69]	3.37 [1.78, 10.07]
B2	6	spend share 0.894	+0.63	[−1.15, 2.44]	0.71 [0.24, 2.11]
B3	—	cost ratio 0.095	−3.63	[−7.00, −1.50]	6.80 [1.67, unbounded]

Test	Reading (pre-registered decision rule)	Shown no worse (Holm-adjusted)	Shown no worse (Holm-adjusted) at Study 1b's secondary 10-point margin	Decision
Test 1	worse by > margin	no	—	worse by > 10 (pre-registered rule)
Test 2	worse by > margin	no	—	worse by > 10 (pre-registered rule)
Test 3	worse by > margin	no	—	worse by > 10 (pre-registered rule; marking rule as run)
Test 4	worse by > margin	no	—	worse by > 10 (pre-registered rule)
Test 5	shown no worse by > margin (Holm-adjusted)	yes	—	shown no worse by > 10 (Holm-adjusted)
B1	inconclusive	no	yes	inconclusive at 5; shown no worse by > 10 (Holm-adjusted)
B2	shown no worse by > margin (Holm-adjusted), in all three views	yes	yes	shown no worse by > 5 (Holm-adjusted)
B3	inconclusive	no	yes	inconclusive at 5; shown no worse by > 10 (Holm-adjusted)

Table S12: (continued)

Test	Label
Test 1	the test: confirmatory, read by the pre-registered decision rule on the unadjusted interval (not Holm-adjusted); error-rate ratio: pre-specified descriptive, reported beside the test; k: set by the pre-registered rule for a cost-matched vote, not tested; spend share: pre-specified descriptive, not tested
Test 2	the test: confirmatory, read by the pre-registered decision rule on the unadjusted interval (not Holm-adjusted); error-rate ratio: pre-specified descriptive, reported beside the test; k: set by the pre-registered rule for a cost-matched vote, not tested; spend share: pre-specified descriptive, not tested
Test 3	the test: confirmatory, read by the pre-registered decision rule on the unadjusted interval (not Holm-adjusted); error-rate ratio: pre-specified descriptive, reported beside the test; k: set by the pre-registered rule for a cost-matched vote, not tested; spend share: pre-specified descriptive, not tested
Test 4	the test: confirmatory, read by the pre-registered decision rule on the unadjusted interval (not Holm-adjusted); error-rate ratio: pre-specified descriptive, reported beside the test; cost ratio: post hoc arithmetic on pre-specified means, not tested
Test 5	the test: confirmatory, shown no worse (Holm-adjusted across Study 1's tests); error-rate ratio: pre-specified descriptive, reported beside the test; cost ratio: exploratory, no test (intervals two-sided, unadjusted)
B1	the test: confirmatory, read by the pre-registered decision rule on the unadjusted interval (not Holm-adjusted); not shown no worse (Holm-adjusted) at the primary 5-point margin; error-rate ratio: pre-specified descriptive, reported beside the test; cost ratio: pre-specified descriptive, not tested
B2	the test: confirmatory, shown no worse (Holm-adjusted across Study 1b's tests) at the primary 5-point margin; error-rate ratio: pre-specified descriptive, reported beside the test; k: set by the pre-registered rule for a cost-matched vote, not tested; spend share: pre-specified descriptive, not tested
B3	the test: confirmatory, read by the pre-registered decision rule on the unadjusted interval (not Holm-adjusted); not shown no worse (Holm-adjusted) at the primary 5-point margin; error-rate ratio: pre-specified descriptive, reported beside the test; cost ratio: pre-specified descriptive, not tested

In the reading columns, "> margin" stands for "by more than the margin": "worse by > margin" is worse by more than the margin, as the other tables of this supplement write it. Holm's correction runs within each study only; the two studies are never pooled. At Study 1's 10-point margin (its only confirmatory margin; a secondary 5-point view is in Table S16) only Test 5 was shown no worse (Holm-adjusted), and at Study 1b's primary 5-point margin only B2. Tests 1–4, B1 and B3 carry the pre-registered decision rule's reading of the unadjusted 95% interval, which is never a Holm-adjusted result. At Study 1b's pre-registered secondary 10-point margin, quoted only as the secondary, B1, B2 and B3 were each shown no worse than Opus-medium by more than the margin (Holm-adjusted; B1: $p = 0.00027$, below its Holm threshold of 0.0125; B2: $p \leq 0.00011$, below its Holm threshold of 0.00833; B3: $p \leq 0.00108$, below its Holm threshold of 0.025). $\leq$ marks a p at the smallest value the bootstrap can return (no resample reached the margin). Under the original pre-registered marking rule, which Deviation 2 kept as a sensitivity check (fixed in Lock 3, before any main-run data; not a pre-registered secondary view), Sonnet-medium's chain score is 55.00% instead of 80.00% (format compliance 65.00% instead of 97.50%). Test 3 would read inconclusive ($d = +0.30$ points, 95% interval $[-13.58, 14.95]$) instead of worse by more than the margin. Test 5's reading is unchanged ($d = +41.25$ points, 95% interval $[28.75, 52.50]$). Sonnet ran at medium effort, one level below its default (high), in Tests 3 and 5. Study 1's tests were pre-registered at 24 steps and run at 6 steps (Deviation 1). B2's accuracy is the expected accuracy of a vote of $k = 6$ Haiku-max samples, computed from the 10 designed samples per puzzle and shape (no vote of that size was run separately); k was measured from the two settings' mean list costs per sample. Per shape, the vote costs \$0.0736 per answer on chains against Opus-medium's \$0.0674, and \$0.0605 on pieces against \$0.0825 (list price). d (points): the challenger (named first) minus the comparator, in accuracy points, paired over puzzles (in Study 1b both shapes of a puzzle are pooled); a positive d means the challenger scored higher. Shapes: 1×6 is the chain form of a puzzle and 6×1 the pieces form. k (Tests 1–3): the size of the cost-matched Haiku-medium vote, the most samples whose combined mean cost fits within the comparator's mean cost per puzzle; the k computation prices each call at the Batch rate, exactly half the list price, so k is the same at list prices (B2's k is in the note above). Test 1's k is capped at 40, the number of samples drawn per puzzle and shape, from the cost-matched 44. k, spend share and cost ratio (none is tested; k is defined in the notes above). Each cell of the spend share or cost ratio column names its quantity; spend shares and the cost ratios of Test 4, B1 and B3 print to three decimals and Test 5's cost ratio to two, as Table S17 prints it. Spend share: Tests 1–3, the vote's realised spend over the comparator's at list price (pre-specified descriptive); B2, the vote's list cost per answer over Opus-medium's, as reported beside the test (pre-specified descriptive). Cost ratio: Test 4, step voting's mean list cost per puzzle over Opus-medium's, computed for this table from the two reported means (post hoc arithmetic on pre-specified means); Test 5, Opus-low's list cost per answer over Sonnet-medium's with its 95% interval, labelled exploratory, no test (intervals two-sided, unadjusted); B1 and B3, the challenger's list cost per answer over Opus-medium's, as reported beside each test (pre-specified descriptive). The file's own words for one Haiku-max sample's cost are, for B1, "about 1/6 of the cost per answer (k measured)" and, for B3, "about 1/10 of the cost per answer (k measured)". Error-rate ratio (pre-specified descriptive, reported beside the test; not part of any test's reading): the challenger's errors over the comparator's. An unbounded upper end comes from bootstrap draws in which the comparator made no errors. Test 1's error-rate ratio is undefined: the comparator made no errors, so its error rate is at most 4.51%, the upper end of the exact 95% interval.

Table S13: Study 1's Tests 1–5 in full (confirmatory; Study 1, main run, 6 steps, 80 puzzles per test): the columns of the paper's version, plus the spend share or cost ratio, the vote size before the cap (raw k), the one-sided p-value with its Holm threshold, the error-rate ratio with its 95% interval, the tie-decided share and the near-ceiling label (both defined in the note). Test 5 passed Holm's correction (Holm-adjusted); Tests 1–4 are read by the pre-registered decision rule on unadjusted intervals, not Holm-adjusted. d in points with 95% intervals from a paired BCa bootstrap (10,000 resamples) against the 10-point margin. The cap on k is the same for every vote and is given in the note. The spend share or cost ratio is the challenger's list-price cost over the comparator's and is not tested; for Test 5 it is a cost ratio (exploratory): Opus-low cost 2.48 times as much as Sonnet-medium per answer. The error-rate ratios are pre-specified descriptive, reported beside each test.

Test	Shape	Challenger	Comparator	k	Raw k	Spend share or cost ratio (challenger / comparator)
Test 1	6×1	Haiku-medium vote	Opus-medium	40	44	spend share 0.909
Test 2	1×6	Haiku-medium vote	Opus-medium	36	36	spend share 0.984
Test 3	1×6	Haiku-medium vote	Sonnet-medium	13	13	spend share 0.970
Test 4	1×6	Step voting	Opus-medium	—	—	0.088; cost ratio, post hoc arithmetic on pre-specified means, not tested
Test 5	1×6	Opus-low	Sonnet-medium	—	—	2.48 [2.36, 2.59]; cost ratio, exploratory, no test (intervals two-sided, unadjusted)

Test	Challenger	Comparator	Challenger accuracy (%)	Challenger format compliance (%)	Comparator accuracy (%)	Comparator format compliance (%)
Test 1	Haiku-medium vote	Opus-medium	66.88	99.72	100.00	100.00
Test 2	Haiku-medium vote	Opus-medium	58.61	98.59	96.25	97.50
Test 3	Haiku-medium vote	Sonnet-medium	55.30	98.59	80.00	97.50
Test 4	Step voting	Opus-medium	53.75	—	96.25	97.50
Test 5	Opus-low	Sonnet-medium	97.50	98.75	80.00	97.50

Test	d (points)	95% interval	p (one-sided)	Holm threshold	Shown no worse (Holm-adjusted)	Reading (pre-registered decision rule)
Test 1	−33.13	[−43.75, −23.75]	0.99996	0.025	no	worse by more than the margin
Test 2	−37.64	[−48.82, −26.96]	0.99993	0.0125	no	worse by more than the margin
Test 3	−24.70	[−36.12, −12.20]	0.98913	0.00625	no	worse by more than the margin
Test 4	−42.50	[−53.75, −30.00]	0.99987	0.00833	no	worse by more than the margin
Test 5	+17.50	[8.75, 27.50]	$\leq$ 0.00004	0.005	yes	shown no worse than Sonnet-medium by more than the margin (Holm-adjusted)

Table S13: (continued)

Test	Error-rate ratio	Error-rate ratio 95% interval	Tie-decided share (%)	Near-ceiling label
Test 1	undefined	comparator error rate at most 4.51 (exact 95%)	1.25	near ceiling (Amendment 1)
Test 2	11.04	[4.75, unbounded]	3.61	near ceiling (Amendment 1)
Test 3	2.23	[1.47, 3.92]	6.19	—
Test 4	12.33	[5.17, unbounded]	—	—
Test 5	0.13	[0.00, 0.40]	—	—

Study 1's tests were pre-registered at 24 steps and run at 6 steps (Deviation 1). Sonnet ran at medium effort, one level below its default, in Tests 3 and 5. Under the original pre-registered marking rule, which Deviation 2 kept as a sensitivity check, Sonnet-medium's chain score is 55.00% instead of 80.00%, and Test 3 would read inconclusive (could not show) instead of worse by more than the margin ($d = +0.30$ points, 95% interval $[-13.58, 14.95]$). Under the same rule, Test 5's reading is unchanged: shown no worse than Sonnet-medium by more than the margin ($d = +41.25$ points, 95% interval [28.75, 52.50]). Columns. k is the number of Haiku-medium samples in the vote; raw k is the cost-matched k before the cap (the comparator's mean cost per puzzle over Haiku-medium's mean cost per sample, rounded down); the cap is the most samples a vote may use, 40 for every vote. p (one-sided) is the bootstrap p of the one-sided test of d against the margin (the no-worse-than reading); the Holm threshold is the level that Holm's order sets for that test. The error-rate ratio is the challenger's errors over the comparator's, counted over the puzzles (for a vote, its expected errors); it is pre-specified descriptive, reported beside the test, not part of its reading. The tie-decided share is the mean over puzzles of the share of the vote's draws of k samples in which a tie that included the right answer decided the credit. Near ceiling (Amendment 1): a label that the pre-registration's Amendment 1 turned on for Tests 1 and 2 because their comparator, Opus-medium, was right on 40 of 40 answers in the length sweep at 6 steps, a pooled whole-answer accuracy of 100.00%, at or above the pre-set 95%; it marks a comparator at or near a perfect score. With it, the error-rate ratio (the challenger's errors over Opus-medium's) is reported with its interval next to the no-worse-than reading; where Opus-medium made no errors the ratio is undefined, and its exact 95% upper error rate, in %, is given instead. Format compliance is the share of samples in the required answer format; for Tests 1–3 it is that of the Haiku-medium samples the vote is drawn from. No format-compliance figure is stamped for step voting (Test 4's challenger), so that cell is a dash. $\leq$ marks a p at the smallest value the bootstrap can return (no resample reached the margin).

Table S14: Every Study 1 setting at 6 steps: accuracy, format compliance and cost (pre-specified descriptive; Study 1, main run, 80 puzzles per setting and shape). The 95% intervals come from a cluster bootstrap over puzzles for Haiku and step voting, and are Clopper–Pearson intervals for the one-sample Opus and Sonnet settings. Accuracy is the mean over puzzles, with the median in the last column of the table's first part; cost per sample and per correct answer at list price, with the Batch price at half.

Setting	Shape	Role	Used in	Samples per puzzle	Accuracy (%)	Accuracy 95% interval	Interval type	Median over puzzles (%)
Haiku, thinking off	1×6	exploratory	—	40	22.00	[16.97, 27.31]	cluster bootstrap	15.00
Haiku, thinking off	6×1	exploratory	—	40	23.50	[18.81, 28.56]	cluster bootstrap	18.75
Haiku-low	1×6	exploratory	—	40	39.22	[32.28, 46.16]	cluster bootstrap	32.50
Haiku-low	6×1	exploratory	—	40	46.50	[40.22, 53.00]	cluster bootstrap	50.00
Haiku-medium	1×6	confirmatory	Test 2, Test 3	40	43.72	[36.44, 50.72]	cluster bootstrap	40.00
Haiku-medium	6×1	confirmatory	Test 1	40	51.00	[44.28, 57.25]	cluster bootstrap	50.00
Haiku-max	1×6	exploratory	—	3	97.50	[95.42, 99.17]	cluster bootstrap	100.00
Haiku-max	6×1	exploratory	—	3	95.00	[91.25, 97.92]	cluster bootstrap	100.00
Step voting	1×6	confirmatory	Test 4	—	53.75	[42.50, 63.75]	cluster bootstrap	—
Opus-low	1×6	confirmatory	Test 5	1	97.50	[91.26, 99.70]	Clopper–Pearson (exact)	—
Opus-low	6×1	five-test setting, in no test at this shape	—	1	100.00	[95.49, 100.00]	Clopper–Pearson (exact)	—
Opus-medium	1×6	confirmatory	Test 2, Test 4	1	96.25	[89.43, 99.22]	Clopper–Pearson (exact)	—
Opus-medium	6×1	confirmatory	Test 1	1	100.00	[95.49, 100.00]	Clopper–Pearson (exact)	—
Sonnet-low	1×6	exploratory	—	1	71.25	[60.05, 80.82]	Clopper–Pearson (exact)	—
Sonnet-low	6×1	exploratory	—	1	76.25	[65.42, 85.05]	Clopper–Pearson (exact)	—
Sonnet-medium	1×6	confirmatory	Test 3, Test 5	1	80.00	[69.56, 88.11]	Clopper–Pearson (exact)	—
Sonnet-medium	6×1	five-test setting, in no test at this shape	—	1	75.00	[64.06, 84.01]	Clopper–Pearson (exact)	—

Table S14: (continued)

Setting	Shape	Format compliance (%)	List cost per sample ($)	Batch cost per sample ($)	Cost per correct answer ($)	Cost per correct answer, lower end of the 95% interval ($)	Cost per correct answer, upper end of the 95% interval ($)
Haiku, thinking off	1×6	81.72	0.000890	0.000445	0.00405	0.00322	0.00523
Haiku, thinking off	6×1	99.56	0.000956	0.000478	0.00407	0.00334	0.00517
Haiku-low	1×6	98.13	0.00103	0.000516	0.00263	0.00220	0.00323
Haiku-low	6×1	99.66	0.00103	0.000513	0.00221	0.00191	0.00258
Haiku-medium	1×6	98.59	0.00110	0.000550	0.00252	0.00214	0.00301
Haiku-medium	6×1	99.72	0.00109	0.000547	0.00214	0.00189	0.00246
Haiku-max	1×6	100.00	0.00441	0.00220	0.00452	0.00433	0.00471
Haiku-max	6×1	99.58	0.00407	0.00203	0.00428	0.00408	0.00453
Step voting	1×6	—	0.00356	0.00178	0.00662	0.00537	0.00851
Opus-low	1×6	98.75	0.0365	0.0183	0.0375	0.0352	0.0399
Opus-low	6×1	100.00	0.0449	0.0224	0.0449	0.0438	0.0458
Opus-medium	1×6	97.50	0.0402	0.0201	0.0418	0.0395	0.0447
Opus-medium	6×1	100.00	0.0481	0.0241	0.0481	0.0471	0.0491
Sonnet-low	1×6	100.00	0.0142	0.00708	0.0199	0.0174	0.0231
Sonnet-low	6×1	100.00	0.0158	0.00792	0.0208	0.0185	0.0238
Sonnet-medium	1×6	97.50	0.0147	0.00737	0.0184	0.0166	0.0209
Sonnet-medium	6×1	100.00	0.0161	0.00805	0.0215	0.0191	0.0247

Role: confirmatory settings are those a test uses at that shape; a five-test setting is a setting of the confirmatory run (the file's own list) that no test uses at that shape; the rest are exploratory. Haiku-medium enters Tests 1–3 as a vote of k samples; the row here is its single sample, and the votes are in the Study 1 test tables. Samples per puzzle is the stamped samples divided by the stamped puzzles. Step voting has no designed number of calls, so its samples-per-puzzle cell is a dash; it made 1,086 calls. Refusals and truncations: 0 in every row; the one exception is step voting's refusal count, which its row does not stamp (the file stamps all 1,086 of its calls as answered, with no refusal category). Cost per sample: for the Haiku settings, one sample's mean list cost (a vote of k samples costs k times this); for step voting, all its calls on a puzzle; for Opus and Sonnet, their one answer. The Batch column is the same at the Batch price (half the list price). Cost per correct answer is the total cost of one sample per puzzle over the expected number of those samples that are right, at list price. Costs are in US dollars. Interval type: the Haiku settings' and step voting's accuracy intervals are cluster bootstraps over puzzles; the one-sample Opus and Sonnet settings' are Clopper–Pearson intervals, which are exact: they come from the binomial distribution itself, with no resampling.

The vote curve's details. On chains, the difference between the curve's peak (58.75% at 33 samples) and its value at 40 samples (57.50%) is 1.25 points, within sampling noise. Because of the tie rule, a 2-sample vote scores the single-sample rate, apart from a small gain when one sample is unreadable, so the first real step comes at 3 samples (Haiku-medium on chains: 43.72%, 44.24% and 47.53%).

Checks fixed in advance. Three pre-specified checks bear on how the puzzles were built (Table S15). Flag R did not fire: when two Haiku-medium samples slipped at the same step from the same input, they wrote the same balances in 89.23% of pairs on general steps (90% interval 83.01 to 94.91; 11,972 pairs) and in 66.57% on the rule kinds tuned against Haiku (59.42 to 73.26; 25,082 pairs). The flag fires only if the tuned-kind rate is at least twice the general one; it guards against tuned kinds making Haiku's slips repeat, which would work against the vote; the data lean the other way. Haiku-medium's step slips do cluster on two "look again" rule kinds, partly by design, since these kinds were added where Haiku slipped: on chains, repeat steps slipped 1,105 of 3,700 times (29.86%) and re-pick steps 621 of 2,766 (22.45%), against 728 of 12,470 (5.84%) on general steps. The pre-registered fairness check (exploratory) did not fire either: its targeting index was 0.82 (90% interval 0.43 to 1.37; a value above 1 would mean the tuned kinds hit Haiku harder). Opus made only 1 slip in 948 placed steps, so the comparison is in effect Haiku against Sonnet, and the interval cannot exclude a moderate excess targeting of Haiku.

Effort inside Study 1 (exploratory). Paying for more Opus effort bought no accuracy gain these data could detect: Opus-low minus Opus-medium was +1.25 points (95% interval −3.75 to +7.50) on chains and 0.00 on pieces (both 100.00%), at about 6.77–9.22% less per puzzle; Opus is near its ceiling, so the data cannot separate "effort does not help Opus" from "these puzzles are too easy for Opus". In paired data, Haiku-low scored 4.50 points below Haiku-medium on both shapes (chains −4.50, 95% interval −6.91 to −2.13; pieces −4.50, −7.84 to −1.09) for about 6.23% less cost. Haiku, thinking off, scored 22.00% on chains (16.97 to 27.31) and 23.50% on pieces (18.81 to 28.56), against Haiku-medium's 43.72% and 51.00%, and saved about 19.09% and 12.51% of the cost. The reply format asks only for the value after each step, with no worked reasoning, so this setting measures answering in that format with thinking disabled; on chains, 18.28% of its replies also failed the format, and on the final answer alone it reached 27.97% instead of 22.00%.

Shape. The two shapes did not separate any setting at 6 steps: every setting's pieces-minus-chains interval included zero (Haiku-medium +7.28, 95% interval −1.47 to +15.38; Opus-medium +3.75, 0.00 to +8.75; Sonnet-medium −5.00, −18.75 to +8.75; Haiku-max −2.50, −6.25 to +0.83), and for Tests 1–2 the vote's gap to Opus-medium was not shown to differ between shapes (+4.52, −9.85 to +18.23). In the length sweep (20 puzzles each, one sample), Sonnet-medium scored higher on chains than on pieces at 4 of 5 lengths. In Study 1b, Haiku-max scored lower on pieces than on chains at 6 steps (94.25% against 97.25%), but higher on pieces at 12 steps (95.13% against 90.38%) and at 16 steps (93.00% against 92.25%). The shape effect did not keep one direction across sets and lengths (pre-specified descriptive).

Table S15: Pre-specified checks and flags, which are not tests (Study 1, main run; the format flags also cover Study 1b, and the drift check between runs comes from the top-up run). The fairness check is exploratory. The fairness check and Flag R carry 90% intervals from a percentile bootstrap, the fairness check's over 80 puzzles; the other intervals are 95% intervals.

Check	Quantity	Value	Interval	Interval level (%)	Count	Rule	Stamped reading	Fired
Flag R	Haiku-medium same-slip pairwise agreement on T steps, both shapes (%)	66.57	[59.42, 73.26]	90	25,082 pairs	—		—
	Haiku-medium same-slip pairwise agreement on G steps, both shapes (%)	89.23	[83.01, 94.91]	90	11,972 pairs	—		—
	Haiku-medium same-slip pairwise agreement on T steps, 1×6 (%)	62.09	[53.61, 69.50]	90	12,700 pairs	—		—
	Haiku-medium same-slip pairwise agreement on G steps, 1×6 (%)	88.62	[76.63, 98.25]	90	4,596 pairs	—		—
	Haiku-medium same-slip pairwise agreement on T steps, 6×1 (%)	71.16	[61.43, 79.88]	90	12,382 pairs	—		—
	Haiku-medium same-slip pairwise agreement on G steps, 6×1 (%)	89.61	[81.82, 95.94]	90	7,376 pairs	—		—
	Flag R: agreement on T steps against G steps (pooled)	—	—	—	25,082 T pairs, 11,972 G pairs	fires when T-step agreement is at least 2 times G-step agreement, with at least 10 pairs in each (as the analysis code applies it)	does not fire	no
Fairness check	Targeting index (ratio) against Opus and Sonnet pooled	0.82	[0.43, 1.37]	90	—	fires when the index is assessable and its point is at least 2.0, with its interval's lower end above 1.0 (as the analysis code applies it, which also sets the fewest slips and steps an index needs to be assessable)	assessable	no
	Targeting index, house variant (ratio; the comparison's ratio zero-slip corrected) against Opus and Sonnet pooled	0.78	[0.14, 1.83]	90	—	fires when the index is assessable and its point is at least 2.0, with its interval's lower end above 1.0 (as the analysis code applies it, which also sets the fewest slips and steps an index needs to be assessable)	not assessable: Opus and Sonnet pooled had 0 exposed slips in 457 exposed steps, fewer than the 3 the index needs	no

Table S15: (continued)

Check	Quantity	Value	Interval	Interval level (%)	Count	Rule	Stamped reading	Fired
Fairness check	Flag K (either index fires)	—	—	—	—	—	does not fire (the targeting index does not fire; the house variant is not assessable)	no
	Slip-rate ratio, T steps over G steps: Haiku (ratio)	4.03	[3.23, 5.10]	90	—	—		—
	Slip-rate ratio, T steps over G steps: Opus (ratio; zero-slip corrected)	0.65	[0.28, 2.16]	90	—	—		—
	Slip-rate ratio, T steps over G steps: Sonnet (ratio)	5.37	[3.09, 10.70]	90	—	—		—
	Slip-rate ratio, T steps over G steps: Opus and Sonnet pooled (ratio)	4.91	[2.98, 9.36]	90	—	—		—
	Slip-rate ratio, house variant (missed house-rule steps over G steps): Haiku (ratio)	0.09	[0.02, 0.20]	90	—	—		—
	Slip-rate ratio, house variant (missed house-rule steps over G steps): Opus (ratio; zero-slip corrected)	0.91	[0.39, 2.88]	90	—	—		—
	Slip-rate ratio, house variant (missed house-rule steps over G steps): Sonnet (ratio; zero-slip corrected)	0.13	[0.09, 0.25]	90	—	—		—
	Slip-rate ratio, house variant (missed house-rule steps over G steps): Opus and Sonnet pooled (ratio; zero-slip corrected)	0.12	[0.08, 0.21]	90	—	—		—
	Opus: slips on T steps, of steps checked (counts)	0 of 322	—	—	—	—		—
	Opus: slips on G steps, of steps checked (counts)	1 of 626	—	—	—	—		—

Table S15: (continued)

Check	Quantity	Value	Interval	Interval level (%)	Count	Rule	Stamped reading	Fired
Fairness check	Opus: house-rule slips missed, of steps that should fire (counts)	0 of 230	—	—	—	—		—
Low-information flag	Haiku-medium single-sample accuracy at 6×1 (pieces), mean over puzzles (%)	51.00	—	—	—	fires when that accuracy is above 60.0% (as the analysis code applies it)		no
	Median over puzzles of the same accuracy (%)	50.00	—	—	—	—		—
Near-ceiling label	Opus-medium pooled accuracy, main run (%) (label shown beside Test 1, Test 2)	98.13	[95.63, 100.00]	95	80 puzzles	set on the length sweep at 6 steps: Opus-medium right on 20 of 20 chains and 20 of 20 pieces	near ceiling (Amendment 1)	yes
Rewording check	Reworded prompt minus Haiku-medium on the same puzzles, both shapes (points)	+8.75	[0.00, 17.25]	95	20 puzzles in each shape	—		—
	Reworded prompt minus Haiku-medium on the same puzzles, 1×6 (points)	+11.50	[0.00, 23.50]	95	20 puzzles	—		—
	Reworded prompt minus Haiku-medium on the same puzzles, 6×1 (points)	+6.00	[−5.00, 17.00]	95	20 puzzles	—		—
Drift check (top-up)	Top-up minus main run, Haiku-medium, both shapes (points)	+1.50	[−8.00, 10.00]	95	20 puzzles in each shape	—		—
	Top-up minus main run, Haiku-medium, 1×6 (points)	+6.00	[−3.00, 15.00]	95	20 puzzles	—		—
	Top-up minus main run, Haiku-medium, 6×1 (points)	−3.00	[−14.00, 8.00]	95	20 puzzles	—		—
	Mean output tokens, relative change top-up against main (%)	+0.29	—	—	—	—		no
	Accuracy condition	—	—	—	—	—		no
	Drift-check flag	—	—	—	—	fires if the pooled 95% interval excludes 0 and $\|d\| > 10$ points, or mean output tokens move by more than 25%; a flag, not a gate	not fired	no

Table S15: (continued)

Panel 1: Format flags, Study 1

Setting	Shape	Answered samples	Format ok	Non-compliance (%)	Fired
Haiku, thinking off	1×6	3,200	2,615	18.28	yes
Haiku, thinking off	6×1	3,200	3,186	0.44	no
Haiku-low	1×6	3,200	3,140	1.88	no
Haiku-low	6×1	3,200	3,189	0.34	no
Haiku-medium	1×6	3,200	3,155	1.41	no
Haiku-medium	6×1	3,200	3,191	0.28	no
Haiku-max	1×6	240	240	0.00	no
Haiku-max	6×1	240	239	0.42	no
Haiku-medium (rewording check)	1×6	200	200	0.00	no
Haiku-medium (rewording check)	6×1	200	200	0.00	no
Opus-low	1×6	80	79	1.25	no
Opus-low	6×1	80	80	0.00	no
Opus-medium	1×6	80	78	2.50	no
Opus-medium	6×1	80	80	0.00	no
Sonnet-low	1×6	80	80	0.00	no
Sonnet-low	6×1	80	80	0.00	no
Sonnet-medium	1×6	80	78	2.50	no
Sonnet-medium	6×1	80	80	0.00	no
Haiku-medium, real time	1×6	25	25	0.00	no
Haiku-medium, real time	6×1	25	25	0.00	no
Opus-medium, real time	1×6	5	5	0.00	no
Opus-medium, real time	6×1	5	5	0.00	no
Sonnet-medium, real time	1×6	5	5	0.00	no
Sonnet-medium, real time	6×1	5	5	0.00	no

Table S15: (continued)

Panel 2: Format flags, Study 1b

Setting	Shape	Answered samples	Format ok	Non-compliance (%)	Fired
Haiku, thinking off	1×6	200	167	16.50	yes
Haiku, thinking off	6×1	200	199	0.50	no
Haiku, thinking off	1×16	400	112	72.00	yes
Haiku, thinking off	16×1	400	313	21.75	yes
Haiku-low	1×6	200	197	1.50	no
Haiku-low	6×1	200	200	0.00	no
Haiku-low	1×16	400	375	6.25	yes
Haiku-low	16×1	400	400	0.00	no
Haiku-medium	1×6	200	198	1.00	no
Haiku-medium	6×1	200	198	1.00	no
Haiku-medium	1×16	399	381	4.51	no
Haiku-medium	16×1	400	400	0.00	no
Haiku-high	1×6	200	200	0.00	no
Haiku-high	6×1	200	200	0.00	no

Table S15: (continued)

Setting	Shape	Answered samples	Format ok	Non-compliance (%)	Fired
Haiku-high	1×16	400	375	6.25	yes
Haiku-high	16×1	400	400	0.00	no
Haiku-xhigh	1×6	200	197	1.50	no
Haiku-xhigh	6×1	200	199	0.50	no
Haiku-xhigh	1×16	400	390	2.50	no
Haiku-xhigh	16×1	400	400	0.00	no
Haiku-max	1×6	400	400	0.00	no
Haiku-max	6×1	400	400	0.00	no
Haiku-max	1×12	800	799	0.13	no
Haiku-max	12×1	800	800	0.00	no
Haiku-max	1×16	800	796	0.50	no
Haiku-max	16×1	800	799	0.13	no
Opus-low	1×6	40	39	2.50	no
Opus-low	6×1	40	40	0.00	no
Opus-medium	1×6	80	80	0.00	no
Opus-medium	6×1	80	80	0.00	no
Opus-medium	1×12	80	79	1.25	no
Opus-medium	12×1	80	80	0.00	no
Opus-medium	1×16	160	160	0.00	no
Opus-medium	16×1	160	160	0.00	no
Opus-high	1×6	40	38	5.00	no
Opus-high	6×1	40	40	0.00	no
Opus-xhigh	1×6	40	38	5.00	no
Opus-xhigh	6×1	40	40	0.00	no
Opus-max	1×6	40	40	0.00	no
Opus-max	6×1	40	40	0.00	no
Sonnet-low	1×6	40	40	0.00	no
Sonnet-low	6×1	40	40	0.00	no
Sonnet-medium	1×6	40	39	2.50	no
Sonnet-medium	6×1	40	40	0.00	no
Sonnet-high	1×6	40	40	0.00	no
Sonnet-high	6×1	40	40	0.00	no
Sonnet-xhigh	1×6	40	40	0.00	no
Sonnet-xhigh	6×1	40	40	0.00	no
Sonnet-max	1×6	40	40	0.00	no
Sonnet-max	6×1	40	40	0.00	no

Checks and flags, not tests. Each Quantity cell names its unit: % for shares and accuracies, points for differences, ratios and counts as named. Dashes mark cells with no stamped value for that column. Terms. A slip is a step whose written balances differ from that step's rule applied to the model's own previous balances. T steps are steps of the targeted rule kinds: kinds added, or made more frequent, where Haiku slipped while the puzzles were tuned; G steps are steps of the other kinds. Same-slip pairwise agreement (Flag R): among pairs of Haiku-medium samples that slip at the same step of the same puzzle (on a chain only its first slip counts, on pieces every slipped piece), the share that wrote the same wrong balances there. Slip-rate ratio: a model's slip rate on T steps divided by its slip rate on G steps. House variant: the house rule is a puzzle-wide rule that should act on some steps; the ratio's numerator is the share of those steps (the exposed steps) where the model missed it (the exposed slips). Targeting index: Haiku-medium's ratio divided by the comparison's ratio, here Opus-medium and Sonnet-medium pooled; near one means the T kinds are no harder for Haiku in particular. Zero-slip corrected: the model had no slips in one group, so half a slip was added to both groups' counts before dividing, which pulls the ratio towards one. Rule column: the rule text is stamped for the flag of the drift check between runs and for the near-ceiling label's source (given in plain words), and the format flag's rule is stamped and given under the format panels. For Flag R, the targeting indices behind Flag K and the low-information flag, the result file stamps no threshold: the Rule cell gives each threshold as the Study 1 analysis code that produced the result file applies it (the fairness check's code and the analysis driver, at the code commit Table S34 lists for the Study 1 analysis). The thresholds printed in those Rule cells are

the ones the pre-registered plan (Flag R and Flag K) and Study 1's reporting definitions (the low-information flag) state. The stamped reading column repeats the files' own reason strings, in plain words where they hold code names. Assessability: the fewest slips and steps an index needs to be assessable are as the analysis code sets them. For the house variant it asks the comparison (Opus and Sonnet pooled) for at least 3 exposed slips, the number in its reading; the pre-registered plan states no such number, and applies its general rule to the house index, which asks the comparison for at least 5 slips in each group, with its house-rule slips as one group. With 0 exposed slips, the house variant is not assessable under either number. The rewording check's lower interval ends at zero are floating-point zeros in the file (exact zeros up to rounding error), printed as zero without a sign. The rewording check (a robustness check planned in the pre-registered design, outside the confirmatory tests; no test): its wording, puzzles and comparison were fixed after Lock 3, before any result was opened. It was checked on Haiku-medium single samples only. Near ceiling (Amendment 1): a label that the pre-registration's Amendment 1 turned on for Tests 1 and 2 because their comparator, Opus-medium, was right on 40 of 40 answers in the length sweep at 6 steps, a pooled whole-answer accuracy of 100.00%, at or above the pre-set 95%; it marks a comparator at or near a perfect score. With it, the error-rate ratio (the challenger's errors over Opus-medium's) is reported with its interval next to the no-worse-than reading (in Table S13); where Opus-medium made no errors the ratio is undefined, and its exact 95% upper error rate, in %, is given instead. The near-ceiling label was set from the length sweep (Amendment 1) and was not re-decided afterwards; the main run's Opus-medium pooled score (the near-ceiling row) is shown for information only. The fairness check's first-slip variant is a separate stamped sensitivity and is not shown here. Flag rule, the same in both studies: the flag fires when non-compliance is above 5% of answered samples.

S4.2 The tests under other views

Table S16 gives Tests 1–5 under every sensitivity and secondary view. **Test 3's pieces counterpart (exploratory).** It is outside Holm's correction and not one of Tests 1–5. At Sonnet-medium's price on pieces (a 14-sample vote, spending 0.951 of Sonnet-medium's cost per puzzle), the Haiku-medium vote read inconclusive against one Sonnet-medium answer: $d = -9.82$ points (95% interval -21.94 to $+2.84$; $p = 0.49$). It is one exploratory line beside Test 3, never a second test. Test 4 has no main-run pieces counterpart, because step voting did not run on pieces there. The top-up run's step voting on pieces later supplied one from a separate run: $d = -48.75$ (95% interval -60.00 to -37.50) against main-run Opus-medium. That comparison is unadjusted, exploratory and outside the pre-registered family, and the drift check between runs did not fire.

Table S16: Tests 1–5 under every sensitivity and secondary view (each column's label is in the note), and under the original marking rule that Deviation 2 replaced; none of these views is a test (Study 1, main run, 80 puzzles per test; paired BCa bootstrap, 10,000 resamples, unadjusted 95% intervals; list price, with the cached view re-priced at cache rates). A panel gives each setting's format compliance under both marking rules.

Test	Shape	Challenger	Comparator	As deployed	Final answer	Copy slips forgiven
Test 1	6×1	Haiku-medium vote	Opus-medium	−33.13 [−43.75, −23.75] (worse by more than the margin)	−33.13 [−43.75, −23.75] (worse by more than the margin)	−33.13 [−43.75, −23.75] (worse by more than the margin)
Test 2	1×6	Haiku-medium vote	Opus-medium	−37.64 [−48.82, −26.96] (worse by more than the margin)	−38.91 [−49.50, −29.23] (worse by more than the margin)	−35.51 [−46.69, −25.13] (worse by more than the margin)
Test 3	1×6	Haiku-medium vote	Sonnet-medium	−24.70 [−36.12, −12.20] (worse by more than the margin)	−25.38 [−36.52, −13.46] (worse by more than the margin)	−22.20 [−33.44, −9.70] (inconclusive, could not show)
Test 4	1×6	Step voting	Opus-medium	−42.50 [−53.75, −30.00] (worse by more than the margin)	−45.00 [−56.25, −33.75] (worse by more than the margin)	−42.50 [−53.75, −30.00] (worse by more than the margin)
Test 5	1×6	Opus-low	Sonnet-medium	+17.50 [8.75, 27.50] (shown no worse than Sonnet-medium by more than the margin)	+16.25 [8.75, 26.25] (shown no worse than Sonnet-medium by more than the margin)	+17.50 [8.75, 27.50] (shown no worse than Sonnet-medium by more than the margin)

Test	Shape	Refusals out	Truncations out	Ties as wrong	Original marking rule	Superiority sign-flip p
Test 1	6×1	−33.13 [−43.75, −23.75] (worse by more than the margin)	−33.13 [−43.75, −23.75] (worse by more than the margin)	−33.75 [−45.00, −23.75] (worse by more than the margin)	−33.13 [−43.75, −23.75] (worse by more than the margin)	1.0000
Test 2	1×6	−37.64 [−48.82, −26.96] (worse by more than the margin)	−37.64 [−48.82, −26.96] (worse by more than the margin)	−39.37 [−50.50, −28.30] (worse by more than the margin)	−37.64 [−48.82, −26.96] (worse by more than the margin)	1.0000
Test 3	1×6	−24.70 [−36.12, −12.20] (worse by more than the margin)	−24.70 [−36.12, −12.20] (worse by more than the margin)	−27.43 [−38.92, −14.84] (worse by more than the margin)	+0.30 [−13.58, 14.95] (inconclusive, could not show)	0.9999
Test 4	1×6	−42.50 [−53.75, −30.00] (worse by more than the margin)	−42.50 [−53.75, −30.00] (worse by more than the margin)	−42.50 [−53.75, −30.00] (worse by more than the margin)	−42.50 [−53.75, −30.00] (worse by more than the margin)	1.0000
Test 5	1×6	+17.50 [8.75, 27.50] (shown no worse than Sonnet-medium by more than the margin)	+17.50 [8.75, 27.50] (shown no worse than Sonnet-medium by more than the margin)	+17.50 [8.75, 27.50] (shown no worse than Sonnet-medium by more than the margin)	+41.25 [28.75, 52.50] (shown no worse than Sonnet-medium by more than the margin)	0.0006

Table S16: (continued)

Test	Shape	5-point margin	k (cached re-pricing)	Cached re-pricing	Equal-token view
Test 1	6×1	−33.13 [−43.75, −23.75] (worse by more than the margin)	40	−33.13 [−43.75, −23.75] (worse by more than the margin)	−49.00 [−55.66, −42.66] (worse by more than the margin)
Test 2	1×6	−37.64 [−48.82, −26.96] (worse by more than the margin)	39	−38.21 [−49.39, −27.04] (worse by more than the margin)	−52.53 [−60.34, −44.16] (worse by more than the margin)
Test 3	1×6	−24.70 [−36.12, −12.20] (worse by more than the margin)	14	−24.39 [−35.90, −11.80] (worse by more than the margin)	−36.28 [−46.31, −24.47] (worse by more than the margin)
Test 4	1×6	−42.50 [−53.75, −30.00] (worse by more than the margin)	—	−42.50 [−53.75, −30.00] (worse by more than the margin)	−42.50 [−53.75, −30.00] (worse by more than the margin)
Test 5	1×6	+17.50 [8.75, 27.50] (shown no worse than Sonnet-medium by more than the margin)	—	+17.50 [8.75, 27.50] (shown no worse than Sonnet-medium by more than the margin)	+17.50 [8.75, 27.50] (shown no worse than Sonnet-medium by more than the margin)

Table S16: (continued)

Panel 1: Format compliance per setting under both rules

Setting	Shape	Samples	Format compliance, official rule (%)	Format compliance, original marking rule (%)
Haiku-low	1×6	3,200	98.13	98.09
Haiku-low	6×1	3,200	99.66	—
Haiku-medium	1×6	3,200	98.59	98.56
Haiku-medium	6×1	3,200	99.72	—
Haiku-max	1×6	240	100.00	100.00
Haiku-max	6×1	240	99.58	—
Haiku, thinking off	1×6	3,200	81.72	79.34
Haiku, thinking off	6×1	3,200	99.56	—
Opus-low	1×6	80	98.75	97.50
Opus-low	6×1	80	100.00	—
Opus-medium	1×6	80	97.50	97.50
Opus-medium	6×1	80	100.00	—
Sonnet-low	1×6	80	100.00	77.50
Sonnet-low	6×1	80	100.00	—
Sonnet-medium	1×6	80	97.50	65.00
Sonnet-medium	6×1	80	100.00	—

Every view cell reads: d in points (challenger minus comparator), its 95% interval, and its reading by the pre-registered decision rule against that view's margin. Margin in every view except the 5-point view: −10 points; the 5-point view's margin: −5 points. Stamped view labels, in the file's own words. 5-point margin: "secondary, 5-point margin"; cached re-pricing: "cached view (offline re-pricing)"; sign-flip p: "sensitivity (superiority, $d = 0$)" (a superiority check against no difference, not against the margin). Refusals out and truncations out, as stamped: "no missing items in the five tests' arms" (the file's "arms" are this paper's settings). Tests 4 and 5 have no vote size k, so no k is shown for them in the cached re-pricing view. In the equal-token view k is 1 for every vote (Tests 1–3), so it is given here and is not a column. Label of each column. As deployed: the confirmatory reading (as in Table S13). Final answer and copy slips forgiven: pre-registered secondary views. Refusals out, truncations out and ties as wrong: pre-registered sensitivity checks. Original marking rule: the sensitivity check Deviation 2 fixed in Lock 3, before any main-run data (not a pre-registered secondary view). 5-point margin, cached re-pricing and equal-token view: reporting views fixed after Lock 3, before any result was opened. Superiority sign-flip p: a pre-specified sensitivity check against no difference. None of these views is a test. What each view changes. Refusals out and truncations out: blocked or truncated calls are left out instead of counted wrong. Final answer: a sample is marked, and the vote counted, by its final answer only, not its step lines. Copy slips forgiven: a sample whose step lines differ from the key at one line only, with the next line right, counts as right. Ties as wrong: a vote whose top answers tie, the right one among them, scores nothing instead of an equal share. Cached re-pricing: the vote's repeated input is priced at cache rates, so the same money can buy a larger k. Equal-token view: k is the number of Haiku-medium samples whose mean total tokens fit within the comparator's (at least one), instead of its money. Sonnet ran at medium effort, one level below its default, in Tests 3 and 5. The original marking rule has a stamped format figure for the 1×6 chains only; the pieces cells are dashes. The original-rule figures for Haiku-low; Haiku-max; Haiku, thinking off; and Sonnet-low come from the exploratory settings' own rate blocks, those for the other four settings from the secondary views' rates; the analysis file's appendix copy of the format block repeats the official values and is not used.

Cached re-pricing (a labelled offline re-pricing view). Pricing the vote's repeated input at cache rates moves k to 40 for Test 1 (unchanged), to 39 for Test 2 (from 36) and to 14 for Test 3 (from 13). It changes no reading, and headline costs stay uncached list prices:

- Test 2: $d = -38.21$ (95% interval −49.39 to −27.04);
- Test 3: $d = -24.39$ (−35.90 to −11.80);
- Tests 1, 4 and 5: as deployed.

S4.3 The weaker Haiku settings, Sonnet-low and Opus against Sonnet (exploratory)

The weaker Haiku settings (no Holm correction, no verdict; Table S17). At their own cost-matched vote sizes, they also stayed far below one Opus-medium answer:

- **Haiku-low's vote** (39 samples on chains; 40 on pieces, capped from 46) scored 49.38% and 62.50%: $d = -46.88$ (95% interval -58.13 to -35.63) and -37.50 (-48.13 to -27.50).
- **The vote of Haiku, thinking off** (40 samples, capped from 45 and 50) scored 42.75% and 40.00%: $d = -53.50$ (-64.38 to -42.00) and -60.00 (-70.00 to -48.75).

Turning thinking off saved money per call but raised the cost per correct answer, at list price: from \$0.00252 to \$0.00405 on chains, and from \$0.00214 to \$0.00407 on pieces.

Sonnet-low (no Holm correction, no verdict; paired over 80 puzzles) scored 71.25% on chains and 76.25% on pieces.

- It was not shown to differ from Sonnet-medium: chains -8.75 (95% interval -21.25 to $+3.75$); pieces $+1.25$ (-10.00 to $+12.50$).
- It scored below Opus-low: chains -26.25 (-36.25 to -17.50); pieces -23.75 (-33.75 to -15.00).
- Its chain score depends on Deviation 2: 52.5% under the original rule.

Low and medium Sonnet were not separated in either study: both scored 80.0% in Study 1b at 6 steps. Sonnet's large gain came at high effort, which only Study 1b ran.

Opus-medium against Sonnet-medium (no test). Opus-medium minus Sonnet-medium was $+16.25$ points on chains (95% interval $+6.25$ to $+26.25$) and $+25.00$ on pieces ($+16.25$ to $+35.00$). It cost 2.73 and 2.99 times as much per puzzle, and 2.27 (1.95 to 2.59) and 2.24 (1.96 to 2.52) times as much per correct answer. Sonnet here ran one level below its default.

Table S17: Study 1's exploratory[‡] paired comparisons, unadjusted and outside Holm's correction (Study 1, main run, 80 puzzles per pair; list price), in blocks: each Haiku setting's own cost-matched vote against Opus-medium and against Sonnet-medium, Sonnet-low against Sonnet-medium and Opus-low, the pieces counterparts of Tests 3 and 5, the low-against-medium effort pairs with their cost differences, and Opus against Sonnet. The 95% intervals come from a paired BCa bootstrap in every block except Opus against Sonnet, which uses a paired cluster bootstrap over puzzles; where both settings of a pair sit at the ceiling, each setting's accuracy also carries its exact Clopper–Pearson interval (post hoc[†], a presentation only).

Group	Challenger	Comparator	Shape	k	Raw k	Capped by	Spend share	Status (stamped)
Own-k vote vs Opus-medium	Haiku-low vote	Opus-medium	1×6	39	39		0.9998	exploratory (no Holm, no verdict)
Own-k vote vs Sonnet-medium	Haiku-low vote	Sonnet-medium	1×6	14	14		0.980	exploratory (no Holm, no verdict)
Own-k vote vs Opus-medium	Haiku-low vote	Opus-medium	6×1	40	46	cap on k	0.853	exploratory (no Holm, no verdict)
Own-k vote vs Sonnet-medium	Haiku-low vote	Sonnet-medium	6×1	15	15		0.956	exploratory (no Holm, no verdict)
Own-k vote vs Opus-medium	Haiku, thinking off vote	Opus-medium	1×6	40	45	cap on k	0.885	exploratory (no Holm, no verdict)
Own-k vote vs Sonnet-medium	Haiku, thinking off vote	Sonnet-medium	1×6	16	16		0.966	exploratory (no Holm, no verdict)
Own-k vote vs Opus-medium	Haiku, thinking off vote	Opus-medium	6×1	40	50	cap on k	0.795	exploratory (no Holm, no verdict)
Own-k vote vs Sonnet-medium	Haiku, thinking off vote	Sonnet-medium	6×1	16	16		0.950	exploratory (no Holm, no verdict)
Own-k vote vs Opus-medium	Haiku-max vote	Opus-medium	1×6	3	9	samples drawn	0.328	exploratory (no Holm, no verdict)
Own-k vote vs Sonnet-medium	Haiku-max vote	Sonnet-medium	1×6	3	3		0.896	exploratory (no Holm, no verdict)
Own-k vote vs Opus-medium	Haiku-max vote	Opus-medium	6×1	3	11	samples drawn	0.254	exploratory (no Holm, no verdict)
Own-k vote vs Sonnet-medium	Haiku-max vote	Sonnet-medium	6×1	3	3		0.758	exploratory (no Holm, no verdict)
Sonnet-low vs Sonnet-medium and Opus-low	Sonnet-low	Sonnet-medium	1×6	1	—		—	exploratory (no Holm, no verdict)
	Sonnet-low	Opus-low	1×6	1	—		—	exploratory (no Holm, no verdict)
	Sonnet-low	Sonnet-medium	6×1	1	—		—	exploratory (no Holm, no verdict)
	Sonnet-low	Opus-low	6×1	1	—		—	exploratory (no Holm, no verdict)
Pieces counterpart of Test 3	Haiku-medium vote	Sonnet-medium	6×1	14	14		0.951	unadjusted (exploratory, outside Holm)
Pieces counterpart of Test 5	Opus-low	Sonnet-medium	6×1	—	—		—	unadjusted (exploratory, outside Holm)
Effort pair	Haiku-low	Haiku-medium	1×6	1	—		—	exploratory: no Holm, no verdict
	Haiku-low	Haiku-medium	6×1	1	—		—	exploratory: no Holm, no verdict
	Opus-low	Opus-medium	1×6	1	—		—	exploratory: no Holm, no verdict
	Opus-low	Opus-medium	6×1	1	—		—	exploratory: no Holm, no verdict
Opus against Sonnet	Opus-low	Sonnet-medium	1×6	—	—		—	exploratory, no test (intervals two-sided, unadjusted)
	Opus-medium	Sonnet-medium	1×6	—	—		—	exploratory, no test (intervals two-sided, unadjusted)
	Opus-low	Sonnet-medium	6×1	—	—		—	exploratory, no test (intervals two-sided, unadjusted)
	Opus-medium	Sonnet-medium	6×1	—	—		—	exploratory, no test (intervals two-sided, unadjusted)

Table S17: (continued)

Group	Challenger	Comparator	Shape	Challenger accuracy (%)	Challenger format compliance (%)	Comparator accuracy (%)	Comparator format compliance (%)	d (points)	d 95% interval
Own-k vote vs Opus-medium	Haiku-low vote	Opus-medium	1×6	49.38	98.13	96.25	97.50	−46.88	[−58.13, −35.63]
Own-k vote vs Sonnet-medium	Haiku-low vote	Sonnet-medium	1×6	48.26	98.13	80.00	97.50	−31.74	[−43.32, −19.14]
Own-k vote vs Opus-medium	Haiku-low vote	Opus-medium	6×1	62.50	99.66	100.00	100.00	−37.50	[−48.13, −27.50]
Own-k vote vs Sonnet-medium	Haiku-low vote	Sonnet-medium	6×1	61.08	99.66	75.00	100.00	−13.92	[−26.63, −0.88]
Own-k vote vs Opus-medium	Haiku, thinking off vote	Opus-medium	1×6	42.75	81.72	96.25	97.50	−53.50	[−64.38, −42.00]
Own-k vote vs Sonnet-medium	Haiku, thinking off vote	Sonnet-medium	1×6	40.41	81.72	80.00	97.50	−39.59	[−51.52, −26.32]
Own-k vote vs Opus-medium	Haiku, thinking off vote	Opus-medium	6×1	40.00	99.56	100.00	100.00	−60.00	[−70.00, −48.75]
Own-k vote vs Sonnet-medium	Haiku, thinking off vote	Sonnet-medium	6×1	39.05	99.56	75.00	100.00	−35.95	[−48.17, −23.30]
Own-k vote vs Opus-medium	Haiku-max vote	Opus-medium	1×6	100.00	100.00	96.25	97.50	+3.75	[1.25, 10.00]
Own-k vote vs Sonnet-medium	Haiku-max vote	Sonnet-medium	1×6	100.00	100.00	80.00	97.50	+20.00	[12.50, 30.00]
Own-k vote vs Opus-medium	Haiku-max vote	Opus-medium	6×1	98.75	99.58	100.00	100.00	−1.25	[−7.50, 0.00]
Own-k vote vs Sonnet-medium	Haiku-max vote	Sonnet-medium	6×1	98.75	99.58	75.00	100.00	+23.75	[15.00, 35.00]

Table S17: (continued)

Group	Challenger	Comparator	Shape	Challenger accuracy (%)	Challenger format compliance (%)	Comparator accuracy (%)	Comparator format compliance (%)	d (points)	d 95% interval
Sonnet-low vs Sonnet-medium and Opus-low	Sonnet-low	Sonnet-medium	1×6	71.25	100.00	80.00	97.50	−8.75	[−21.25, 3.75]
	Sonnet-low	Opus-low	1×6	71.25	100.00	97.50	98.75	−26.25	[−36.25, −17.50]
	Sonnet-low	Sonnet-medium	6×1	76.25	100.00	75.00	100.00	+1.25	[−10.00, 12.50]
	Sonnet-low	Opus-low	6×1	76.25	100.00	100.00	100.00	−23.75	[−33.75, −15.00]
Pieces counterpart of Test 3	Haiku-medium vote	Sonnet-medium	6×1	65.18	99.72	75.00	100.00	−9.82	[−21.94, 2.84]
Pieces counterpart of Test 5	Opus-low	Sonnet-medium	6×1	100.00	100.00	75.00	100.00	+25.00	[16.25, 35.00]
Effort pair	Haiku-low	Haiku-medium	1×6	39.22	98.13	43.72	98.59	−4.50	[−6.91, −2.13]
	Haiku-low	Haiku-medium	6×1	46.50	99.66	51.00	99.72	−4.50	[−7.84, −1.09]
	Opus-low	Opus-medium	1×6	97.50	98.75	96.25	97.50	+1.25	[−3.75, 7.50]
	Opus-low	Opus-medium	6×1	100.00 [95.49, 100.00] post hoc† (exact)	100.00	100.00 [95.49, 100.00] post hoc† (exact)	100.00	0.00	[0.00, 0.00] (collapsed)
Opus against Sonnet	Opus-low	Sonnet-medium	1×6	97.50	98.75	80.00	97.50	—	Test 5: see Table S13
	Opus-medium	Sonnet-medium	1×6	96.25	97.50	80.00	97.50	+16.25	[6.25, 26.25]
	Opus-low	Sonnet-medium	6×1	100.00	100.00	75.00	100.00	—	see the pieces counterpart of Test 5 row
	Opus-medium	Sonnet-medium	6×1	100.00	100.00	75.00	100.00	+25.00	[16.25, 35.00]

Table S17: (continued)

Group	Challenger	Comparator	Shape	Cost ratio (challenger / comparator) [95% interval]	Cost per correct answer ratio [95% interval]	Cost difference ($ per sample)
Effort pair	Haiku-low	Haiku-medium	1×6	—	—	−0.0000686
	Haiku-low	Haiku-medium	6×1	—	—	−0.0000672
	Opus-low	Opus-medium	1×6	—	—	−0.00371
	Opus-low	Opus-medium	6×1	—	—	−0.00326
Opus against Sonnet	Opus-low	Sonnet-medium	1×6	2.48 [2.36, 2.59]	2.03 [1.76, 2.32]	—
	Opus-medium	Sonnet-medium	1×6	2.73 [2.62, 2.84]	2.27 [1.95, 2.59]	—
	Opus-low	Sonnet-medium	6×1	2.79 [2.72, 2.85]	2.09 [1.81, 2.34]	—
	Opus-medium	Sonnet-medium	6×1	2.99 [2.92, 3.05]	2.24 [1.96, 2.52]	—

Every comparison is exploratory, unadjusted and outside the Holm family: no verdict is printed. The status column repeats each block's stamped label. The own-k votes use each Haiku setting's own cost-matched k against Opus-medium and against Sonnet-medium; where a k is capped, the Raw k column gives the cost-matched k before the cap. Capped, all against Opus-medium: Haiku-low on 6×1, held at the cap on k of 40 (raw k 46); Haiku, thinking off on 1×6 and 6×1, held at the cap on k of 40 (raw k 45 and 50); Haiku-max on 1×6 and 6×1, held at its 3 samples drawn (raw k 9 and 11). Against Sonnet-medium no k is capped. The spend share of a vote is its spend over the comparator's (the file's "spend ratio"), a point with no interval. Spend shares print at three decimals; 0.9998 (Haiku-low vote against Opus-medium at 1×6) keeps a fourth decimal, because at three it would print as 1.000 and read as equal to the comparator's spend. The pieces counterparts repeat Test 3's and Test 5's comparisons at 6×1. Test 4's comparison was not computed at 6×1 in the main run, because step voting did not run on pieces there. Step voting on pieces against one Opus-medium answer (the pieces counterpart of Test 4; unadjusted, exploratory, outside the pre-registered family) ran later, in the top-up run, and is reported in Table S1. Neither pieces counterpart is shown no worse (Holm-adjusted): both are outside the Holm family. The file's own "declared" field on the counterpart of Test 5 means its p fell below the one-sided level for that comparison alone, not under Holm, and is not shown. Format compliance is the share of each side's samples in the required answer format; for a vote, that of the setting's samples the vote is drawn from. Effort pairs are low effort minus medium effort at one sample; the cost difference is the stamped difference of the two mean list costs of one sample (in US dollars; for Opus, its one answer). Where both settings of an effort pair are at the ceiling, the bootstrap interval of d collapses to one point (marked collapsed). The difference itself has no exact interval in free analysis 4, which gives each setting's exact 95% interval instead (Clopper–Pearson over puzzles), printed in that setting's accuracy cell as post hoc† (exact): a presentation only, not a test. Opus-low against Sonnet-medium at 1×6 is Test 5's comparison. Its confirmatory d and interval are in Table S13; the Opus-against-Sonnet block's separate exploratory interval for the same pair is not printed, and that row keeps the block's accuracies and cost ratios only. At 6×1 the same pair is the pieces counterpart of Test 5: its d and interval are printed once, in that row, and the Opus-against-Sonnet row keeps its accuracies and cost ratios only. Dashes mark cells that do not apply (a spend share and a raw k belong to the votes, a cost ratio to the Opus-against-Sonnet rows), cells the files do not stamp (the k of the Opus-against-Sonnet rows and of the pieces counterpart of Test 5, and the cost difference in dollars of the Opus-against-Sonnet rows) and the d cells of the two Opus-against-Sonnet rows whose comparison is printed elsewhere. The cost columns are in a part of their own that prints only the rows with a cost figure, so the dashes of the other rows are not printed there; the accuracy columns are in two parts, one for the own-k votes and one for the other rows. Sonnet-medium ran at medium effort, one level below Sonnet's default, in every row where it appears.

Table S18: The planned descriptive analyses, computed after the confirmatory results were opened (Study 1, main run, 80 puzzles; list price for the effort pairs). The main block gives each setting's correlation between its scores on the two shapes and its pieces-minus-chains difference. Four panels give the shape-by-setup interaction for Tests 1 and 2, the difficulty bins and the crossover, wrong-sample agreement against vote gain, and the effort pairs with their cost differences, which are exploratory (no Holm, no verdict). The 95% intervals come from the shared cluster bootstrap for the correlations, the shape differences, the interaction and the agreement; from a paired BCa bootstrap for Tests 1 and 2 (their confirmatory intervals) and for the effort pairs; and, where both settings of a pair sit at the ceiling, exact Clopper–Pearson per setting (post hoc†).

Setting	C2 statistic	C2 correlation, 6×1 against 1×6	C2 95% interval	C2 undefined where
Haiku, thinking off	Pearson	+0.20	[−0.09, 0.45]	
Haiku-low	Pearson	+0.26	[0.05, 0.45]	
Haiku-medium	Pearson	+0.20	[−0.02, 0.40]	
Haiku-max	Pearson	+0.12	[−0.09, 0.44]	
Opus-low	phi (Pearson on 0/1)	undefined (no variation)	—	6×1 has no variation
Opus-medium	phi (Pearson on 0/1)	undefined (no variation)	—	6×1 has no variation
Sonnet-low	phi (Pearson on 0/1)	+0.03	[−0.19, 0.25]	
Sonnet-medium	phi (Pearson on 0/1)	−0.14	[−0.31, 0.05]	

Setting	C3 accuracy 1×6 (%)	Format compliance 1×6 (%)	C3 accuracy 6×1 (%)	Format compliance 6×1 (%)	C3 *d*: 6×1 minus 1×6 (points)	C3 95% interval
Haiku, thinking off	22.00	81.72	23.50	99.56	+1.50	[−4.81, 8.34]
Haiku-low	39.22	98.13	46.50	99.66	+7.28	[−0.94, 15.16]
Haiku-medium	43.72	98.59	51.00	99.72	+7.28	[−1.47, 15.38]
Haiku-max	97.50	100.00	95.00	99.58	−2.50	[−6.25, 0.83]
Opus-low	97.50	98.75	100.00	100.00	+2.50	[0.00, 6.25]
Opus-medium	96.25	97.50	100.00	100.00	+3.75	[0.00, 8.75]
Sonnet-low	71.25	100.00	76.25	100.00	+5.00	[−8.75, 18.75]
Sonnet-medium	80.00	97.50	75.00	100.00	−5.00	[−18.75, 8.75]

Panel 1: C5: shape-by-setup interaction (Tests 1 and 2)

Quantity	*k*	Value (points)	95% interval
Test 1 (6×1): Haiku-medium vote minus Opus-medium (confirmatory, as in Table S13)	40	−33.13	[−43.75, −23.75]
Test 2 (1×6): Haiku-medium vote minus Opus-medium (confirmatory, as in Table S13)	36	−37.64	[−48.82, −26.96]
Shape-by-setup interaction: Test 1 minus Test 2	—	+4.52	[−9.85, 18.23]

S4.4 Where the vote helped, and the planned descriptive analyses

These descriptive analyses were fixed in advance and computed after the test results were opened; Table S18 lists them.

Table S18: (continued)

Panel 2: C7: difficulty bins and the crossover

Shape	k_O	Single-sample accuracy bin (%)	Puzzles	Haiku-medium single sample (%)	Haiku-medium vote at k_O (%)	Opus-medium (%)	Vote reaches Opus within k_O	Smallest k reaching Opus
1×6	36	[0,10)	16	3.59	0.15	93.75	no	none up to 40
1×6	36	[10,20)	8	14.06	30.10	100.00	no	none up to 40
1×6	36	[20,30)	6	22.08	29.20	100.00	no	none up to 40
1×6	36	[30,40)	8	32.81	48.58	87.50	no	none up to 40
1×6	36	[40,50)	9	43.61	66.88	100.00	no	none up to 40
1×6	36	[50,60)	8	53.44	97.45	100.00	no	39
1×6	36	[60,70)	3	64.17	100.00	66.67	yes	3
1×6	36	[70,80)	7	73.93	100.00	100.00	yes	22
1×6	36	[80,90)	4	84.38	100.00	100.00	yes	10
1×6	36	[90,100]	11	96.82	100.00	100.00	yes	6
6×1	40	[0,10)	8	2.19	0.00	100.00	no	none up to 40
6×1	40	[10,20)	6	12.08	0.00	100.00	no	none up to 40
6×1	40	[20,30)	9	24.44	22.22	100.00	no	none up to 40
6×1	40	[30,40)	3	31.67	0.00	100.00	no	none up to 40
6×1	40	[40,50)	12	43.96	79.17	100.00	no	none up to 40
6×1	40	[50,60)	8	53.13	100.00	100.00	yes	39
6×1	40	[60,70)	7	62.14	100.00	100.00	yes	31
6×1	40	[70,80)	9	74.72	100.00	100.00	yes	21
6×1	40	[80,90)	9	82.78	100.00	100.00	yes	16
6×1	40	[90,100]	9	96.67	100.00	100.00	yes	7

Panel 3: C9: wrong-sample agreement against vote gain

Shape	k_O	Spearman rho, wrong-sample agreement against vote gain	95% interval	Puzzles kept	Puzzles left out (fewer than two wrong samples)
1×6	36	−0.18	[−0.40, 0.04]	74	6
6×1	40	−0.30	[−0.50, −0.09]	75	5

Table S18: (continued)

Panel 4: C10: effort pairs with cost differences

Pair (low minus medium effort)	Shape	Low-effort accuracy (%)	Low-effort format compliance (%)	Medium-effort accuracy (%)	Medium-effort format compliance (%)
Haiku-low minus Haiku-medium	1×6	39.22	98.13	43.72	98.59
Haiku-low minus Haiku-medium	6×1	46.50	99.66	51.00	99.72
Opus-low minus Opus-medium	1×6	97.50	98.75	96.25	97.50
Opus-low minus Opus-medium	6×1	100.00 [95.49, 100.00] post hoc† (exact)	100.00	100.00 [95.49, 100.00] post hoc† (exact)	100.00

Panel 4: (continued)

Pair (low minus medium effort)	Shape	d (points)	d 95% interval	Low-effort list cost per sample ($)	Medium-effort list cost per sample ($)	Cost difference ($ per sample)
Haiku-low minus Haiku-medium	1×6	−4.50	[−6.91, −2.13]	0.00103	0.00110	−0.0000686
Haiku-low minus Haiku-medium	6×1	−4.50	[−7.84, −1.09]	0.00103	0.00109	−0.0000672
Opus-low minus Opus-medium	1×6	+1.25	[−3.75, 7.50]	0.0365	0.0402	−0.00371
Opus-low minus Opus-medium	6×1	0.00	[0.00, 0.00] (collapsed)	0.0449	0.0481	−0.00326

C2 is the correlation, across the puzzles, of each puzzle's accuracy at 6×1 and at 1×6 (Pearson for the Haiku settings, phi for one-sample settings); C3 is the paired difference 6×1 minus 1×6. A correlation is undefined where a setting has no variation at one shape (the cell shows the stamped text). C2 definition, as stamped: "Per arm, across the 80 base puzzles: the Pearson correlation of per-puzzle accuracy at 6×1 and at 1×6. For the Haiku arms that is each puzzle's share right of its tries; for one-try arms it is 0/1, which gives the phi coefficient. Shared cluster bootstrap 95%. Descriptive". C3 definition, as stamped: "Per arm: the paired difference in whole-answer accuracy, 6×1 minus 1×6, over the 80 bases, with the shared cluster bootstrap 95%. Descriptive. This is also the 'structure effect'". In the stamped texts (the C2 and C3 definitions and the C7 sentences), "arm" means setting, "base puzzle" and "bases" mean puzzle and puzzles, and "try" and "tries" mean sample and samples. C2 and C3: 80 puzzles in every row. Format compliance is the share of the setting's answered samples in the required answer format at that shape (from the main analysis), beside each C3 accuracy. Haiku-max's C2 interval leaves out the 9 of 2,000 bootstrap draws in which the correlation was undefined (0.45% of the draws). k_O is the Haiku-medium vote's k against Opus-medium at that shape, the k of Test 1 (6×1) and of Test 2 (1×6) in Table S13: its cost-matched k, held at the cap of 40 on 6×1 (raw k 44). Sign: Test 1 (6×1) minus Test 2 (1×6), each the Haiku-medium vote at its k_O minus Opus-medium. Tests 1 and 2's rows repeat their confirmatory d and interval (paired BCa bootstrap, as in Table S13); the interaction's interval is the shared cluster bootstrap. C5 panel: 80 puzzles in every row. 1×6, as stamped: "the lowest single-try accuracy bin at which a vote within k_O reaches Opus: [60,70) ($k_O = 36$)". 6×1, as stamped: "the lowest single-try accuracy bin at which a vote within k_O reaches Opus: [50,60) ($k_O = 40$)". The file's own words for the unit: "vote gain in points; agreement a share"; the puzzles left out are counted, not used in the correlation. Costs are the mean list cost of one sample (for Opus, its one answer); the cost difference is low effort minus medium effort. The file's own words for the status: "exploratory: no Holm, no verdict". Where both settings are at the ceiling, the bootstrap interval of d collapses to one point (marked collapsed). The difference itself has no exact interval in free analysis 4, which gives each setting's exact 95% interval instead (Clopper–Pearson over puzzles), printed in that setting's accuracy cell as post hoc† (exact): a presentation only, not a test.

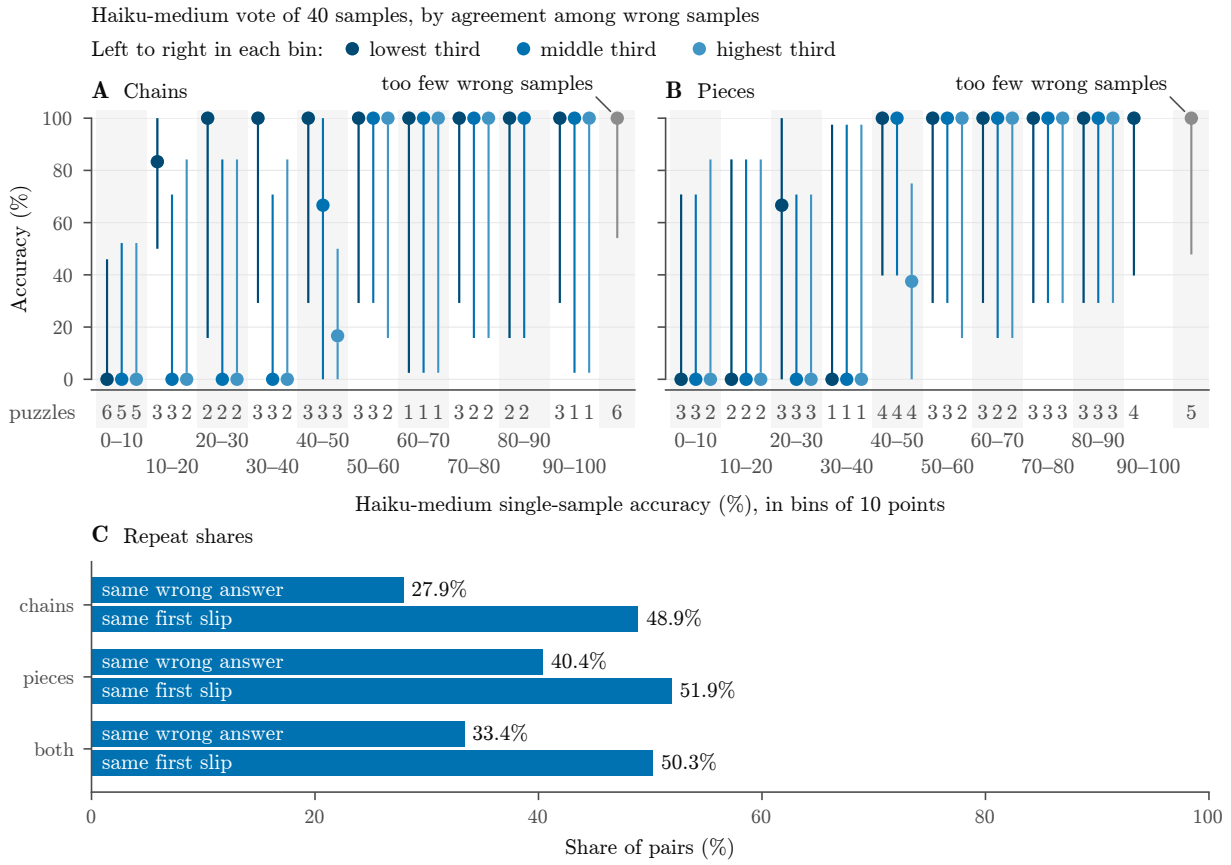


Figure S10: Where the vote helped (pre-specified descriptive; no test and no cause): Haiku-medium's 40-sample vote on Study 1's 6-step puzzles, by single-sample accuracy bin (A chains, B pieces) and by how often a puzzle's wrong samples repeat each other (dots left to right, lowest third to highest). The number under each dot is its cell's puzzle count; cells are small, so no line joins the dots. No value prints beside a dot; the cells neither at zero nor at the top are, on chains, 83.33% (10–20% bin, lowest third) and 66.67% and 16.67% (40–50% bin, middle and highest thirds), and on pieces 66.67% (20–30% bin, lowest third) and 37.5% (40–50% bin, highest third). Bars are cluster-bootstrap 95% intervals, exact where a cell is all right or all wrong (post hoc†). C: how often pairs of wrong samples share an answer or a first slip (no interval).

Where the vote helped (Figure S10; pre-specified descriptive). The lookup chart splits each single-sample difficulty bin into thirds by how often the wrong samples agree. It reports the 40-sample vote (the cost-matched size is 36 on chains and 40 on pieces), over the puzzles with at least 2 wrong samples. On chains, in the 10–20%, 20–30% and 30–40% bins the lowest-agreement third scored 83.33%, 100.0% and 100.0% while both higher thirds scored 0.0%; the 40–50% bin gave 100.0%, 66.67% and 16.67%. On pieces the pattern is weaker: 66.67%, 0.0% and 0.0% in the 20–30% bin, 100.0%, 100.0% and 37.5% in the 40–50% bin, and 0.0% in every third of the 10–20% and 30–40% bins. Below 10% single-sample accuracy the vote scored 0.0% in every third (16 chain and 8 piece puzzles). Cells hold 1 to 6 puzzles, and there is no test. In post hoc counts at the cost-matched vote size, the vote scored below the puzzle's own single-sample rate on 28 of 74 chain puzzles and 22 of 75 piece puzzles, above it on 41 and 49, and equal on 5 and 4; it scored 100% on 34 and 48 of them.

The crossover. This is the lowest single-sample bin in which a vote within the cost-matched size first scores no lower than Opus-medium on point estimates: on chains the 60–70% bin, within the cost-matched size of 36 samples; on pieces the 50–60% bin, within the cost-matched size of 40. Two qualifiers apply:

- The chain bin holds 3 puzzles, on which Opus-medium itself scored 66.67%, so the vote's standing there partly reflects Opus's misses.
- In the 8-puzzle chain bin below it (50–60%), the 36-sample vote scored 97.45%, against 100.00% for Opus-medium, and first scores no lower only at 39 samples, beyond the cost-matched size. On pieces the 50–60% bin also first does so at 39 samples, inside the cost-matched size of 40.

Bins hold 3 to 16 puzzles. These are pre-specified descriptive results, computed after the confirmatory results were opened; there is no test.

How alike the two shapes are. The pairing of each puzzle across the two shapes asked for the cross-shape correlation to be reported. The correlations are weak: $r = 0.20$ for Haiku-medium (-0.02 to 0.40), 0.26 for Haiku-low (0.05 to 0.45), 0.12 for Haiku-max (-0.09 to 0.44), 0.20 for Haiku, thinking off (-0.09 to 0.45), -0.14 for Sonnet-medium (-0.31 to 0.05) and 0.03 for Sonnet-low (Table S18 gives its interval).

It is undefined for Opus, which shows no variation. Our reading, not a test: the pairing bought little extra precision for the shape contrasts, which helps explain why the pieces-minus-chains intervals are wide.

Sonnet's shape order is not stable (descriptive and untested; these are readings across puzzle sets).

- **The length sweep** (20 puzzles per length, one sample per puzzle, Deviation 2's rule): Sonnet-medium scored higher on chains than on pieces at 4 of 5 lengths. The exception is 4 steps (chains 80%, pieces 95%).
- **Under the original marking rule:** the count is 3 of 5 (at 6 steps, chains 70% and pieces 75%). Deviation 2 replaced that rule after these same Sonnet replies had been seen.
- **Study 1b's 6-step set** (40 puzzles, one sample each): Sonnet-medium scored 77.50% on chains (61.55 to 89.16) and 82.50% on pieces (67.22 to 92.66), and Sonnet-low scored the same.
- **Study 1** (80 puzzles): Sonnet-medium scored 80.00% on chains and 75.00% on pieces (pieces minus chains -5.00, -18.75 to $+8.75$), but Sonnet-low scored 71.25% and 76.25% ($+5.00$, -8.75 to $+18.75$).

At 6 to 16 steps, Sonnet-medium's length-sweep chains sit above the reference line (the per-step rate on pieces, compounded over the steps). The line's q comes from pieces only (94.67%), and Sonnet's per-step accuracy on chains (98.13–99.17%) is higher than on pieces (93.33–95.83%). Predicted from its own chain step rate, the figures are close: 77.73% against 75.0% observed at 16 steps, and 90.45% against 95.0% at 12.

S4.5 Other views and checks

The per-piece view. The pre-registration names this as the reader-facing secondary view (descriptive, no test).

- **Haiku-medium:** a single sample got 89.52% of individual pieces right on pieces (95% interval 87.76 to 91.23), at \$0.000182 per piece, about 1/44 of Opus-medium's cost per piece (\$0.00802; cost ratio 44.03).
- **Haiku-max:** 99.03% (98.33 to 99.58) at \$0.000678, about 1/12 of Opus-medium's cost (cost ratio 11.83).
- **Opus-medium and Opus-low** got every piece right.
- **A 40-sample vote taken piece by piece**, the cost-matched size against Opus-medium on pieces, got 93.75% (91.46 to 95.73) at \$0.00729 per piece (0.909 of Opus-medium's).

The whole-answer pieces score is close to the per-piece rate compounded six times: 89.52% compounded over six pieces gives 51.45%, against an observed 51.00% (post hoc arithmetic).

The escalate-to-Opus cascade (a pre-specified exploratory offline simulation; Table S19). It takes a vote of the first k_c Haiku-medium samples ($k_c = 5$, 10 or 20). If the winning answer's share of that vote is below a threshold, it uses Opus-medium's answer instead. Each threshold was chosen on the odd-numbered puzzles and scored on the 40 even-numbered ones.

At $k_c = 10$:

- **On chains** (threshold 0.8) it saved nothing measurable. It scored 90.00% against Opus-medium's 92.50% on the same puzzles: $d = -2.50$ (-7.50 to 0.00), escalating 70.00% of puzzles, at \$0.0408 against \$0.0416. The cost difference was -0.000822 US dollars per puzzle (-0.00719 to $+0.00490$), which spans zero.
- **On pieces** (threshold 0.7) it traded 7.50 points for a 24.99% saving. It scored 92.50% against 100.00%: $d = -7.50$ (-15.00 to 0.00), escalating 52.50%, at \$0.0366 against \$0.0489. The cost difference was -0.0122 US dollars per puzzle (-0.0195 to -0.00470).

At $k_c = 5$ the accuracy loss was larger: chains -10.0 (-20.0 to -2.5); pieces -12.5 (-22.5 to -2.5). A cascade on Haiku-max samples was not simulated.

Table S19: The escalate-to-Opus cascade, an offline simulation (pre-specified exploratory‡; Study 1, main run). Each threshold was chosen on the odd-numbered puzzles and scored on the 40 even-numbered ones; bootstrap 95% intervals over puzzles; list price per puzzle. On pieces, Opus-medium was right on every even-numbered puzzle, so its interval is the exact two-sided Clopper–Pearson interval from free analysis 4 (post hoc†).

Shape	k_c	τ	Rule for τ	Escalation rate (%)	Cascade accuracy (%)	Cascade accuracy 95% interval	Opus-medium accuracy (%)	Opus-medium accuracy 95% interval	d: cascade minus Opus (points)	d 95% interval
1×6	5	0.7	"none within 10 points: the highest odd-base accuracy"	62.50	82.50	[70.00, 92.50]	92.50	[82.50, 100.00]	−10.00	[−20.00, −2.50]
1×6	10	0.8	"cheapest tau within 10 points of Opus-medium on the odd bases"	70.00	90.00	[80.00, 97.50]	as above	as above	−2.50	[−7.50, 0.00]
1×6	20	0.7	"cheapest tau within 10 points of Opus-medium on the odd bases"	67.50	90.00	[80.00, 97.50]	as above	as above	−2.50	[−7.50, 0.00]
6×1	5	0.7	"cheapest tau within 10 points of Opus-medium on the odd bases"	47.50	87.50	[77.50, 97.50]	100.00	[91.19, 100.00] post hoc† (exact)	−12.50	[−22.50, −2.50]
6×1	10	0.7	"cheapest tau within 10 points of Opus-medium on the odd bases"	52.50	92.50	[85.00, 100.00]	as above	as above	−7.50	[−15.00, 0.00]
6×1	20	0.6	"cheapest tau within 10 points of Opus-medium on the odd bases"	45.00	90.00	[80.00, 97.50]	as above	as above	−10.00	[−20.00, −2.50]

Table S19: (continued)

Shape	k_c	Cascade cost per puzzle ($)	Cascade cost 95% interval ($)	Opus-medium cost per puzzle ($)	Opus-medium cost 95% interval ($)	Cost difference ($ per puzzle)	Cost difference, lower end of the 95% interval ($)	Cost difference, upper end of the 95% interval ($)
1×6	5	0.0323	[0.0260, 0.0389]	0.0416	[0.0397, 0.0436]	−0.00937	−0.0155	−0.00327
1×6	10	0.0408	[0.0340, 0.0471]	as above	as above	−0.000822	−0.00719	0.00490
1×6	20	0.0509	[0.0440, 0.0578]	as above	as above	+0.00930	0.00287	0.0151
6×1	5	0.0290	[0.0213, 0.0369]	0.0489	[0.0478, 0.0499]	−0.0199	−0.0272	−0.0124
6×1	10	0.0366	[0.0293, 0.0447]	as above	as above	−0.0122	−0.0195	−0.00470
6×1	20	0.0441	[0.0370, 0.0518]	as above	as above	−0.00472	−0.0117	0.00266

Offline simulation: the cascade takes a vote of k_c Haiku-medium samples and escalates to Opus-medium when the top answer's share is below τ. τ is chosen on the odd-numbered puzzles; the cascade and Opus-medium are compared on the even-numbered puzzles. Every column except τ and its rule is on the even-numbered puzzles. The file's own label, in its words: "escalate-to-Opus cascade (offline, exploratory)"; its units: "points; USD, list price". The Rule for τ column is the file's own text, in quotation marks (its "bases" are puzzles): where no τ came within the margin on the odd-numbered puzzles, the highest accuracy on them was taken. Opus-medium's accuracy on the odd-numbered puzzles is 100.00% in every row. No test and no verdict: d is a point difference with an unadjusted 95% interval. Opus-medium's accuracy and cost on the even-numbered puzzles are the same in every k_c row of a shape, so they are printed once per shape, with the interval from that shape's first row; each row's own bootstrap gives slightly different intervals for them, which are not printed again ('as above'). At 6×1 Opus-medium was right on every even-numbered puzzle, so its bootstrap interval collapses to one point; the exact two-sided 95% (Clopper–Pearson) interval over the even-numbered puzzles from free analysis 4 is printed in its place, labelled post hoc† (exact). No format-compliance figure is stamped for the cascade; the format compliance of Haiku-medium's and Opus-medium's samples over all puzzles is in Table S14.

Table S20: Real time against Batch, and latency (pre-specified descriptive; the Pilot 1 panel is exploratory‡, at 24 steps on its own puzzles, scored under the original marking rule). Study 1's real-time subset: 50 Haiku-medium calls and 10 calls each for Opus-medium and Sonnet-medium; counts, medians and quartiles, with no intervals.

Setting	Calls	Accuracy, real time (%)	Accuracy, Batch (%)	Difference, real time minus Batch (points)	Mean output tokens per call, real time	Mean output tokens per call, Batch
Haiku-medium, real time	50	54.00	58.00	−4.00	1,791.3	1,812.7
Opus-medium, real time	10	90.00	100.00	−10.00	1,797.4	1,932.9
Sonnet-medium, real time	10	70.00	90.00	−20.00	1,280.4	1,285.3

Setting	Latency median (s)	Latency quartiles (s)	Latency p95 (s)	Latency max (s)
Haiku-medium, real time	6.91	[6.51, 7.34]	8.00	8.39
Opus-medium, real time	16.20	[13.78, 16.89]	17.84	18.16
Sonnet-medium, real time	7.77	[7.13, 8.30]	8.63	8.74

Panel 1: Pilot 1 real time next to Batch (exploratory)

Model	Mode	24×1 whole (%)	24×1 per piece (%)	1×24 whole (%)	1×24 final (%)	1×24 lines = key (%)	1×24 local (%)	1×24 format ok (%)
Haiku-medium	"real time (25+25 tries)"	0.0	79.2	0.0	0.0	17.7	75.1	92.0
Haiku-medium	"Batch (50+50 tries)"	0.0	81.1	0.0	0.0	10.0	76.5	71.4
Opus-medium	"real time (5+5 tries)"	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Opus-medium	"Batch (5+5 tries)"	100.0	100.0	60.0	80.0	80.2	97.9	80.0
Sonnet-medium	"real time (5+5 tries)"	20.0	96.7	40.0	40.0	52.5	93.3	100.0
Sonnet-medium	"Batch (5+5 tries)"	40.0	95.0	20.0	20.0	26.7	95.8	100.0

Vote times are computed from single calls, not measured: all at once is the slowest of a puzzle's real-time calls, one after another is their sum; the figures are medians over puzzles. Only Haiku-medium has repeated real-time calls, so only it has vote times: a vote of its real-time calls takes a median of 7.30 s all at once and 34.51 s one after another. The real-time subset is a small check next to Batch; accuracy differences are of a few puzzles and carry no test. Format compliance of Study 1's real-time calls is 100.00% for all three settings: the format-ok share of their answered samples, both shapes together (the per-shape counts are in Table S15's Study 1 format panel). No format figure is stamped for the Batch samples on this subset; the settings' full Batch figures are in Table S14. Latency is the time of answered calls; quartiles are the first and third quartile; p95 is the time that 95% of the calls did not exceed, and max is the slowest call's. Pilot 1 is exploratory (24 steps, a different puzzle set). The cells are for the 5 Pilot 1 puzzles of its real-time check, each as a chain and as pieces (10 in all). As the ledger words it: "Real time: Haiku ×5, Opus ×1, Sonnet ×1 per puzzle; Batch: the same puzzles' Batch tries (Haiku ×10, Opus ×1, Sonnet ×1)." The measures, as Pilot 1's ledger defines them. 24×1 is the puzzle as separate pieces and 1×24 as one chain. Whole: the whole-answer accuracy under the original marking rule. Per piece (24×1): the share of pieces right. Final (1×24): the accuracy on the answer line alone (at 24×1 it equals the whole answer). Lines = key (1×24): the share of step lines equal to the key, over samples whose lines can be placed. Local (1×24): the per-step accuracy with each step judged from the model's own previous line, over samples whose lines can be placed. Format ok (1×24): the share of samples in the required answer format. Pilot 1's results file gives percentages to one decimal; the cells are copied from it without the percent sign. Calls per cell are the samples named in the Mode column, which keeps the ledger's own wording in quotation marks (its "tries" are samples). Marking rule: Pilot 1's results file was scored with the scorers that were official at the time, under the original marking rule, before Deviation 2. The paper's other tables use Deviation 2's rule; no re-score of these cells under it is stamped.

Real time against Batch (Table S20; pre-specified descriptive, no test). All headline accuracies were measured in Batch. In the 10-puzzle real-time subset, accuracy was lower in real time than in Batch for all three models:

- **Haiku-medium:** 54.0% against 58.0%, over 50 calls.
- **Opus-medium:** 90.0% against 100.0%, over 10 calls (a gap of one answer).
- **Sonnet-medium:** 70.0% against 90.0%, over 10 calls (a gap of two answers).

Output tokens differed by 1.18% (Haiku), 0.38% (Sonnet) and 7.01% (Opus; 7.54% against the real-time setting). Pilot 1's real-time check (exploratory; 24 steps, 5 puzzles in both shapes, original marking rule) pointed in mixed directions:

- **Opus-medium chains:** 100% in real time against 60% in Batch (5 samples each; pieces 100% in both).
- **Sonnet-medium:** pieces 20% against 40%, chains 40% against 20%.
- **Haiku-medium:** 0.0% whole answer in both (79.2% against 81.1% per piece).

The subsets are far too small to tell whether there is a real difference in either direction.

The low-information flag did not fire. Haiku-medium scored 51.0% on pieces, and the flag fires if its single-sample score on pieces is above 60%. Haiku-medium's per-puzzle single-sample scores spread to both ends: on chains, 16 of 80 puzzles were below 10% and 11 at 90% or more. The median over puzzles was 40.0% on chains and 50.0% on pieces.

Repeat-share definition. The pairwise repeat share used in the main text is the share of pairs of wrong samples on the same puzzle that give the same wrong answer. A second figure in the scoring code, the share of a puzzle's wrong samples that give its most common wrong answer, averaged over puzzles with at least 2 wrong samples, is 0.499 on chains (74 puzzles) and 0.592 on pieces (75). It is descriptive, defined in the scoring code, and not a share of all votes.

S5 Study 1b, further results

S5.1 Further detail on Study 1b

These paragraphs give details that Study 1b's results in Section 4 of the paper leave out.

Study 1b's sets and lock. Study 1b's puzzle sets were Study 1's unused candidate and reserve sets, frozen before any Haiku-max data existed. The record that they were never sent to a model under test covers 3 sets with 200 puzzles (400 prompts, one per shape). Study 1b ran under Lock 4, with its vote size taken from cost-only records.

That B1 did not read "no worse by more than 5 points" was expected for a gap of this size. Study 1b's locked power table said in advance that near Opus-medium's ceiling a 5-point margin can show "no worse" only if Haiku-max's true accuracy is close to Opus-medium's: with Opus-medium at 98%, B1's chance of reading "no worse by more than 5 points" was 0.89–0.99 if Haiku-max were at 98%, 0.23–0.31 at 95% and 0.00–0.01 at 92%. Haiku-max scored 92.63%, so the error-rate ratio beside the test is what carries the size of the gap.

Wrong answers, not wrong formats. At 16 steps, all 320 of Opus-medium's replies were well formatted, so its 7 errors (5 on chains, 2 on pieces) were wrong balances. Only 5 of Haiku-max's 118 wrong samples were format failures, so B1's error-rate ratio of 3.37 is almost entirely wrong balances (post hoc arithmetic). That is unlike Study 1's chains, where 2 of Opus-medium's 3 misses were extra lines (our reading, not a test). Haiku-max kept the format in 796 of 800 chain replies and 799 of 800 piece replies (pre-specified descriptive).

How the Haiku-max vote grows. Figure S3 shows the pooled Haiku-max vote at 16 steps for every vote size (pre-specified descriptive; only the 6-sample vote is tested, as B2): 92.63% with 1 sample ($0.0112 per answer), 92.87% with 2, 97.03% with 3 (95% interval 95.32 to 98.40; 0.447 of Opus-medium's price), 98.30% with 5 (0.745), 98.44% with 6 (0.894) and 99.38% with 10. Opus-medium scored 97.81% (95% interval 96.25

to 99.06) at $0.0750. The 5- and 6-sample votes scored above Opus-medium on point estimates, by 0.49–0.63 points, inside sampling error. The post hoc intervals for the other sizes (Section S5.4) are not tests, so the 3-sample vote at 0.447 of the price is a descriptive point, not a saving.

B2 with ties counted wrong (free analysis 13, post hoc). Under the locked rule, a vote whose top count is shared by m answers, the right one among them, scores $1/m$. A tie at the top decided 1.32% of B2's 6-sample votes, pooled over both shapes, and 3–3 ties alone 1.15%; 11 of B2's 80 puzzles had at least one vote decided by a tie. Counting a vote decided by a tie as wrong, B2's difference from Opus-medium becomes $d = -0.03$ points (95% interval -2.11 to $+1.90$), against the locked $+0.63$. We report it without a p-value or a reading; B2's pre-registered reading is unchanged.

B2 in the worst case. In a post hoc check from the per-sample marks alone, which fix the vote on 148 of the 160 puzzle-shapes, even if every wrong answer to a puzzle had been the same answer (the worst case for a vote), the 6-sample vote would score 97.78% and d would be -0.04 points (95% interval -2.32 to $+1.89$); scoring one random 6-sample subset per puzzle and shape instead, still in the worst case, as a vote of 6 separately drawn samples, kept d between -1.25 and $+0.94$ points in 95% of 10,000 draws, and no draw fell below -2.50, so none reached the 5-point margin.

S5.2 Without the partly shown puzzles

Parts of the answer keys of three test puzzles were displayed in our own Claude Code work sessions, never to a model under test: one in the 16-step set and two in the 6-step set. The pre-registered sensitivity analysis re-ran the whole family of Tests B1–B3 without them, on 79, 79 and 38 puzzles, with the same seeds, margins and Holm's correction (Table S21's second panel). Every test kept its reading in all three views:

- **B1:** $d = -5.19$ points (95% interval -7.85 to -2.66). We could not show a single Haiku-max sample within 5 points of Opus-medium. At the secondary 10-point margin it was shown no worse than Opus-medium by more than 10 points (Holm-adjusted; $p = 0.00014$, below its Holm threshold of 0.0125). Its error rate was 3.34 times Opus-medium's (95% interval 1.76 to 9.80).
- **B2:** $d = +0.64$ points (-1.20 to $+2.50$). It was shown no worse than Opus-medium by more than 5 points (Holm-adjusted; $p \leq 0.00009$, no resample reached the margin; below its Holm threshold of 0.00833). Its error-rate ratio was 0.71 (0.24 to 2.13).
- **B3:** $d = -3.82$ points (-7.50 to -1.45). We could not show it within 5 points. At the secondary margin it was shown no worse by more than 10 points (Holm-adjusted; $p = 0.0016$, below its Holm threshold of 0.025). Its error rate was 6.8 times Opus-medium's (1.67 to unbounded).

The descriptive outputs were not re-run without them; the check never replaces the primary readings.

Table S21: At 16 steps a small Haiku-max vote (its accuracy estimated from 10 samples) was shown no worse than one Opus-medium answer by more than 5 points (Holm-adjusted), while a single Haiku-max sample could not be shown to be within 5 points at either length. Study 1b's Tests B1–B3 (confirmatory, with Holm's correction within Study 1b only): B1 and B2 at 16 steps on 80 puzzles, B3 at 6 steps on 40. The challenger is one Haiku-max sample in B1 and B3 and the 6-sample vote in B2; d is its accuracy minus Opus-medium's, in points, and leaving out refusals or truncations changes no test. Readings at the primary 5-point margin and, quoted only as the secondary, the 10-point margin. d in points with 95% intervals from a paired BCa bootstrap (10,000 resamples); error-rate ratios with 95% intervals from a paired bootstrap over the same puzzles (separate draws; pre-specified descriptive, not tested); accuracy and format compliance for each side. It also gives list-price costs per answer, each test's k and a sensitivity check without the puzzles exposed before the run.

Test	Challenger	Steps	Puzzles	Challenger accuracy (%)	Opus-medium accuracy (%)	Format compliance, challenger, chains / pieces (%)
Test B1	One Haiku-max sample	16	80	92.63	97.81	99.50 / 99.88
Test B2	Haiku-max vote	16	80	98.44	97.81	99.50 / 99.88
Test B3	One Haiku-max sample	6	40	95.75	99.38	100.00 / 100.00

Test	d (points)	d 95% interval	5-point margin: p	5-point margin: Holm threshold	5-point margin: reading	5-point margin: shown no worse (Holm-adjusted)
Test B1	−5.19	[−7.75, −2.69]	0.56595	0.025	"inconclusive (could not show)"	no
Test B2	+0.63	[−1.15, 2.44]	≤ 0.00011	0.00833	"holds (no worse in all three views)"	yes
Test B3	−3.63	[−7.00, −1.50]	0.20571	0.0125	"inconclusive (could not show)"	no

Test	10-point margin: p	10-point margin: Holm threshold	Error-rate ratio R	Error-rate ratio 95% interval
Test B1	0.00027	0.0125	3.37	[1.78, 10.07]
Test B2	≤ 0.00011	0.00833	0.71	[0.24, 2.11]
Test B3	≤ 0.00108	0.025	6.80	[1.67, unbounded]

Panel 1: Costs beside the tests, list price

Test	k (measured)	Wording	Challenger list cost per answer ($)	Opus-medium list cost per answer ($)	Spend share or cost ratio (challenger / Opus-medium)
Test B1	6	"about 1/6 of the cost per answer (k measured)"	0.0112	0.0750	0.149
Test B2	6	"the vote of $k = 6$ Haiku-max tries, about Opus-medium's price"	0.0670	0.0750	0.894
Test B3	10	"about 1/10 of the cost per answer (k measured)"	0.00419	0.0441	0.095

Panel 1: (continued)

Test	Challenger list cost, chain ($)	Challenger list cost, pieces ($)	Opus-medium list cost, chain ($)	Opus-medium list cost, pieces ($)
Test B1	0.0123	0.0101	0.0674	0.0825
Test B2	0.0736	0.0605	0.0674	0.0825
Test B3	0.00433	0.00405	0.0406	0.0476

Table S21: (continued)

Panel 2: Sensitivity: exposed puzzles left out

Test	Puzzles (exposed ones left out)	d (points)	d 95% interval	5-point margin: p	5-point margin: reading	Error-rate ratio R	Error-rate ratio 95% interval
Test B1	79	−5.19	[−7.85, −2.66]	0.56397	"inconclusive (could not show)"	3.34	[1.76, 9.80]
Test B2	79	+0.64	[−1.20, 2.50]	$\leq$ 0.00009	"holds (no worse in all three views)"	0.71	[0.24, 2.13]
Test B3	38	−3.82	[−7.50, −1.45]	0.25586	"inconclusive (could not show)"	6.80	[1.67, unbounded]

Readings in quotation marks are the Study 1b file's own template words: "holds" only if the test is no worse in all three views; an inconclusive test reads "inconclusive (could not show)" and is never called worse. The ruled reading, shown no worse (Holm-adjusted), is the column "5-point margin: shown no worse (Holm-adjusted)" of this table. Panel 2 has no such column: at the 5-point margin Test B2 is shown no worse (Holm-adjusted) and Tests B1 and B3 are not. Holm's correction runs within Study 1b only; Study 1 and Study 1b are never pooled. Shown no worse (Holm-adjusted) is yes where p is below the test's Holm threshold at that margin, in Holm's step-down order: a test is shown no worse (Holm-adjusted) only if every test before it in Holm's order is too. d is the same at both margins: Haiku-max (or its vote) minus Opus-medium, in points, paired over puzzles. A positive d means Haiku-max scored higher. Error-rate ratio R is the challenger's error rate divided by Opus-medium's, over the same puzzles (an error rate is the share of answers not right, the complement of the accuracy); R above one means the challenger made more errors. R is pre-specified descriptive, reported beside the test: Study 1b's specification, frozen with Lock 4, reports it beside every test, and it is not part of any test's reading. Its 95% interval is a bootstrap over puzzles; an unbounded upper end means Opus-medium made no error in too many resamples for the interval to have a finite top. The B2 challenger is the vote of k Haiku-max samples, $k = 6$ (cost-matched with Opus-medium, by the pre-registered cost-matched k rule). For Test B2 the challenger's format compliance is that of the single Haiku-max samples the vote is taken over. Costs: list price per answer; the vote at k costs k answers. Each test's cost wording, k and per-shape costs (B2's included) are in Panel 1 (costs). Its ratio column holds Test B2's spend share (the vote's cost over its cost-matched comparator's) and the cost ratios of Tests B1 and B3. Panel 2 (sensitivity) leaves out the puzzles that were exposed before the run (the file lists them under excluded); it is a sensitivity, not a test. Panel 1's wordings are the Study 1b file's own, in quotation marks; its "tries" are the paper's samples. Columns that print one value in every row are left out. In every test the comparator is Opus-medium, whose format compliance is 100.00% / 100.00% (chains / pieces); at the 10-point margin every test reads "holds (no worse in all three views)" and is shown no worse (Holm-adjusted). Panel 2's 10-point readings are the same. $\leq$ marks a p at the smallest value the bootstrap can return (no resample reached the margin).

S5.3 Each test by shape (Table S22)

The tests pool the two shapes by design, so no per-shape result is a test; Table S22 gives each test by shape (descriptive).

B3 ran on 40 puzzles. On B3's pieces, Opus-medium's bootstrap interval collapses to zero width, so the exact Clopper–Pearson bound replaces it. The shape pattern is not stable across lengths and puzzle sets.

S5.4 The Haiku-max vote at every size (Table S23, post hoc)

For every vote size, the pooled Haiku-max vote's paired difference from Opus-medium at 16 steps, with its 95% interval and its cost as a share of Opus-medium's price (post hoc; not tests):

- 2 samples: −4.94 points (−7.44 to −2.38; 0.298);
- 3 samples: −0.79 (−2.83 to +1.21; 0.447);
- 4 samples: −0.36 (−2.24 to +1.59; 0.596);
- 5 samples: +0.49 (−1.37 to +2.31; 0.745);
- 6 samples, the B2 vote: +0.63 (−1.15 to +2.44; 0.894; B2, pre-registered);
- 7 samples: +0.99 (−0.89 to +2.81; 1.043);
- 8 samples: +1.18 (−0.76 to +2.95; 1.192);
- 9 samples: +1.41 (−0.78 to +3.28; 1.341);
- 10 samples: +1.56 (−0.63 to +3.44; 1.490).

At each shape's own cost-matched size, 5 samples on chains and 8 on pieces (0.909 and 0.978 of Opus-medium's price), the differences were +1.02 points (−2.22 to +4.76) and +0.56 (−0.90 to +3.26). This is a sensitivity analysis: no size is singled out as a saving, and it never replaces B2.

Table S22: Study 1b's tests by shape (B1 and B2 at 16 steps on 80 puzzles per shape, B3 at 6 steps on 40). The per-shape blocks are a descriptive breakdown of the confirmatory tests, with a cluster bootstrap over puzzles (2,000 resamples) for the accuracies and a paired bootstrap for the per-puzzle differences. Its first panel gives each pooled test in the three views (as deployed, refusals out, truncations out) with its reading at each margin, and one post hoc† row for Test B2 with ties counted wrong (free analysis 13), which has no p-value and no reading; the first panel's intervals come from paired BCa bootstraps with 10,000 resamples: the tests' own, and free analysis 13's for the ties row.

Test	Shape	Challenger	Challenger accuracy (%)	Challenger 95% interval	Opus-medium accuracy (%)	Opus-medium 95% interval	d (points)	d 95% interval	Format compliance, challenger (%)
Test B1	chains (1×16)	One Haiku-max sample	92.25	[89.13, 94.88]	96.88	[93.75, 99.38]	−4.63	[−8.75, −0.38]	99.50
	pieces (16×1)	One Haiku-max sample	93.00	[90.38, 95.38]	98.75	[96.88, 100.00]	−5.75	[−8.63, −2.63]	99.88
Test B2	chains (1×16)	Haiku-max vote	98.04	[95.65, 99.60]	96.88	[93.75, 99.38]	+1.16	[−2.22, 4.72]	99.50
	pieces (16×1)	Haiku-max vote	98.85	[97.54, 99.90]	98.75	[96.88, 100.00]	+0.10	[−1.80, 2.23]	99.88
Test B3	chains (1×6)	One Haiku-max sample	97.25	[94.75, 99.25]	98.75	[96.25, 100.00]	−1.50	[−4.75, 1.50]	100.00
	pieces (6×1)	One Haiku-max sample	94.25	[90.75, 97.25]	100.00	[91.19, 100.00] (exact, post hoc; 40 puzzles)	−5.75	[−9.25, −2.75]	100.00

Table S22: (continued)

Panel 1: Pooled tests in the three views

Test	View	Puzzles used	d (points)	d 95% interval	5-point margin: p	5-point margin: reading in this view	10-point margin: p	10-point margin: reading in this view
Test B1	as deployed	80	−5.19	[−7.75, −2.69]	0.56595	inconclusive at 5	0.00027	shown no worse by > 10 (Holm-adjusted)
	refusals out	80	−5.19	[−7.75, −2.69]	0.56595	inconclusive at 5	0.00027	shown no worse by > 10 (Holm-adjusted)
	truncations out	80	−5.19	[−7.75, −2.69]	0.56595	inconclusive at 5	0.00027	shown no worse by > 10 (Holm-adjusted)
Test B2	as deployed	80	+0.63	[−1.15, 2.44]	≤ 0.00011	shown no worse by > 5 (Holm-adjusted)	≤ 0.00011	shown no worse by > 10 (Holm-adjusted)
	refusals out	80	+0.63	[−1.15, 2.44]	≤ 0.00011	shown no worse by > 5 (Holm-adjusted)	≤ 0.00011	shown no worse by > 10 (Holm-adjusted)
	truncations out	80	+0.63	[−1.15, 2.44]	≤ 0.00011	shown no worse by > 5 (Holm-adjusted)	≤ 0.00011	shown no worse by > 10 (Holm-adjusted)
	ties counted wrong (post hoc)	80	−0.03	[−2.11, 1.90]	—	—	—	—
Test B3	as deployed	40	−3.63	[−7.00, −1.50]	0.20571	inconclusive at 5	≤ 0.00108	shown no worse by > 10 (Holm-adjusted)
	refusals out	40	−3.63	[−7.00, −1.50]	0.20571	inconclusive at 5	≤ 0.00108	shown no worse by > 10 (Holm-adjusted)
	truncations out	40	−3.63	[−7.00, −1.50]	0.20571	inconclusive at 5	≤ 0.00108	shown no worse by > 10 (Holm-adjusted)

Per-shape numbers are descriptive: the tests are the pooled ones in Table S21. A chain is one puzzle whose answer carries all the steps; pieces are separate one-step puzzles. The B2 challenger is the vote of k Haiku-max samples, $k = 6$; B1 and B3 use one Haiku-max sample. Intervals are cluster bootstrap over puzzles (accuracies) and paired bootstrap (d). For Test B2 the challenger's format compliance is that of the single Haiku-max samples the vote is taken over. The per-test intervals here and Table S25's intervals for the same setting and shape (for the B2 vote, Figure S3's) come from separate bootstrap draws over the same puzzles, with the same points, and differ only by resampling, so their ends can differ slightly; where both collapsed to one value, both show the same exact interval. "(exact, post hoc; ... puzzles)" marks a bootstrap interval that collapsed to one value (every puzzle right, or every one wrong), replaced by the exact two-sided 95% Clopper–Pearson interval from free analysis 4 (post hoc; it changes no test). The count in the mark is the number of puzzles, not of samples (the conservative unit). Panel 1 gives the pooled tests in the three views; its readings are per view, and a test holds only if it is no worse in all three views (Table S21). The three views agree to every printed digit because no refusal or truncation touched either side of a test. Panel 1's ties-counted-wrong row for Test B2 is post hoc (free analysis 13): the vote is scored as deployed, except that a vote whose top answers tie, the right one among them, scores nothing instead of an equal share; d and its paired bootstrap interval only, with no p and no reading, and B2's reading is unchanged. A tie that included the right answer decided 1.32% of B2's 6-sample votes (a mean over puzzles, each over all of its possible votes; three-against-three ties alone 1.15%); such a tie decided at least one vote in 11 of the 80 puzzles (in either shape). Table S13 gives the tie-decided share for Tests 1–3. Columns that print one value in every row are left out: Opus-medium's format compliance is 100.00% for every test and shape, and Panel 1's number of puzzles left out is 0 in every row. ≤ marks a p at the smallest value the bootstrap can return (no resample reached the margin).

Table S23: Test B2's Haiku-max vote at every size above a single sample (the single sample is Test B1's challenger), 16 steps (Study 1b, 80 puzzles; 10 designed Haiku-max samples and 2 Opus-medium samples per puzzle). The vote rows' d and cost ratio are a post hoc† sensitivity analysis, with no p-value, no Holm correction and no reading against the margin; the Test B2 row's pooled d and interval are the confirmatory result, unchanged. d from paired BCa bootstraps (10,000 resamples); vote accuracies from a cluster bootstrap over puzzles; list price per answer, a vote of k samples costing k answers. Its first panel gives each shape's own cost-matched size with its paired difference and interval from the same post hoc paired bootstrap, and B2's per-shape cost.

Setting	Samples voted (k)	Accuracy, pooled (%)	Accuracy 95% interval	Format compliance of the samples, 1×16 / 16×1 (%)	List cost per answer, pooled ($)
One Opus-medium answer (the comparator)	—	97.81	[96.25, 99.06]	100.00 / 100.00	0.0750
Haiku-max vote	2	92.87	[90.62, 94.81]	99.50 / 99.88	0.0223
Haiku-max vote	3	97.03	[95.32, 98.40]	99.50 / 99.88	0.0335
Haiku-max vote	4	97.45	[95.95, 98.65]	99.50 / 99.88	0.0447
Haiku-max vote	5	98.30	[96.93, 99.36]	99.50 / 99.88	0.0559
Haiku-max vote: Test B2	6	98.44	[97.08, 99.46]	99.50 / 99.88	0.0670
Haiku-max vote	7	98.80	[97.40, 99.74]	99.50 / 99.88	0.0782
Haiku-max vote	8	98.99	[97.50, 99.90]	99.50 / 99.88	0.0894
Haiku-max vote	9	99.22	[97.66, 100.00]	99.50 / 99.88	0.101
Haiku-max vote	10	99.38	[98.13, 100.00]	99.50 / 99.88	0.112

Setting	Samples voted (k)	Spend share or cost ratio (vote / Opus-medium), pooled	d pooled: vote minus Opus-medium (points)	d pooled 95% interval (paired)	Error-rate ratio R (vote / Opus-medium) [95% interval]
One Opus-medium answer (the comparator)	—	—	—	—	—
Haiku-max vote	2	0.298	−4.94	[−7.44, −2.38]	—
Haiku-max vote	3	0.447	−0.79	[−2.83, 1.21]	—
Haiku-max vote	4	0.596	−0.36	[−2.24, 1.59]	—
Haiku-max vote	5	0.745	+0.49	[−1.37, 2.31]	—
Haiku-max vote: Test B2	6	0.894	+0.63	[−1.15, 2.44]	0.71 [0.24, 2.11]
Haiku-max vote	7	1.043	+0.99	[−0.89, 2.81]	—
Haiku-max vote	8	1.192	+1.18	[−0.76, 2.95]	—
Haiku-max vote	9	1.341	+1.41	[−0.78, 3.28]	—
Haiku-max vote	10	1.490	+1.56	[−0.63, 3.44]	—

Table S23: (continued)

Setting	Samples voted (k)	Label
One Opus-medium answer (the comparator)	—	pre-specified descriptive (comparator of Tests B1 and B2)
Haiku-max vote	2	post hoc descriptive sensitivity: d and cost ratio (free analysis 3); accuracy, format compliance and cost pre-specified descriptive (vote curve)
Haiku-max vote	3	post hoc descriptive sensitivity: d and cost ratio (free analysis 3); accuracy, format compliance and cost pre-specified descriptive (vote curve)
Haiku-max vote	4	post hoc descriptive sensitivity: d and cost ratio (free analysis 3); accuracy, format compliance and cost pre-specified descriptive (vote curve)
Haiku-max vote	5	post hoc descriptive sensitivity: d and cost ratio (free analysis 3); accuracy, format compliance and cost pre-specified descriptive (vote curve)
Haiku-max vote: Test B2	6	confirmatory: the pooled d and its interval (Test B2, 5-point margin), stamped values unchanged. Error-rate ratio: pre-specified descriptive, reported beside the test, not part of it. Accuracy, format compliance and costs: pre-specified descriptive
Haiku-max vote	7	post hoc descriptive sensitivity: d and cost ratio (free analysis 3); accuracy, format compliance and cost pre-specified descriptive (vote curve)
Haiku-max vote	8	post hoc descriptive sensitivity: d and cost ratio (free analysis 3); accuracy, format compliance and cost pre-specified descriptive (vote curve)
Haiku-max vote	9	post hoc descriptive sensitivity: d and cost ratio (free analysis 3); accuracy, format compliance and cost pre-specified descriptive (vote curve)
Haiku-max vote	10	post hoc descriptive sensitivity: d and cost ratio (free analysis 3); accuracy, format compliance and cost pre-specified descriptive (vote curve)

Panel 1: Each shape's own cost-matched k, and B2's vote's per-shape cost

Shape	Own cost-matched k	Samples per comparator answer (Opus-medium cost per answer / one Haiku-max sample)	One Haiku-max sample: list cost per answer ($)	Vote at own k: list cost per answer ($)	Opus-medium: list cost per answer ($)
1×16	5	5.50	0.0123	0.0613	0.0674
16×1	8	8.18	0.0101	0.0807	0.0825

Table S23: (continued)

Panel 1: (continued)

Shape	d: vote at own k minus Opus-medium (points)	d 95% interval (paired)	B2's vote: list cost per answer ($)	B2's vote / Opus-medium spend share (derived)
1×16	+1.02	[−2.22, 4.76]	0.0736	1.091
16×1	+0.56	[−0.90, 3.26]	0.0605	0.733

d is the vote of k Haiku-max samples minus one Opus-medium answer, in points, paired over the same puzzles; a positive d means the vote scored higher. The Test B2 row is the pre-registered test at B2's own $k = 6$ (Study 1b's rule for a cost-matched vote). Its pooled d and interval are Test B2's stamped confirmatory values, unchanged: $p \leq 0.00011$, below its Holm threshold of 0.00833; reading: shown no worse by $>$ margin (Holm-adjusted), in all three views. Its accuracy, format compliance and costs are pre-specified descriptive, not part of the test. One comparison, one interval: free analysis 3's own interval for this vote is not printed. Error-rate ratio R: the vote's errors over Opus-medium's on the same puzzles. Near the ceiling it sits beside d; on the Test B2 row it is Study 1b's stamped value for Test B2, pre-specified descriptive, reported beside the test and not part of it. The post hoc rows have none: free analysis 3 stamps no error-rate ratio. The other vote rows are post hoc descriptive sensitivity: Study 1b's spec set no sensitivity analysis over k, so their d and cost ratio carry no p-value, no Holm correction and no reading against the margin, and they leave B2 unchanged. Their accuracy, format compliance and cost are Study 1b's pre-specified vote curve. One Haiku-max sample ($k = 1$) is Test B1's challenger and is reported with the Study 1b tests. Cost: US dollars per answer at list price (item 9b); a vote of k samples costs k times one sample's list cost per answer. Every value in the spend share or cost ratio column is the vote's pooled list cost per answer over Opus-medium's: a cost ratio on the post hoc rows and, on the Test B2 row, B2's spend share, as Table S12 prints it. The samples per comparator answer that set a cost-matched k (Opus-medium's cost per answer over one Haiku-max sample's, as in Panel 1) are printed to two decimals; a vote's cost over Opus-medium's, as reported for Test B2 with the Study 1b tests, to three. Format compliance: the Haiku-max samples the votes are taken over, and Opus-medium's answers on its row. Panel 1: per-shape intervals are printed only here (one comparison, one interval): free analysis 3 also stamps per-shape intervals at each k, drawn with other seeds, and they are not printed. $\leq$ marks a p at the smallest value the bootstrap can return (no resample reached the margin). Label of every row: post hoc (free analysis 3) for own k, samples per comparator answer, vote cost and d; pre-specified descriptive for the one-sample costs and B2's vote cost; derived for B2's spend share. Post hoc (free analysis 3): each shape's own cost-matched k is the floor of Opus-medium's list cost per answer over one Haiku-max sample's on that shape, held to the designed samples; the vote's cost at that k and its paired d against Opus-medium are free analysis 3's. Pre-specified descriptive: one Haiku-max sample's and Opus-medium's list cost per answer on each shape (Study 1b's settings and costs beside Test B2), and B2's vote's cost per shape (costs beside Test B2). Derived: B2's vote / Opus-medium spend share per shape, the quotient of those two stamped costs.

S6 Effort and length, further results

S6.1 Deviation 2 and the length curve

Deviation 2 also reshapes Sonnet-medium's length curve (pre-registered descriptive; both marking rules' files are kept; Figure S11 plots the curves under the original rule). Its pooled whole-answer accuracy at 4, 6, 8, 12 and 16 steps was 87.5, 80.0, 75.0, 72.5 and 65.0% under Deviation 2's rule, and 75.0, 72.5, 70.0, 72.5 and 65.0% under the original rule (Table S24).

In the length sweep the rule moves only Sonnet-medium's chains, and only at 4, 6 and 8 steps:

- **4 steps:** chains from 55.0% to 80.0%; chain format compliance from 60.0% to 100.0%.
- **6 steps:** chains from 70.0% to 85.0%; compliance from 75.0% to 95.0%.
- **8 steps:** chains from 75.0% to 85.0%; compliance from 90.0% to 100.0%.

Only Sonnet-medium's chain whole-answer accuracy moved. Haiku-medium's and Opus-medium's whole-answer accuracy is unchanged in the length sweep; Pilot 1's Opus-medium marker falls from 94.0% to 92.0% under the original rule. Haiku-medium's chain format compliance moves by 0.5 points at 8, 12 and 16 steps. Its 16-step chain format flag fires under both rules (non-compliance 6.00% under the original rule and 5.50% under Deviation 2's).

S6.2 The shortest puzzles

In the length sweep, Haiku-medium did not fall steadily from 4 to 8 steps (55.50%, 48.00%, 52.25%). The 4-step chains were the hardest per step (79.58%). Both may reflect differences between the puzzle sets rather than length, since each length had 20 puzzles of its own (post hoc; our reading, not a test).

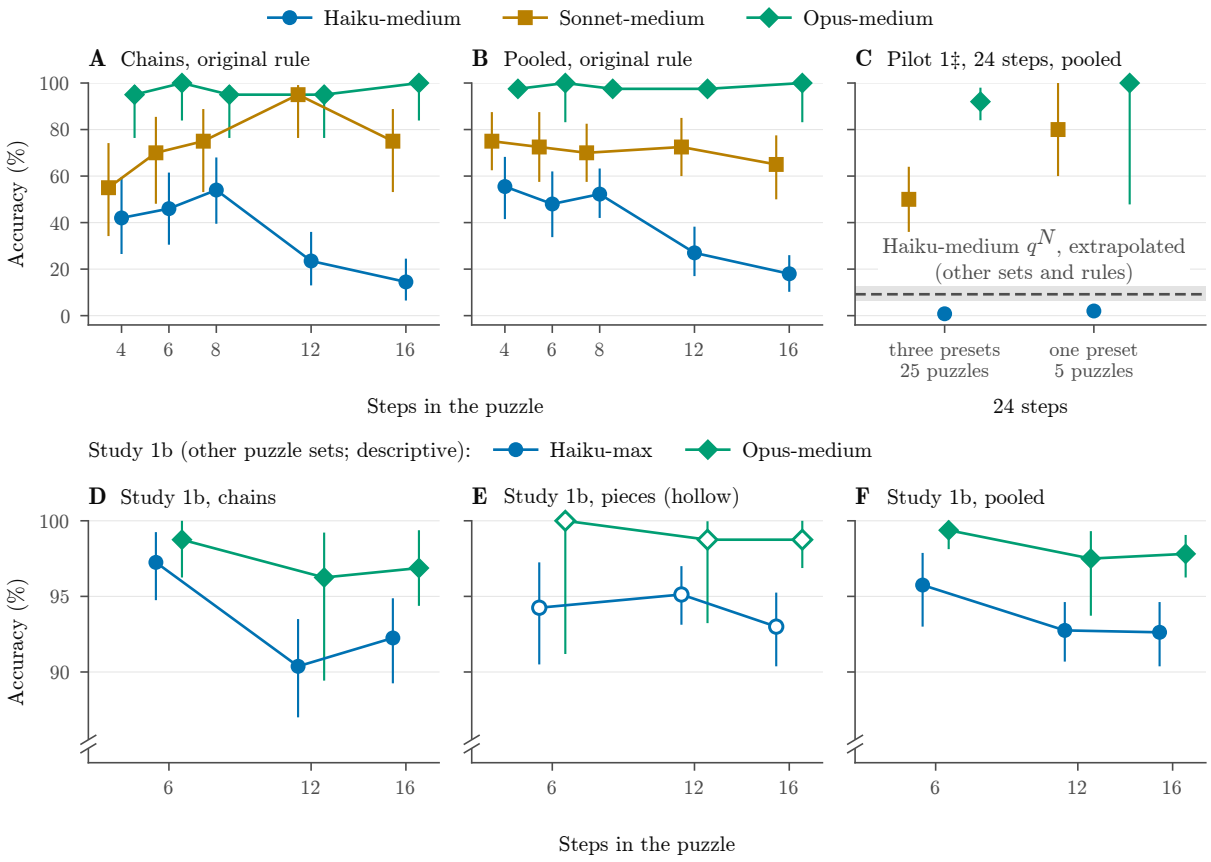


Figure S11: Under the locked original marking rule, Sonnet-medium scores lower on chains at 4 to 8 steps and fails the format check there, the reason for Deviation 2. In the length sweep Haiku-medium's and Opus-medium's whole-answer values do not change; Pilot 1's Opus-medium falls from 94.0% to 92.0% under it. Row one: the length sweep under the original rule (pre-registered descriptive), chains (A) and pooled (B); C shows Pilot 1 at 24 steps, pooled (exploratory‡: other sets and rules), its three rule-set presets left and the one preset used later right, beside the per-piece line extended to 24 steps, an extrapolation. Row two: Study 1b's Haiku-max and Opus-medium (other puzzle sets; descriptive); its accuracy axes start at 84, not at zero. Bars are 95% intervals; exact ones are post hoc†: length-sweep Opus-medium pooled at 6 steps and 16 steps, Study 1b's Opus-medium on pieces at 6 steps, and Pilot 1's Opus-medium; bars shorter than their marker, with neither end reaching a printed point past its edge, are not drawn (the length sweep's pooled Opus-medium at 4, 8 and 12 steps, and Pilot 1's Haiku-medium). Format flags on chains: Sonnet-medium at 4, 6 and 8 steps, Haiku-medium at 16 steps, Opus-medium at 4 steps and 12 steps (both at exactly 5.0%, as in Figure S5) and Pilot 1's Opus-medium.

Table S24: The length sweep (pre-registered descriptive; the 24-step markers are exploratory[‡]): Haiku-medium, Opus-medium and Sonnet-medium at 4, 6, 8, 12 and 16 steps on 20 puzzles per length, a different set at each length, under both marking rules (Deviation 2's, which is official, and the original). Intervals: cluster-bootstrap 95% intervals over puzzles; Wilson intervals for Opus-medium and Sonnet-medium by shape. An interval that collapsed to one value is replaced by the exact Clopper–Pearson 95% interval from free analysis 4 (post hoc[†]), marked as exact; Opus-medium's per-piece rate and the reference built from it have no exact interval and are marked as of zero width.

Model	Steps	Samples per puzzle	Whole answer, chain (%) [95% interval], Deviation 2 rule	Whole answer, pieces (%) [95% interval], Deviation 2 rule	Whole answer, pooled (%) [95% interval], Deviation 2 rule	Format compliance, chain (%), Deviation 2 rule	Format flag fires, Deviation 2 rule
Haiku-medium	4	10	42.00 [26.50, 59.50]	69.00 [54.00, 82.00]	55.50 [41.50, 68.25]	99.00	no
Haiku-medium	6	10	46.00 [30.50, 61.50]	50.00 [34.50, 65.50]	48.00 [33.75, 62.00]	98.00	no
Haiku-medium	8	10	54.00 [39.50, 68.00]	50.50 [39.00, 61.50]	52.25 [42.00, 63.25]	99.00	no
Haiku-medium	12	10	23.50 [13.00, 36.00]	30.50 [15.00, 47.00]	27.00 [17.00, 38.25]	96.50	no
Haiku-medium	16	10	14.50 [6.50, 24.50]	21.50 [13.00, 30.50]	18.00 [10.25, 26.00]	94.50	yes
Opus-medium	4	1	95.00 [76.39, 99.11]	100.00 [83.89, 100.00]	97.50 [92.50, 100.00]	95.00	yes
Opus-medium	6	1	100.00 [83.89, 100.00]	100.00 [83.89, 100.00]	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00	no
Opus-medium	8	1	95.00 [76.39, 99.11]	100.00 [83.89, 100.00]	97.50 [92.50, 100.00]	100.00	no
Opus-medium	12	1	95.00 [76.39, 99.11]	100.00 [83.89, 100.00]	97.50 [92.50, 100.00]	95.00	yes
Opus-medium	16	1	100.00 [83.89, 100.00]	100.00 [83.89, 100.00]	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00	no
Sonnet-medium	4	1	80.00 [58.40, 91.93]	95.00 [76.39, 99.11]	87.50 [77.50, 95.00]	100.00	no
Sonnet-medium	6	1	85.00 [63.96, 94.76]	75.00 [53.13, 88.81]	80.00 [65.00, 90.00]	95.00	yes
Sonnet-medium	8	1	85.00 [63.96, 94.76]	65.00 [43.29, 81.88]	75.00 [62.50, 87.50]	100.00	no
Sonnet-medium	12	1	95.00 [76.39, 99.11]	50.00 [29.93, 70.07]	72.50 [60.00, 85.00]	100.00	no
Sonnet-medium	16	1	75.00 [53.13, 88.81]	55.00 [34.21, 74.18]	65.00 [50.00, 77.50]	100.00	no

Table S24: (continued)

Model	Steps	Whole answer, chain (%) [95% interval], original rule	Whole answer, pieces (%) [95% interval], original rule	Whole answer, pooled (%) [95% interval], original rule	Format compliance, chain (%), original rule	Format flag fires, original rule
Haiku-medium	4	42.00 [26.50, 59.50]	69.00 [54.00, 82.00]	55.50 [41.50, 68.25]	99.00	no
Haiku-medium	6	46.00 [30.50, 61.50]	50.00 [34.50, 65.50]	48.00 [33.75, 62.00]	98.00	no
Haiku-medium	8	54.00 [39.50, 68.00]	50.50 [39.00, 61.50]	52.25 [42.00, 63.25]	98.50	no
Haiku-medium	12	23.50 [13.00, 36.00]	30.50 [15.00, 47.00]	27.00 [17.00, 38.25]	96.00	no
Haiku-medium	16	14.50 [6.50, 24.50]	21.50 [13.00, 30.50]	18.00 [10.25, 26.00]	94.00	yes
Opus-medium	4	95.00 [76.39, 99.11]	100.00 [83.89, 100.00]	97.50 [92.50, 100.00]	95.00	yes
Opus-medium	6	100.00 [83.89, 100.00]	100.00 [83.89, 100.00]	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00	no
Opus-medium	8	95.00 [76.39, 99.11]	100.00 [83.89, 100.00]	97.50 [92.50, 100.00]	100.00	no
Opus-medium	12	95.00 [76.39, 99.11]	100.00 [83.89, 100.00]	97.50 [92.50, 100.00]	95.00	yes
Opus-medium	16	100.00 [83.89, 100.00]	100.00 [83.89, 100.00]	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00	no
Sonnet-medium	4	55.00 [34.21, 74.18]	95.00 [76.39, 99.11]	75.00 [62.50, 87.50]	60.00	yes
Sonnet-medium	6	70.00 [48.10, 85.45]	75.00 [53.13, 88.81]	72.50 [57.50, 87.50]	75.00	yes
Sonnet-medium	8	75.00 [53.13, 88.81]	65.00 [43.29, 81.88]	70.00 [57.50, 82.50]	90.00	yes
Sonnet-medium	12	95.00 [76.39, 99.11]	50.00 [29.93, 70.07]	72.50 [60.00, 85.00]	100.00	no
Sonnet-medium	16	75.00 [53.13, 88.81]	55.00 [34.21, 74.18]	65.00 [50.00, 77.50]	100.00	no

Table S24: (continued)

Panel 1: Per-step accuracy

Model	Steps	Per-step accuracy, chain, Deviation 2 rule (%) [95% interval]	Per-step accuracy, pieces, Deviation 2 rule (%) [95% interval]	Per-step accuracy, pooled, Deviation 2 rule (%) [95% interval]	Per-step accuracy, chain, original rule (%) [95% interval]	Per-step accuracy, pieces, original rule (%) [95% interval]	Per-step accuracy, pooled, original rule (%) [95% interval]
Haiku-medium	4	79.58 [71.46, 87.08]	90.38 [85.13, 94.75]	84.98 [79.19, 90.19]	79.58 [71.46, 87.08]	90.38 [85.13, 94.75]	84.98 [79.19, 90.19]
Haiku-medium	6	86.48 [81.31, 91.40]	89.00 [85.17, 92.92]	87.74 [83.65, 91.84]	86.48 [81.31, 91.40]	89.00 [85.17, 92.92]	87.74 [83.65, 91.84]
Haiku-medium	8	91.60 [88.78, 94.41]	92.75 [90.81, 94.50]	92.17 [90.00, 94.26]	91.56 [88.70, 94.41]	92.75 [90.81, 94.50]	92.15 [89.98, 94.26]
Haiku-medium	12	86.37 [83.42, 89.61]	90.08 [87.25, 92.75]	88.22 [85.56, 90.90]	86.42 [83.45, 89.68]	90.08 [87.25, 92.75]	88.25 [85.58, 90.94]
Haiku-medium	16	88.76 [86.17, 91.22]	90.38 [88.16, 92.56]	89.57 [87.30, 91.72]	88.76 [86.17, 91.22]	90.38 [88.16, 92.56]	89.57 [87.30, 91.72]
Opus-medium	4	100.00 [82.35, 100.00] (exact, post hoc; 19 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [82.35, 100.00] (exact, post hoc; 19 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)
Opus-medium	6	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)
Opus-medium	8	98.75 [96.25, 100.00]	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	99.38 [98.13, 100.00]	98.75 [96.25, 100.00]	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	99.38 [98.13, 100.00]
Opus-medium	12	100.00 [82.35, 100.00] (exact, post hoc; 19 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [82.35, 100.00] (exact, post hoc; 19 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)
Opus-medium	16	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)	100.00 [83.16, 100.00] (exact, post hoc; 20 puzzles)
Sonnet-medium	4	95.00 [90.00, 98.75]	98.75 [96.25, 100.00]	96.88 [94.38, 98.75]	97.92 [93.18, 100.00]	98.75 [96.25, 100.00]	98.13 [95.00, 100.00]
Sonnet-medium	6	98.25 [95.83, 100.00]	95.83 [92.50, 98.33]	97.08 [95.00, 98.75]	98.89 [96.43, 100.00]	95.83 [92.50, 98.33]	97.08 [94.58, 99.17]
Sonnet-medium	8	98.13 [96.25, 100.00]	95.63 [93.13, 98.13]	96.88 [95.31, 98.44]	97.92 [95.59, 100.00]	95.63 [93.13, 98.13]	96.88 [95.31, 98.44]
Sonnet-medium	12	99.17 [97.50, 100.00]	93.33 [89.58, 96.67]	96.25 [94.17, 98.33]	99.17 [97.50, 100.00]	93.33 [89.58, 96.67]	96.25 [94.17, 98.33]
Sonnet-medium	16	98.44 [97.19, 99.38]	93.75 [86.88, 98.13]	96.09 [92.34, 98.44]	98.44 [97.19, 99.38]	93.75 [86.88, 98.13]	96.09 [92.34, 98.44]

Table S24: (continued)

Panel 2: Haiku-medium vote of its samples (not cost-matched)

Steps	Haiku-medium vote of its samples, chain (%) [95% interval]	Haiku-medium vote of its samples, pieces (%) [95% interval]	Haiku-medium vote of its samples, pooled (%) [95% interval]
4	50.00 [30.00, 70.00]	77.50 [57.50, 95.00]	63.75 [48.75, 77.50]
6	60.00 [40.00, 80.00]	57.50 [37.50, 77.50]	58.75 [41.25, 75.00]
8	62.50 [40.00, 82.50]	65.00 [45.00, 82.50]	63.75 [48.75, 77.50]
12	36.13 [16.25, 56.25]	38.50 [18.50, 59.00]	37.31 [23.19, 52.06]
16	17.50 [5.00, 35.00]	42.50 [22.50, 65.00]	30.00 [15.00, 46.25]

Panel 3: Format compliance and unplaceable samples on chains

Model	Steps	Chain samples answered	Format compliance, chain (%), Deviation 2 rule	Format flag fires, Deviation 2 rule	Chain samples not placeable, Deviation 2 rule	Format compliance, chain (%), original rule	Format flag fires, original rule	Chain samples not placeable, original rule
Haiku-medium	4	200	99.00	no	2	99.00	no	2
Haiku-medium	6	200	98.00	no	4	98.00	no	4
Haiku-medium	8	200	99.00	no	2	98.50	no	3
Haiku-medium	12	200	96.50	no	7	96.00	no	8
Haiku-medium	16	200	94.50	yes	11	94.00	yes	12
Opus-medium	4	20	95.00	yes	1	95.00	yes	1
Opus-medium	6	20	100.00	no	0	100.00	no	0
Opus-medium	8	20	100.00	no	0	100.00	no	0
Opus-medium	12	20	95.00	yes	1	95.00	yes	1
Opus-medium	16	20	100.00	no	0	100.00	no	0
Sonnet-medium	4	20	100.00	no	0	60.00	yes	8
Sonnet-medium	6	20	95.00	yes	1	75.00	yes	5
Sonnet-medium	8	20	100.00	no	0	90.00	yes	2
Sonnet-medium	12	20	100.00	no	0	100.00	no	0
Sonnet-medium	16	20	100.00	no	0	100.00	no	0

Panel 4: q per model (pieces only)

Model	q (%): share of pieces right, pieces only, pooled over the lengths	q 95% interval (%)
Haiku-medium	90.53	[89.25, 91.80]
Opus-medium	100.00	[100.00, 100.00] (zero width)
Sonnet-medium	94.67	[92.07, 96.63]

Table S24: (continued)

Panel 5: Independent-pieces reference and observed / q^N

Model	Steps	Reference q^N (%)	q^N band (95% interval)	Observed / q^N, chain, Deviation 2 rule	Observed / q^N, pieces, Deviation 2 rule	Observed / q^N, pooled, Deviation 2 rule	Observed / q^N, chain, original rule	Observed / q^N, pieces, original rule	Observed / q^N, pooled, original rule	Note
Haiku-medium	4	67.18	[63.45, 71.03]	0.63	1.03	0.83	0.63	1.03	0.83	
Haiku-medium	6	55.06	[50.54, 59.87]	0.84	0.91	0.87	0.84	0.91	0.87	
Haiku-medium	8	45.13	[40.26, 50.46]	1.20	1.12	1.16	1.20	1.12	1.16	
Haiku-medium	12	30.32	[25.54, 35.84]	0.78	1.01	0.89	0.78	1.01	0.89	
Haiku-medium	16	20.36	[16.21, 25.46]	0.71	1.06	0.88	0.71	1.06	0.88	
Haiku-medium	24	9.19	[6.53, 12.84]							"dotted extrapolation only"
Opus-medium	4	100.00	[100.00, 100.00] (zero width)	0.95	1.00	0.98	0.95	1.00	0.98	
Opus-medium	6	100.00	[100.00, 100.00] (zero width)	1.00	1.00	1.00	1.00	1.00	1.00	
Opus-medium	8	100.00	[100.00, 100.00] (zero width)	0.95	1.00	0.98	0.95	1.00	0.98	
Opus-medium	12	100.00	[100.00, 100.00] (zero width)	0.95	1.00	0.98	0.95	1.00	0.98	
Opus-medium	16	100.00	[100.00, 100.00] (zero width)	1.00	1.00	1.00	1.00	1.00	1.00	
Opus-medium	24	100.00	[100.00, 100.00] (zero width)							"dotted extrapolation only"
Sonnet-medium	4	80.34	[71.84, 87.19]	1.00	1.18	1.09	0.68	1.18	0.93	
Sonnet-medium	6	72.01	[60.89, 81.41]	1.18	1.04	1.11	0.97	1.04	1.01	
Sonnet-medium	8	64.54	[51.61, 76.02]	1.32	1.01	1.16	1.16	1.01	1.08	
Sonnet-medium	12	51.85	[37.08, 66.28]	1.83	0.96	1.40	1.83	0.96	1.40	
Sonnet-medium	16	41.66	[26.64, 57.79]	1.80	1.32	1.56	1.80	1.32	1.56	
Sonnet-medium	24	26.89	[13.75, 43.93]							"dotted extrapolation only"

Table S24: (continued)

Panel 6: Pilot 1 markers (24 steps, exploratory)

Model	Marker	Puzzles	Whole answer, pooled (%) [95% interval], Deviation 2 rule	Whole answer, pooled (%) [95% interval], original rule
Haiku-medium	Pilot 1, the one preset used later	5	2.00 [0.00, 4.00]	2.00 [0.00, 4.00]
Haiku-medium	Pilot 1, three rule-set presets	25	0.80 [0.20, 1.40]	0.80 [0.20, 1.40]
Opus-medium	Pilot 1, the one preset used later	5	100.00 [47.82, 100.00] (exact, post hoc; 5 puzzles)	100.00 [47.82, 100.00] (exact, post hoc; 5 puzzles)
Opus-medium	Pilot 1, three rule-set presets	25	94.00 [88.00, 98.00]	92.00 [84.00, 98.00]
Sonnet-medium	Pilot 1, the one preset used later	5	80.00 [60.00, 100.00]	80.00 [60.00, 100.00]
Sonnet-medium	Pilot 1, three rule-set presets	25	50.00 [36.00, 64.00]	50.00 [36.00, 64.00]

Whole-answer accuracy is the share of samples with every step right; a chain is right only if every step line and the answer line are right. Both marking rules are shown side by side: the Deviation 2 rule, which is official, and the original rule, under which a step line written more than once fails even when the repeats are identical. Format compliance sits beside the whole-answer accuracy: the share of answered chain samples whose format was ok under each marking rule (the files give it for chains only), with the format flag, which fires when more than 5% of the answered chain samples are not ok. Panel 3 adds the chain samples answered and not placeable. Three flags sit at exactly 5.0% non-compliance; the locked check raises them through floating-point rounding, though its rule reads 'more than 5%'; we report them as stamped (Opus-medium's two cells sit at exactly 5.0% under the original rule's columns too, raised the same way). Per-step accuracy (Panel 1): for pieces, the share of a sample's pieces that are right (a missing or unanswered sample scores nothing); for chains, each step is judged from the model's own previous step, and only samples whose step lines could be placed count: unplaceable chain samples are left out (Panel 3 counts them). Each length has its own small set: 20 puzzles. Haiku-medium has 10 samples per puzzle; Opus-medium and Sonnet-medium have 1. Intervals are 95% and unadjusted: cluster bootstrap over puzzles with 2,000 resamples, except Opus-medium and Sonnet-medium per-shape whole-answer intervals, which are Wilson (the file's interval field). "(exact, post hoc; ... puzzles)" marks a bootstrap interval that collapsed to one value (every puzzle right, or every one wrong), replaced by the exact two-sided 95% Clopper–Pearson interval from free analysis 4 (post hoc; it changes no test). The count in the mark is the number of puzzles, not of samples (the conservative unit). In Opus-medium's per-step chain cells the count in the mark is only the puzzles whose chain sample could be placed: the set's puzzles less the unplaceable chain samples (Panel 3). "(zero width)" marks Opus-medium's q interval (Panel 4) and q^N bands (Panel 5), which collapsed to one value; free analysis 4 gives them no exact interval (post hoc): "q pools every piece-step over all lengths and sets (clusters stratified by set), so no single count of base puzzles carries it, and its q^N bands are powers of its interval". Its note on q adds: "the length sweep's q is estimated from pieces only, so its exact counterpart is the pieces per-step row"; Opus-medium's pieces per-step cells carry exact intervals at every length under both marking rules (Panel 1). In the first quotation the file's "base puzzles" are the paper's puzzles. Haiku-medium's vote (Panel 2) is over all 10 of its samples per puzzle and is not cost-matched: its cost is not set equal to that of another model's answer. q is each model's share of pieces right, estimated from the pieces puzzles only and pooled over the lengths; q^N is the independent-pieces reference and observed / q^N compares the observed whole-answer accuracy with it. The 24-step row is an extrapolation only; its label, quoted as stamped, says "dotted", and the figures draw the line dashed. Pilot 1 markers are at 24 steps, exploratory, and ran before Lock 2 with a different rule mix, different sets and different timing; they never enter q. A preset is one setting of the puzzle generator (its rule kinds and dials). One marker pools Pilot 1's sets of three rule-set presets; the other only those of the one preset used later, whose rules the length sweep keeps at every length. The sweep is descriptive (pre-registered as such); no test is run on it. Columns that print one value in every row are left out: the number of truncated calls is 0 in every row.

S6.3 Further detail on effort and length

These paragraphs give details that the effort-and-length results in Section 4 of the paper leave out. Table S25 gives every effort level.

In Study 1b's unscored trial round at 16 steps, Haiku used more output tokens per call at each higher level: 2,566.7 with thinking off, 4,098.9 at low (medium was not run in the trial round), 6,821.0 at high, 10,016.6 at xhigh and 19,382.3 at max (about 1.94 times xhigh). Tokens and accuracy both rise with each level, but the tokens come from different puzzles than the accuracies, and effort, tokens and the token limit move together (post hoc; our reading, not a test).

With thinking off at 16 steps, Haiku scored 0.00% on chains and 1.00% on pieces. Many chain replies also broke the format: only 112 of 400 (28.00%) kept it. But the replies that kept the format also failed: none of the chain replies was right, and on pieces 313 of 400 replies (78.25%) kept the format yet only 4 were right. So the low score is not mostly a format failure.

Opus. With one sample per puzzle on 40 puzzles at 6 steps (two for Opus-medium), the Opus settings scored 96.25–100.00% pooled over both shapes, not rising steadily with effort: 98.75% at low, 99.38% at medium, 96.25% at high, 97.50% at xhigh and 100.00% at max. Opus-high's figure is 3 misses in 80 answers, all on chains, 2 of them extra-line format failures, and Opus-xhigh's 2 chain misses were both extra lines (post hoc reading of the format report). Opus-max cost $0.0888 per answer, 2.01 times Opus-medium's $0.0441. Opus is near its ceiling at every level, so these data cannot separate "effort does not help Opus" from "these puzzles are too easy for Opus".

A reference line, and what the chains do. The dashed line in Figure S5 shows what a whole answer would score if every step were right independently at Haiku-medium's per-piece rate q, the share of single pieces right on the pieces shape pooled over the five lengths: $q = 90.53\%$ (95% interval 89.25% to 91.80%). The plan shows the line without fitting it to the chains, and it is not a test. On pieces, observed accuracy over q^N stayed between 0.91 and 1.12; on chains it was below 1 at 4, 6, 12 and 16 steps (0.63, 0.84, 0.78, 0.71) and above it at 8 steps (1.20). Extended, the line gives about 9.19% at 24 steps (band 6.53% to 12.84%). Pilot 1 at 24 steps scored 0.80% (0.20% to 1.40%), but it used three rule-set presets, a different rule mix, other puzzles and an earlier time, so it cannot judge the line. Chain format compliance was 99.00, 98.00, 99.00, 96.50 and 94.50% at 4 to 16 steps; the 16-step value is flagged (5.50% non-compliant).

The chains' own per-step rate, each step judged from the model's own previous balances, was lower than the pieces' at every length (79.58%, 86.48%, 91.60%, 86.37% and 88.76% against 90.38%, 89.00%, 92.75%, 90.08% and 90.38%). Raised to the power N, the chain rate gives 40.11%, 41.83%, 49.55%, 17.22% and 14.83%, close to the observed 42.00%, 46.00%, 54.00%, 23.50% and 14.50% (post hoc arithmetic). So the chains' shortfall against the pieces' line goes with a lower step rate on chains. Whether that comes from errors breeding errors or from other differences between the shapes, these data do not show (our reading, not a test). For Haiku-max, one sample scored 95.75% at 6 steps and 92.63% at 16. If every step were right independently at the rate the 6-step score implies, 16 steps would give about 89.06% (82.41% to 94.43% across the 6-step interval); the observed value lies inside that range, so the data cannot tell compounding from a smaller decline (post hoc arithmetic, not a test).

Haiku-max's cost and gap by length. Haiku-max's cost per step stayed at about $0.000698 while Opus-medium's fell with length, so Haiku-max's cost per answer rose 2.67 times from 6 to 16 steps ($0.00419 to $0.0112), against 1.70 times for Opus-medium ($0.0441 to $0.0750). A single Haiku-max sample's point gap to Opus-medium was 3.63 points at 6 steps, 4.75 at 12 and 5.19 at 16, but the 6- and 16-step intervals overlap (-7.00 to -1.50 and -7.75 to -2.69) and each length used other puzzles, so no trend in the gap with length is shown.

Other settings at 12 and 16 steps. Pooled over shapes, Opus-medium scored 97.50–100.00% at every length we ran. Study 1b ran Opus-low and every Sonnet effort level only at 6 steps. The only other non-Haiku points at those lengths are the length sweep's Opus-medium and Sonnet-medium (other puzzles, one sample each, not cost-matched: Sonnet-medium 72.50% at 12 steps and 65.00% at 16; Opus-medium 97.50% and 100.00%).

Table S25: Haiku's accuracy rose the most with effort, while Opus scored near its ceiling at every level we ran. The effort sweep (Study 1b; pre-specified descriptive, no tests): accuracy, format compliance, and list-price cost per answer and per correct answer for every model and effort level at 6 steps (40 puzzles), for every Haiku setting and Opus-medium at 16 steps (80), and for Haiku-max and Opus-medium at 12 steps (80). Intervals: cluster bootstrap for settings with several samples, Clopper–Pearson for one-sample settings, and exact Clopper–Pearson (post hoc†) where a bootstrap interval collapsed.

Setting	Steps	Samples per puzzle	Puzzles	Pooled accuracy (%)	Pooled 95% interval (%)	Chain accuracy (%)	Chain 95% interval (%)	Pieces accuracy (%)	Pieces 95% interval (%)
Haiku, thinking off	6	5	40	27.50	[20.00, 35.25]	26.50	[17.50, 36.50]	28.50	[19.00, 38.00]
Haiku, thinking off	16	5	80	0.50	[0.00, 1.13]	0.00	[0.00, 4.51] (exact, post hoc; 80 puzzles)	1.00	[0.00, 2.25]
Haiku-low	6	5	40	52.75	[43.75, 61.25]	52.50	[42.00, 62.50]	53.00	[42.50, 64.50]
Haiku-low	16	5	80	12.75	[9.63, 16.00]	9.50	[6.25, 13.25]	16.00	[11.75, 20.75]
Haiku-medium	6	5	40	54.50	[46.00, 63.25]	52.00	[41.00, 63.00]	57.00	[46.50, 67.50]
Haiku-medium	16	5	80	17.75	[13.75, 21.75]	11.50	[8.00, 15.25]	24.00	[17.25, 30.75]
Haiku-high	6	5	40	60.75	[52.25, 69.50]	59.00	[48.50, 69.50]	62.50	[50.50, 73.50]
Haiku-high	16	5	80	31.25	[26.88, 36.00]	31.00	[25.50, 36.75]	31.50	[25.25, 37.75]
Haiku-xhigh	6	5	40	76.75	[70.00, 83.00]	80.50	[74.00, 86.50]	73.00	[63.50, 82.50]
Haiku-xhigh	16	5	80	58.38	[53.88, 62.88]	57.25	[50.75, 63.75]	59.50	[52.75, 66.25]
Haiku-max	6	10	40	95.75	[93.00, 97.88]	97.25	[94.75, 99.25]	94.25	[90.50, 97.25]
Haiku-max	12	10	80	92.75	[90.69, 94.63]	90.38	[87.00, 93.50]	95.13	[93.13, 97.00]
Haiku-max	16	10	80	92.63	[90.38, 94.63]	92.25	[89.25, 94.88]	93.00	[90.38, 95.25]
Sonnet-low	6	1	40	80.00	[69.56, 88.11]	77.50	[61.55, 89.16]	82.50	[67.22, 92.66]
Sonnet-medium	6	1	40	80.00	[69.56, 88.11]	77.50	[61.55, 89.16]	82.50	[67.22, 92.66]
Sonnet-high	6	1	40	97.50	[91.26, 99.70]	97.50	[86.84, 99.94]	97.50	[86.84, 99.94]
Sonnet-xhigh	6	1	40	97.50	[91.26, 99.70]	95.00	[83.08, 99.39]	100.00	[91.19, 100.00]
Sonnet-max	6	1	40	100.00	[95.49, 100.00]	100.00	[91.19, 100.00]	100.00	[91.19, 100.00]
Opus-low	6	1	40	98.75	[93.23, 99.97]	97.50	[86.84, 99.94]	100.00	[91.19, 100.00]
Opus-medium	6	2	40	99.38	[98.13, 100.00]	98.75	[96.25, 100.00]	100.00	[91.19, 100.00] (exact, post hoc; 40 puzzles)
Opus-medium	12	1	80	97.50	[93.72, 99.31]	96.25	[89.43, 99.22]	98.75	[93.23, 99.97]
Opus-medium	16	2	80	97.81	[96.25, 99.06]	96.88	[94.38, 99.38]	98.75	[96.88, 100.00]
Opus-high	6	1	40	96.25	[89.43, 99.22]	92.50	[79.61, 98.43]	100.00	[91.19, 100.00]
Opus-xhigh	6	1	40	97.50	[91.26, 99.70]	95.00	[83.08, 99.39]	100.00	[91.19, 100.00]
Opus-max	6	1	40	100.00	[95.49, 100.00]	100.00	[91.19, 100.00]	100.00	[91.19, 100.00]

Table S25: (continued)

Setting	Steps	Interval type	Format compliance, chain (%)	Format flag, chain	Format compliance, pieces (%)	Format flag, pieces	List cost per answer ($)	List cost per correct answer ($)	Token limit (max_tokens)	Trial round: largest output (tokens)
Haiku, thinking off	6	cluster bootstrap	83.50	yes	99.50	no	0.000927	0.00337		
Haiku, thinking off	16	cluster bootstrap	28.00	yes	78.25	yes	0.00147	0.293	14,000	3,549
Haiku-low	6	cluster bootstrap	98.50	no	100.00	no	0.00104	0.00196		
Haiku-low	16	cluster bootstrap	93.75	yes	100.00	no	0.00234	0.0184	14,000	4,717
Haiku-medium	6	cluster bootstrap	99.00	no	99.00	no	0.00110	0.00202		
Haiku-medium	16	cluster bootstrap	95.49	no	100.00	no	0.00254	0.0143		
Haiku-high	6	cluster bootstrap	100.00	no	100.00	no	0.00125	0.00206	6,000	2,723
Haiku-high	16	cluster bootstrap	93.75	yes	100.00	no	0.00334	0.0107	32,000	15,600
Haiku-xhigh	6	cluster bootstrap	98.50	no	99.50	no	0.00172	0.00224	11,000	5,356
Haiku-xhigh	16	cluster bootstrap	97.50	no	100.00	no	0.00507	0.00869	39,000	19,018
Haiku-max	6	cluster bootstrap	100.00	no	100.00	no	0.00419	0.00438		
Haiku-max	12	cluster bootstrap	99.88	no	100.00	no	0.00826	0.00890	96,000	29,648
Haiku-max	16	cluster bootstrap	99.50	no	99.88	no	0.0112	0.0121	96,000	27,164
Sonnet-low	6	Clopper–Pearson (exact)	100.00	no	100.00	no	0.0150	0.0188		
Sonnet-medium	6	Clopper–Pearson (exact)	97.50	no	100.00	no	0.0154	0.0192		
Sonnet-high	6	Clopper–Pearson (exact)	100.00	no	100.00	no	0.0188	0.0193	5,000	2,443
Sonnet-xhigh	6	Clopper–Pearson (exact)	100.00	no	100.00	no	0.0228	0.0234	6,000	2,591
Sonnet-max	6	Clopper–Pearson (exact)	100.00	no	100.00	no	0.0475	0.0475	11,000	5,364
Opus-low	6	Clopper–Pearson (exact)	97.50	no	100.00	no	0.0395	0.0399		
Opus-medium	6	cluster bootstrap	100.00	no	100.00	no	0.0441	0.0444		
Opus-medium	12	Clopper–Pearson (exact)	98.75	no	100.00	no	0.0612	0.0628		
Opus-medium	16	cluster bootstrap	100.00	no	100.00	no	0.0750	0.0766		
Opus-high	6	Clopper–Pearson (exact)	95.00	no	100.00	no	0.0455	0.0473	6,000	2,686
Opus-xhigh	6	Clopper–Pearson (exact)	95.00	no	100.00	no	0.0499	0.0512	6,000	2,740
Opus-max	6	Clopper–Pearson (exact)	100.00	no	100.00	no	0.0888	0.0888	11,000	5,411

Format flag: yes means the pre-registered format-compliance flag fires for that setting and shape. Its rule, in the file's words: "non-compliance above 5% of answered tries"; non-compliance is the share of answered samples (the file's "tries") whose format was not ok, one minus the format compliance (checked). Refusals, truncations and missing calls are counted apart in the Study 1b file (not in this table) and do not enter the flag: the answered samples leave them out (checked). Two cells sit at exactly 5.0% and are not flagged, as the rule says 'more than 5%'. Pooled accuracy is each puzzle's mean over its two shapes. Intervals are cluster bootstrap over puzzles for settings with several samples and Clopper–Pearson for one-sample settings (the interval type the file records for each setting, in the Interval type column that opens the table's second part); per-shape intervals are of the same type. Haiku-max's and Opus-medium's intervals here at the lengths of Tests B1–B3 and those in Table S22 (Tests B1–B3 by shape) come from separate bootstrap draws over the same puzzles, with the same points, and differ only by resampling, so their ends can differ slightly; where both collapsed to one value, both show the same exact interval. "(exact, post hoc; ... puzzles)" marks a bootstrap interval that collapsed to one value (every puzzle right, or every one wrong), replaced by the exact two-sided 95% Clopper–Pearson interval from free analysis 4 (post hoc; it changes no test). The count in the mark is the number of puzzles, not of samples (the conservative unit). Token limit and the trial round's largest output are recorded only for the settings whose limit the trial round set; the Study 1b file records no limit for the other settings, so those cells are empty. Format compliance is the share of answered samples whose format was ok, per shape. Cost per correct answer is list-price dollars per correct answer, as pre-registered; no setting in this table is a vote. Sonnet-low and Sonnet-medium at 6 steps have the same accuracies and intervals on both shapes and pooled. They are separate settings: list cost per answer $0.0150 against $0.0154; chain format compliance 100.00% against 97.50%.

Cost per correct answer misleads on its own. At 6 steps in Study 1, cost per correct answer (often called cost-of-pass) at list price was $0.00252 and $0.00214 for Haiku-medium on chains and pieces, $0.00452 and $0.00428 for Haiku-max, $0.0184 and $0.0215 for Sonnet-medium, $0.0375 and $0.0449 for Opus-low, and $0.0418 and $0.0481 for Opus-medium. That measure assumes a wrong answer can be spotted and thrown away for free, which exact tasks without an answer key do not allow, so we compare setups by whole-answer accuracy at an affordable cost, not by cost per correct answer alone. At 16 steps it would make Haiku-xhigh (58.38%, $0.00869 per correct answer) and Haiku-high (31.25%, $0.0107) look cheaper than Haiku-max (92.63%, $0.0121) (post hoc reading). Table S26 sets out the cost of each single call and each cost-matched vote.

The cost-matched vote size does not travel. The number of cheap samples that cost as much as one Opus-medium answer changes with shape (for Haiku-medium in Study 1, 36 on chains, and 44 capped at 40 on pieces) and with length (for Haiku-max in Study 1b, about 10 at 6 steps and 6 at 16). At 16 steps, one Opus-medium answer costs as much as 5.50 Haiku-max samples on chains and 8.18 on pieces (post hoc), so each shape's own cost-matched size differs. It also rests on the list prices of 2026-10-12/13 and on hidden thinking tokens, 81.08% of Haiku-medium's list cost on pieces, and 95.19–96.51% of Haiku-max's (post hoc; thinking tokens as the API reports them).

In the length sweep, the 10-sample Haiku-medium vote scored 63.75%, 58.75%, 63.75%, 37.31% and 30.00% at 4, 6, 8, 12 and 16 steps.

Total tokens at medium effort (pre-specified descriptive). On chains, one Sonnet-medium answer used 2,248.3 total tokens; on pieces, one Haiku-medium sample used 3,048.9 against Opus-medium's 3,269.0.

Where the Haiku dollar goes (post hoc; thinking tokens as the API reports them). Thinking was 95.37% of Haiku-max's list cost on chains (a mean of 8,402.6 thinking tokens per call), 81.32% of Haiku-medium's on chains (mean 1,789.3), 37.83–43.51% of Opus-medium's over both shapes (mean 875.5 on chains) and 0–4.20% of Sonnet-medium's, which thought only on some chain calls and on none of its piece calls. At max effort the picture changes. Haiku-max wrote 3.61 to 4.74 times Opus-medium's output tokens per answer at 6 steps (Study 1) and 5.34 to 7.98 times at 16 steps (Study 1b; 6.53 pooled). It still cost about 1/9 to 1/12 as much per answer at 6 steps and about 1/7 as much at 16 steps (cost ratios 9.14, 11.83 and 6.71), because each Haiku token costs 1/40 as much as an Opus token at list price. Figure S12 sets the API's thinking counts (post hoc) against Study 1's pre-specified estimate from visible characters, which overstates the thinking share of list cost in every setting.

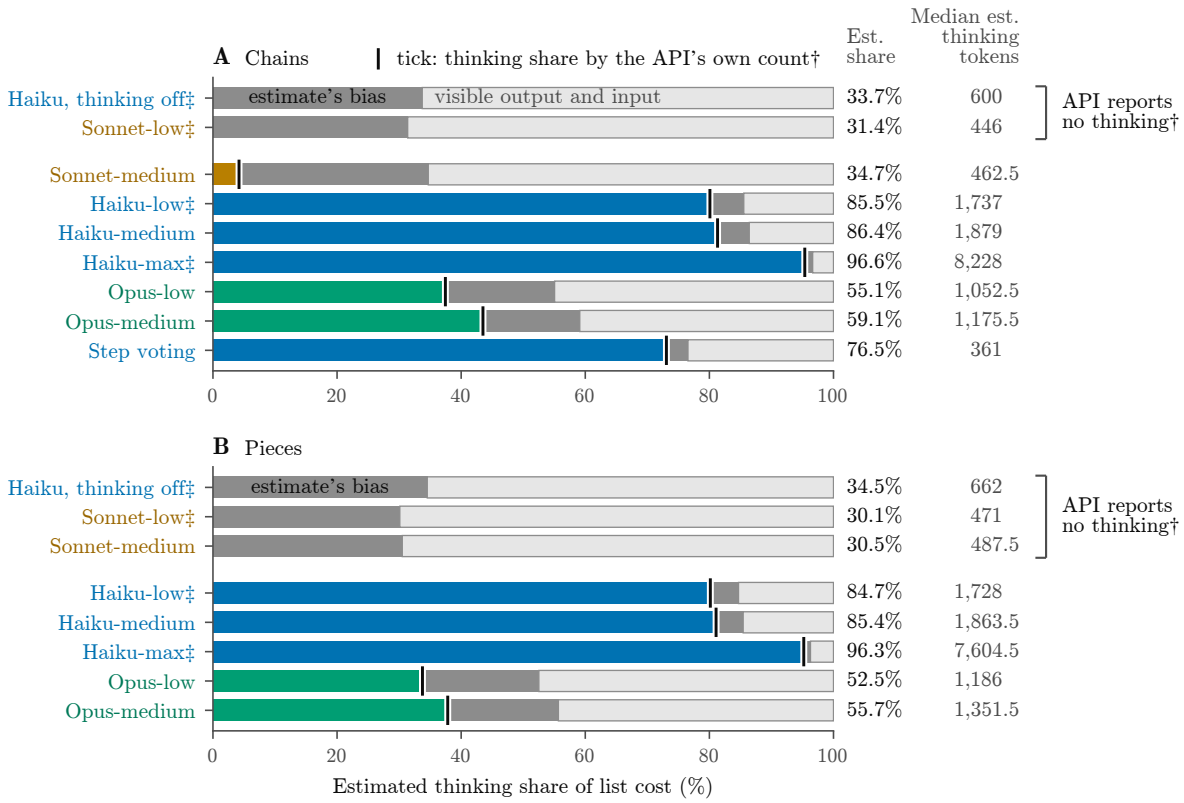


Figure S12: Study 1's estimate overstates the thinking share of list cost in every setting, most where little or no thinking happens. Pre-specified descriptive estimates from visible characters (Study 1, main run, 6 steps; list price). The dark ticks are the API's own thinking counts: each bar is in its model's colour up to its tick and grey beyond it, the estimate's bias, and the bracketed rows are the settings for which the API reports no thinking tokens (all post hoc†); ‡ marks Study 1's exploratory settings. The format check flags Haiku, thinking off, on chains. Step voting, plotted on chains only, where it ran, is in no table. The main text uses the API's counts.

Table S26: Where the money goes, in full: list and Batch costs, costs per correct answer, cost ratios and spend shares, calls, mean input and output tokens, the thinking share of output tokens, and the reporting definitions' thinking estimates from visible characters beside the API's counts. The estimates are labelled as biased: they overstate thinking in every setting (Figure S12). Study 1's settings at 6 steps (80 puzzles) and Study 1b's effort sweep at 16 steps (80); list price; point values only. Four panels follow. The first gives the cost-matched vote sizes, with the raw size, the cap and the spend share. Study 1's three sizes and Study 1b's pooled size at 16 steps come from the cost-only records fixed before any score was seen; the per-shape sizes at 16 steps and the pooled sizes at 6 and 12 steps are post hoc† (the 6-step size is the one measured beside Test B3; no vote was tested there). The second gives Haiku-max's and Opus-medium's mean output tokens per call and list cost per answer, per shape and pooled, at 6 steps in Study 1 and at 6, 12 and 16 steps in Study 1b: the pooled 6- and 16-step cost ratios are the ones stamped beside Tests B3 and B1, the pooled 12-step ratio is computed from the two stamped costs (post hoc†), and the per-shape ratios are post hoc†. The third gives the token structure of Study 1b's other settings, every model and effort level at 6 steps and Haiku-max and Opus-medium at 12 steps, per shape (post hoc†; thinking tokens as the API reports them). The fourth gives the trial round's tokens and list cost by setting (Study 1b; unscored, descriptive).

Study	Steps	Setting	Shape	Accuracy (%)	Format compliance (%)	List cost per answer ($)	Batch cost per answer ($)	List cost per correct answer ($)	Batch cost per correct answer ($)	Cost ratio (see note)	Spend share of a vote (see note)
Study 1	6	Haiku, thinking off	chains (1×6)	22.00	81.72	0.000890	0.000445	0.00405	0.00202	0.022	—
Study 1	6	Haiku, thinking off	pieces (6×1)	23.50	99.56	0.000956	0.000478	0.00407	0.00203	0.020	—
Study 1	6	Haiku-low	chains (1×6)	39.22	98.13	0.00103	0.000516	0.00263	0.00132	0.026	—
Study 1	6	Haiku-low	pieces (6×1)	46.50	99.66	0.00103	0.000513	0.00221	0.00110	0.021	—
Study 1	6	Haiku-medium	chains (1×6)	43.72	98.59	0.00110	0.000550	0.00252	0.00126	0.027	—
Study 1	6	Haiku-medium	pieces (6×1)	51.00	99.72	0.00109	0.000547	0.00214	0.00107	0.023	—
Study 1	6	Haiku-max	chains (1×6)	97.50	100.00	0.00441	0.00220	0.00452	0.00226	0.109	—
Study 1	6	Haiku-max	pieces (6×1)	95.00	99.58	0.00407	0.00203	0.00428	0.00214	0.085	—
Study 1	6	Sonnet-low	chains (1×6)	71.25	100.00	0.0142	0.00708	0.0199	0.00993	0.352	—
Study 1	6	Sonnet-low	pieces (6×1)	76.25	100.00	0.0158	0.00792	0.0208	0.0104	0.329	—
Study 1	6	Sonnet-medium	chains (1×6)	80.00	97.50	0.0147	0.00737	0.0184	0.00921	0.366	—
Study 1	6	Sonnet-medium	pieces (6×1)	75.00	100.00	0.0161	0.00805	0.0215	0.0107	0.334	—
Study 1	6	Opus-low	chains (1×6)	97.50	98.75	0.0365	0.0183	0.0375	0.0187	0.908	—
Study 1	6	Opus-low	pieces (6×1)	100.00	100.00	0.0449	0.0224	0.0449	0.0224	0.932	—
Study 1	6	Opus-medium	chains (1×6)	96.25	97.50	0.0402	0.0201	0.0418	0.0209	—	—
Study 1	6	Opus-medium	pieces (6×1)	100.00	100.00	0.0481	0.0241	0.0481	0.0241	—	—
Study 1	6	Haiku-medium vote, $k = 36$ (cost-matched with Opus-medium)	chains (1×6)	58.61	98.59	0.0396	0.0198	0.0676	0.0338	0.984	0.984
Study 1	6	Haiku-medium vote, $k = 40$ (capped; cost-matched $k = 44$ with Opus-medium)	pieces (6×1)	66.88	99.72	0.0437	0.0219	0.0654	0.0327	0.909	0.909
Study 1	6	Haiku-medium vote, $k = 13$ (cost-matched with Sonnet-medium)	chains (1×6)	55.30	98.59	0.0143	0.00715	0.0259	0.0129	0.355	0.970

Table S26: (continued)

Study	Steps	Setting	Shape	Accuracy (%)	Format compliance (%)	List cost per answer ($)	Batch cost per answer ($)	List cost per correct answer ($)	Batch cost per correct answer ($)	Cost ratio (see note)	Spend share of a vote (see note)
Study 1b	16	Haiku, thinking off	chains (1×16)	0.00	28.00	0.00133		"undefined (no correct answer)"	"undefined (no correct answer)"	0.020	—
Study 1b	16	Haiku, thinking off	pieces (16×1)	1.00	78.25	0.00161		0.161	0.0804	0.019	—
Study 1b	16	Haiku, thinking off	pooled	0.50		0.00147		0.293		0.020	—
Study 1b	16	Haiku-low	chains (1×16)	9.50	93.75	0.00234		0.0246	0.0123	0.035	—
Study 1b	16	Haiku-low	pieces (16×1)	16.00	100.00	0.00235		0.0147	0.00733	0.028	—
Study 1b	16	Haiku-low	pooled	12.75		0.00234		0.0184		0.031	—
Study 1b	16	Haiku-medium	chains (1×16)	11.50	95.49	0.00256		0.0223	0.0111	0.038	—
Study 1b	16	Haiku-medium	pieces (16×1)	24.00	100.00	0.00252		0.0105	0.00525	0.031	—
Study 1b	16	Haiku-medium	pooled	17.75		0.00254		0.0143		0.034	—
Study 1b	16	Haiku-high	chains (1×16)	31.00	93.75	0.00373		0.0120	0.00602	0.055	—
Study 1b	16	Haiku-high	pieces (16×1)	31.50	100.00	0.00295		0.00935	0.00468	0.036	—
Study 1b	16	Haiku-high	pooled	31.25		0.00334		0.0107		0.045	—
Study 1b	16	Haiku-xhigh	chains (1×16)	57.25	97.50	0.00556		0.00972	0.00486	0.083	—
Study 1b	16	Haiku-xhigh	pieces (16×1)	59.50	100.00	0.00459		0.00771	0.00385	0.056	—
Study 1b	16	Haiku-xhigh	pooled	58.38		0.00507		0.00869		0.068	—
Study 1b	16	Haiku-max	chains (1×16)	92.25	99.50	0.0123	0.00613	0.0133	0.00665	0.182	—
Study 1b	16	Haiku-max	pieces (16×1)	93.00	99.88	0.0101	0.00504	0.0108	0.00542	0.122	—
Study 1b	16	Haiku-max	pooled	92.63		0.0112	0.00559	0.0121		0.149	—
Study 1b	16	Haiku-max vote, $k = 6$ (cost-matched with Opus-medium)	chains (1×16)	98.04	99.50	0.0736	0.0368	0.0750		1.091	1.091
Study 1b	16	Haiku-max vote, $k = 6$ (cost-matched with Opus-medium)	pieces (16×1)	98.85	99.88	0.0605	0.0303	0.0612		0.733	0.733
Study 1b	16	Haiku-max vote, $k = 6$ (cost-matched with Opus-medium)	pooled	98.44		0.0670	0.0335	0.0681		0.894	0.894
Study 1b	16	Opus-medium	chains (1×16)	96.88	100.00	0.0674	0.0337	0.0696	0.0348	—	—
Study 1b	16	Opus-medium	pieces (16×1)	98.75	100.00	0.0825	0.0413	0.0836	0.0418	—	—
Study 1b	16	Opus-medium	pooled	97.81		0.0750	0.0375	0.0766		—	—

Table S26: (continued)

Study	Steps	Setting	Shape	Calls (usage records)	Mean input tokens per answer (see note)	Mean output tokens per answer (see note)	Mean thinking tokens per call (API count, post hoc)	Thinking share of output tokens (%, API count, post hoc)	Thinking share of list cost (%, API count, post hoc)	Thinking share of list cost (%), estimate (biased, see note)	Median thinking tokens, estimate (biased, see note)
Study 1	6	Haiku, thinking off	chains (1×6)	3,200	965.6	1,587.2	0.0	0.00	0.00	33.72	600
Study 1	6	Haiku, thinking off	pieces (6×1)	3,200	1,076.6	1,697.2	0.0	0.00	0.00	34.53	662
Study 1	6	Haiku-low	chains (1×6)	3,200	967.6	1,869.6	1,652.3	88.38	80.09	85.54	1,737
Study 1	6	Haiku-low	pieces (6×1)	3,200	1,078.6	1,836.0	1,644.5	89.57	80.15	84.70	1,728
Study 1	6	Haiku-medium	chains (1×6)	3,200	967.6	2,006.8	1,789.3	89.16	81.32	86.43	1,879
Study 1	6	Haiku-medium	pieces (6×1)	3,200	1,078.6	1,970.3	1,772.4	89.96	81.08	85.44	1,863.5
Study 1	6	Haiku-max	chains (1×6)	240	967.6	8,616.6	8,402.6	97.52	95.37	96.65	8,228
Study 1	6	Haiku-max	pieces (6×1)	240	1,078.6	7,918.1	7,744.3	97.81	95.21	96.29	7,604.5
Study 1	6	Sonnet-low	chains (1×6)	80	967.6	1,222.0	0.0	0.00	0.00	31.38	446
Study 1	6	Sonnet-low	pieces (6×1)	80	1,078.6	1,367.5	0.0	0.00	0.00	30.11	471
Study 1	6	Sonnet-medium	chains (1×6)	80	967.6	1,280.7	61.9	4.84	4.20	34.70	462.5
Study 1	6	Sonnet-medium	pieces (6×1)	80	1,078.6	1,394.0	0.0	0.00	0.00	30.51	487.5
Study 1	6	Opus-low	chains (1×6)	80	967.6	1,633.0	684.2	41.90	37.46	55.06	1,052.5
Study 1	6	Opus-low	pieces (6×1)	80	1,078.6	2,027.5	757.4	37.36	33.76	52.52	1,186
Study 1	6	Opus-medium	chains (1×6)	80	967.6	1,818.5	875.5	48.14	43.51	59.11	1,175.5
Study 1	6	Opus-medium	pieces (6×1)	80	1,078.6	2,190.4	910.3	41.56	37.83	55.67	1,351.5
Study 1	6	Haiku-medium vote, $k = 36$ (cost-matched with Opus-medium)	chains (1×6)	—	34,834.5	72,245.5	—	—	—	—	—
Study 1	6	Haiku-medium vote, $k = 40$ (capped; cost-matched $k = 44$ with Opus-medium)	pieces (6×1)	—	43,143.0	78,813.6	—	—	—	—	—
Study 1	6	Haiku-medium vote, $k = 13$ (cost-matched with Sonnet-medium)	chains (1×6)	—	12,579.1	26,088.6	—	—	—	—	—

Table S26: (continued)

Study	Steps	Setting	Shape	Calls (usage records)	Mean input tokens per answer (see note)	Mean output tokens per answer (see note)	Mean thinking tokens per call (API count, post hoc)	Thinking share of output tokens (%, API count, post hoc)	Thinking share of list cost (%, API count, post hoc)
Study 1b	16	Haiku, thinking off	chains (1×16)	400	1,701.7	2,310.6	0.0	0.00	0.00
Study 1b	16	Haiku, thinking off	pieces (16×1)	400	2,152.6	2,785.7	0.0	0.00	0.00
Study 1b	16	Haiku, thinking off	pooled	800					
Study 1b	16	Haiku-low	chains (1×16)	400	1,703.7	4,340.8	3,822.7	88.06	81.65
Study 1b	16	Haiku-low	pieces (16×1)	400	2,154.6	4,261.8	3,726.9	87.45	79.42
Study 1b	16	Haiku-low	pooled	800					
Study 1b	16	Haiku-medium	chains (1×16)	400	1,703.7	4,777.7	4,261.9	89.20	83.27
Study 1b	16	Haiku-medium	pieces (16×1)	400	2,154.6	4,606.3	4,096.7	88.94	81.33
Study 1b	16	Haiku-medium	pooled	800					
Study 1b	16	Haiku-high	chains (1×16)	400	1,703.7	7,125.1	6,607.2	92.73	88.50
Study 1b	16	Haiku-high	pieces (16×1)	400	2,154.6	5,460.7	4,914.6	90.00	83.42
Study 1b	16	Haiku-high	pooled	800					
Study 1b	16	Haiku-xhigh	chains (1×16)	400	1,703.7	10,785.4	10,270.1	95.22	92.31
Study 1b	16	Haiku-xhigh	pieces (16×1)	400	2,154.6	8,739.9	8,229.5	94.16	89.74
Study 1b	16	Haiku-xhigh	pooled	800					
Study 1b	16	Haiku-max	chains (1×16)	800	1,703.7	24,180.2	23,666.3	97.87	96.51
Study 1b	16	Haiku-max	pieces (16×1)	800	2,154.6	19,741.9	19,287.9	97.70	95.61
Study 1b	16	Haiku-max	pooled	1,600					
Study 1b	16	Haiku-max vote, $k = 6$ (cost-matched with Opus-medium)	chains (1×16)	—			—	—	—
Study 1b	16	Haiku-max vote, $k = 6$ (cost-matched with Opus-medium)	pieces (16×1)	—			—	—	—
Study 1b	16	Haiku-max vote, $k = 6$ (cost-matched with Opus-medium)	pooled	—			—	—	—
Study 1b	16	Opus-medium	chains (1×16)	160	1,703.7	3,030.8	2,176.5	71.81	64.56
Study 1b	16	Opus-medium	pieces (16×1)	160	2,154.6	3,694.8	2,197.4	59.47	53.26
Study 1b	16	Opus-medium	pooled	320					

Table S26: (continued)

Panel 1: Cost-matched k

Study	Steps	Challenger	Comparator	Shape	k	Raw k	Cap	Capped	Samples per comparator answer (comparator cost per answer / one challenger sample)	Spend share, challenger vote / comparator	Label
Study 1	6	Haiku-medium	Opus-medium	chains (1×6)	36	36	40	no	36.58	0.984	pre-registered cost-matched k rule
Study 1	6	Haiku-medium	Opus-medium	pieces (6×1)	40	44	40	yes	44.03	0.909	pre-registered cost-matched k rule
Study 1	6	Haiku-medium	Sonnet-medium	chains (1×6)	13	13	40	no	13.40	0.970	pre-registered cost-matched k rule
Study 1b	16	Haiku-max	Opus-medium	pooled	6	6	10	no	6.71	0.894	pre-registered cost-matched k rule (Study 1b)
Study 1b	16	Haiku-max	Opus-medium	chains (1×16)	5				5.50	0.909	post hoc (free analysis 3): the shape's own cost-matched k
Study 1b	16	Haiku-max	Opus-medium	pieces (16×1)	8				8.18	0.978	post hoc (free analysis 3): the shape's own cost-matched k
Study 1b	6	Haiku-max	Opus-medium	pooled	10		10		10.52	0.950	post hoc (free analysis 2), not a test; the same k as measured beside Test B3
Study 1b	12	Haiku-max	Opus-medium	pooled	7		10		7.42	0.944	post hoc (free analysis 2), not a test

Table S26: (continued)

Panel 2: Haiku-max against Opus-medium by length and shape (output tokens and list cost per answer)

Study	Steps	Shape	Haiku-max mean output tokens per call (post hoc)	Opus-medium mean output tokens per call (Study 1: pre-specified; Study 1b: post hoc)	Output-token ratio, Haiku-max / Opus-medium (post hoc)	Haiku-max list cost per answer ($)	Opus-medium list cost per answer ($)	List-cost ratio, Haiku-max / Opus-medium	Ratio source
Study 1	6	chains (1×6)	8,616.6	1,818.5	4.74	0.00441	0.0402	0.109	free analysis 10 (post hoc)
Study 1	6	pieces (6×1)	7,918.1	2,190.4	3.61	0.00407	0.0481	0.085	free analysis 10 (post hoc)
Study 1b	6	chains (1×6)	8,458.5	1,834.7	4.61	0.00433	0.0406	0.107	free analysis 10 (post hoc)
Study 1b	6	pieces (6×1)	7,891.8	2,164.5	3.65	0.00405	0.0476	0.085	free analysis 10 (post hoc)
Study 1b	6	pooled				0.00419	0.0441	0.095	recorded beside Test B3
Study 1b	12	chains (1×12)	17,418.5	2,439.7	7.14	0.00885	0.0544	0.163	free analysis 10 (post hoc)
Study 1b	12	pieces (12×1)	14,978.0	3,057.3	4.90	0.00766	0.0680	0.113	free analysis 10 (post hoc)
Study 1b	12	pooled				0.00826	0.0612	0.135	computed from the two recorded costs (post hoc arithmetic); free analysis 2 gives the inverse ratio (post hoc; Panel 1)
Study 1b	16	chains (1×16)	24,180.2	3,030.8	7.98	0.0123	0.0674	0.182	free analysis 10 (post hoc)
Study 1b	16	pieces (16×1)	19,741.9	3,694.8	5.34	0.0101	0.0825	0.122	free analysis 10 (post hoc)
Study 1b	16	pooled				0.0112	0.0750	0.149	recorded beside Test B1

Table S26: (continued)

Panel 3: Token structure of the other Study 1b settings (6 steps and 12 steps; post hoc, thinking tokens as the API reports them)

Steps	Setting	Shape	Calls (usage records)	Mean input tokens per call	Mean output tokens per call	Mean thinking tokens per call (API count)	Thinking share of output tokens (%, API count)	Thinking share of list cost (%, API count)
6	Haiku, thinking off	chains (1×6)	200	968.4	1,583.6	0.0	0.00	0.00
6	Haiku, thinking off	pieces (6×1)	200	1,079.3	1,716.4	0.0	0.00	0.00
6	Haiku-low	chains (1×6)	200	970.4	1,891.8	1,676.9	88.64	80.39
6	Haiku-low	pieces (6×1)	200	1,081.3	1,843.6	1,649.7	89.48	80.08
6	Haiku-medium	chains (1×6)	200	970.4	1,989.7	1,774.1	89.16	81.24
6	Haiku-medium	pieces (6×1)	200	1,081.3	2,013.3	1,808.3	89.82	81.11
6	Haiku-high	chains (1×6)	200	970.4	2,359.1	2,144.8	90.91	84.00
6	Haiku-high	pieces (6×1)	200	1,081.3	2,244.8	2,031.9	90.52	82.56
6	Haiku-xhigh	chains (1×6)	200	970.4	3,435.7	3,220.8	93.75	88.73
6	Haiku-xhigh	pieces (6×1)	200	1,081.3	3,041.9	2,850.3	93.70	87.48
6	Haiku-max	chains (1×6)	400	970.4	8,458.5	8,244.4	97.47	95.28
6	Haiku-max	pieces (6×1)	400	1,081.3	7,891.8	7,717.8	97.80	95.19
6	Sonnet-low	chains (1×6)	40	970.4	1,239.5	0.0	0.00	0.00
6	Sonnet-low	pieces (6×1)	40	1,081.3	1,354.5	0.0	0.00	0.00
6	Sonnet-medium	chains (1×6)	40	970.4	1,289.6	127.6	9.89	8.60
6	Sonnet-medium	pieces (6×1)	40	1,081.3	1,371.9	0.0	0.00	0.00
6	Sonnet-high	chains (1×6)	40	970.4	1,414.6	1,013.4	71.64	63.00
6	Sonnet-high	pieces (6×1)	40	1,081.3	1,942.4	1,056.0	54.36	48.92
6	Sonnet-xhigh	chains (1×6)	40	970.4	1,909.2	1,344.6	70.43	63.93
6	Sonnet-xhigh	pieces (6×1)	40	1,081.3	2,248.0	1,286.0	57.21	52.19
6	Sonnet-max	chains (1×6)	40	970.4	4,501.6	3,731.0	82.88	79.46
6	Sonnet-max	pieces (6×1)	40	1,081.3	4,595.8	3,616.7	78.70	75.16
6	Opus-low	chains (1×6)	40	970.4	1,571.8	651.9	41.47	36.92
6	Opus-low	pieces (6×1)	40	1,081.3	1,962.9	713.7	36.36	32.75
6	Opus-medium	chains (1×6)	80	970.4	1,834.7	864.7	47.13	42.62
6	Opus-medium	pieces (6×1)	80	1,081.3	2,164.5	898.5	41.51	37.74
6	Opus-high	chains (1×6)	40	970.4	1,922.5	948.2	49.32	44.80
6	Opus-high	pieces (6×1)	40	1,081.3	2,218.7	960.0	43.27	39.42
6	Opus-xhigh	chains (1×6)	40	970.4	2,112.1	1,158.5	54.85	50.23
6	Opus-xhigh	pieces (6×1)	40	1,081.3	2,468.3	1,177.0	47.69	43.84
6	Opus-max	chains (1×6)	40	970.4	4,283.7	3,376.4	78.82	75.40
6	Opus-max	pieces (6×1)	40	1,081.3	4,182.3	3,001.3	71.76	68.23
12	Haiku-max	chains (1×12)	800	1,407.9	17,418.5	17,024.4	97.74	96.18
12	Haiku-max	pieces (12×1)	800	1,722.8	14,978.0	14,636.0	97.72	95.52
12	Opus-medium	chains (1×12)	80	1,407.9	2,439.7	1,593.7	65.33	58.57
12	Opus-medium	pieces (12×1)	80	1,722.8	3,057.3	1,648.8	53.93	48.47

Table S26: (continued)

Panel 4: Trial round (Study 1b): tokens and list cost by setting

Steps	Setting	Input tokens	Output tokens	Mean output tokens per call	List cost, all calls ($)
6	Haiku-high	20,450	44,136	2,206.8	0.0241
6	Haiku-xhigh	20,450	63,348	3,167.4	0.0337
6	Sonnet-high	20,450	35,584	1,779.2	0.397
6	Sonnet-xhigh	20,450	40,278	2,013.9	0.444
6	Sonnet-max	20,450	88,427	4,421.4	0.925
6	Opus-high	20,450	41,046	2,052.3	0.903
6	Opus-xhigh	20,450	45,317	2,265.9	0.988
6	Opus-max	20,450	86,359	4,318.0	1.81
12	Haiku-max	31,135	346,304	17,315.2	0.176
16	Haiku, thinking off	38,389	51,334	2,566.7	0.0295
16	Haiku-low	38,429	81,977	4,098.9	0.0448
16	Haiku-high	38,429	136,420	6,821.0	0.0721
16	Haiku-xhigh	38,429	200,332	10,016.6	0.104
16	Haiku-max	38,429	387,645	19,382.3	0.198

Price basis: list price per answer. Study 1: the pre-registered price table, dated 2026-10-12. Study 1b: its file gives the unit as US dollars per answer at list price and no price-table date. The Batch cost we paid is given where the files give it and is empty otherwise. Columns marked API count are counted from the usage records by free analysis 10, labelled "post hoc; thinking tokens as the API reports them". They are post hoc and have no interval. The thinking share of list cost is the thinking tokens priced at the model's list output rate, summed over the calls, divided by the summed list cost. Calls are the pre-specified counts: for Study 1 those recorded with its pre-specified thinking estimates, for Study 1b the Study 1b file's final count; free analysis 10 counts the same calls (checked when the table is built). Mean input and output tokens per answer are Study 1's pre-specified token means only for its Haiku-medium, Opus-medium and Sonnet-medium rows and its votes. Every other row's, including every Study 1b row's, are free analysis 10's (post hoc). Where both files hold a value they are equal (checked when the table is built). The two estimate columns are Study 1's pre-specified thinking estimates. Per call, visible tokens are estimated as the number of characters of the visible reply divided by 3, rounded down, and thinking tokens are the output tokens less that estimate, never below zero (the file's label: "estimates (visible tokens = characters / 3)"). They are biased: Haiku, thinking off gets 33.72% / 34.53% of its list cost as estimated thinking (chains / pieces), while the API counts a mean of 0.0 / 0.0 thinking tokens per call for it. The estimate stays as defined; the API columns are the thinking figures to read. Study 1b rows: format compliance is given per shape only (the files give none for a pooled row). A pooled row's calls are the Study 1b file's pooled count; free analysis 10 gives tokens per shape only, so the pooled rows' token cells are empty. Study 1's pre-specified thinking estimates do not apply to Study 1b (dash; the part of the table that gives Study 1b's token columns leaves those two columns out). Cost ratio is the setting's list cost per answer divided by Opus-medium's on the same shape. For Haiku-max per shape it is free analysis 10's ratio (post hoc; checked against the two recorded costs); for Study 1b's pooled Haiku-max row and the pooled Haiku-max vote it is the ratio recorded beside Tests B1 and B2; elsewhere it is computed from the two recorded costs (post hoc arithmetic). Spend share of a vote: the vote's spend over its cost-matched comparator's. For the Study 1 votes it is the pre-registered k file's, in Batch dollars; for the pooled Study 1b vote it is the ratio recorded beside Test B2, at list price. The Study 1b file gives it for the pooled vote only, so its per-shape cells are the same quotient from the two recorded costs (post hoc arithmetic). A capped k is held to the samples drawn, below its cost-matched k; its Setting cell gives both. A vote at k costs k answers; its cost per correct answer and accuracy are those of the vote. A Study 1 vote's Batch cost per answer is its spend in the pre-registered k file (k samples at the pre-specified Batch cost per sample; checked). A Study 1 vote's tokens per answer are those of its k samples together (pre-specified); the Study 1b vote's are not given. Calls and thinking figures are not given for a vote (dash): it reuses its single samples, which are counted in their own rows. The format compliance shown for a vote is that of its single samples. Cells left empty are not given in the result files; a dash means the column does not apply. The vote at the cost-matched k follows the pre-registered k rule of each study. In Panel 1, each shape's own cost-matched k at 16 steps is post hoc (free analysis 3), and so is the cost-matched k at 6 steps and at 12 steps (free analysis 2: its k, the samples it is held to and its samples per comparator answer; the spend share is its vote cost over Opus-medium's, post hoc arithmetic). Neither changes a test, and no trend across lengths is read from them. Panel 2: per shape, the output-token and list-cost ratios are free analysis 10's ratios (post hoc); Study 1's Opus-medium tokens and both studies' costs are the pre-specified recorded values. The pooled rows give the list costs per answer and the ratio recorded beside Tests B3 and B1. At 12 steps the Study 1b file gives no ratio, so it is computed from the two recorded costs (post hoc arithmetic); free analysis 2 gives its inverse (post hoc, Panel 1). No pooled output-token ratio is given. Panel 3: the Study 1b settings at 6 steps and 12 steps, with token figures counted by free analysis 10 (post hoc) and calls from the Study 1b file; the 16-step settings are in the main table. Panel 4: the trial round ran on separate sets; its step counts are those of the limits it set, and its list cost is the total over all of a setting's calls, not per answer. Columns that print one value in every row are left out: every setting made 20 calls in the trial round.

S6.4 Tables and figures

- **Table S25:** every model and effort level in Study 1b's effort sweep: accuracy, format compliance and list-price cost per answer (pre-specified descriptive, no tests).
- **Table S24:** the length sweep in full, under both marking rules. It is built by script from the length-sweep files.
- **Table S26:** where the money goes, in full: costs, tokens and the thinking share, with four panels (point values at list price; the panels' post hoc parts are labelled).
- **Table S27:** Haiku, thinking off, counting only the replies that kept the format (post hoc). At 16 steps, none of the 112 chain replies that kept it was right, and 4 of the 313 piece replies that kept it were (1.28%). At 6 steps the share right was 26.92% on chains and 23.60% on pieces in Study 1 (of 2,615 and 3,186 replies that kept the format), and 31.74% and 28.64% in Study 1b (of 167 and 199). These shares are conditional on keeping the format; they are not whole-answer scores.
- **Table S28:** the wrong-answer repeat share by effort and length (post hoc exploratory; no causal claim). Among pairs of wrong samples on the same puzzle and shape, the share that gave the same wrong answer rose with Haiku's effort. At 16 steps in Study 1b it was 7.46% at medium (86 of 1,153 pairs, from 151 puzzles with at least two wrong samples, counting each shape separately), 12.96% at high (106 of 818), 27.82% at xhigh (101 of 363) and 58.82% at max (60 of 102 pairs, from 31 puzzles). At 6 steps in Study 1b it was 29.41%, 44.78%, 56.0% and 96.15% (25 of 26 pairs, from 7 puzzles), and for Haiku-max at 12 steps 67.89% (74 of 109). Higher effort left fewer pairs of wrong samples, and more of those pairs agreed. This describes the shares; it shows no cause. Study 1's Haiku-max had too few wrong samples for a meaningful share (3 pairs, all on 1 puzzle).
- **Figure S11:** the length curves under the original marking rule, beside Study 1b's Haiku-max and Opus-medium and Pilot 1's 24-step markers (pre-registered descriptive; Pilot 1 exploratory).
- **Figure S12:** where the Haiku dollar goes, plotted from reporting item 19's estimates (visible characters), labelled as estimates with a note on their bias; the main text uses the API's own thinking counts.

Table S27: Haiku, thinking off, separating replies that did not keep the format from wrong answers (Study 1, main run, 80 puzzles, 40 samples per puzzle; Study 1b's 6-step and 16-step sets, 40 and 80 puzzles, 5 samples per puzzle; each row within one study). Study 1's setting is exploratory[‡] and its format columns are pre-specified descriptive; Study 1b's rows are pre-specified descriptive; the recount columns are post hoc[†]. 95% intervals from a cluster bootstrap over puzzles, with exact Clopper–Pearson intervals where marked.

Study	Steps	Shape	Samples	Format ok	Format compliance (%)	Accuracy (%) [95% interval]	Not-ok format statuses	Format not ok (post hoc)	Format ok but wrong (post hoc)	Right (post hoc)	Accuracy on format-ok samples (%; conditional, post hoc)
Study 1	6	1×6	3,200	2,615	81.72	22.00 [16.97, 27.31]	extra line 373; misnumbered 94; missing line 65; non-integer 52; answer mismatch 1	585	1,911	704	26.92
Study 1	6	6×1	3,200	3,186	99.56	23.50 [18.81, 28.56]	non-integer 5; wrong length 5; missing line 4	14	2,434	752	23.60
Study 1b	6	1×6	200	167	83.50	26.50 [17.50, 36.50]	extra line 26; missing line 3; misnumbered 2; non-integer 2	33	114	53	31.74
Study 1b	6	6×1	200	199	99.50	28.50 [19.00, 38.00]	wrong length 1	1	142	57	28.64
Study 1b	16	1×16	400	112	28.00	0.00 [0.00, 4.51] (exact)	missing line 220; extra line 46; misnumbered 16; non-integer 6	288	112	0	0.00
Study 1b	16	16×1	400	313	78.25	1.00 [0.00, 2.25]	missing line 73; no answer 7; non-integer 7	87	309	4	1.28

Post hoc sample counts. Format not ok: the sample's official format status (the marking rule with Deviation 2) is not ok; the kinds are listed under not-ok format statuses. Format ok but wrong: format ok, answer wrong. Right: the official answer is right; every right sample is format ok, because the official rule scores a sample whose format is not ok as wrong. The three counts add up to the samples (both checked when the table is built). The post hoc analysis calls the format-not-ok samples "unparseable", but on chains such a sample can still carry a readable, even right, final answer. Scored on the final answer alone, Study 1's chains reach 27.97% against the official 22.00% (an exploratory setting). On pieces the final answer is the whole answer, so the two scores coincide (checked against Study 1's stamped final-answer scoring); no other chain row has a stamped final-answer accuracy. Accuracy on format-ok samples (the post hoc analysis calls it "accuracy on parseable tries") is the right answers among the format-ok samples divided by the format-ok samples. It is conditional on format and is not a whole-answer accuracy. Labels. In Study 1, Haiku, thinking off, is one of the pre-registered exploratory settings, so its accuracy and 95% interval are exploratory values, read from Study 1's pre-specified table of every setup's results. The format columns (samples, format ok, format compliance and the not-ok statuses) are pre-specified descriptive in both studies (the pre-registered format rules). Study 1b's accuracy and interval are pre-specified descriptive. The columns marked post hoc are a post hoc descriptive recount of the per-sample scores; the recount's samples, format ok, statuses and accuracy on all samples agree with the locked values (checked when the table is built). Study 1 and Study 1b used different puzzle sets and different runs; set side by side, their rows are a reading across studies: our reading, not a test. (exact): the bootstrap interval collapsed to one value; the exact (Clopper–Pearson) 95% interval over puzzles replaces it (post hoc presentation).

Table S28: The wrong-answer repeat share by effort and length (post hoc† exploratory, with no causal claim; Study 1's Haiku-medium rows are pre-specified descriptive): counts and shares, with the puzzles kept and the pairs per row, each row within one study (Study 1, 80 puzzles per shape; Study 1b's 6-step, 12-step and 16-step sets, 40, 80 and 80). A panel gives the Haiku-max vote at 16 steps by Haiku-medium difficulty bin: its binned rows and wrong-sample agreement thirds are post hoc† exploratory, and its unbinned rows are Study 1b's pre-specified descriptive curve. No intervals.

Setting	Study	Steps	Samples per puzzle	Shape	Puzzles kept	Pairs	Agreeing pairs	Repeat share (%)	Label
Haiku-medium	Study 1	6	40	1×6	74	26,026	7,269	27.93	pre-specified descriptive
Haiku-medium	Study 1	6	40	6×1	75	20,234	8,177	40.41	pre-specified descriptive
Haiku-medium	Study 1	6	40	both shapes	—	46,260	15,446	33.39	pre-specified descriptive
Haiku-medium	Study 1b	6	5	1×6	24	132	33	25.00	post hoc exploratory
Haiku-medium	Study 1b	6	5	6×1	22	106	37	34.91	post hoc exploratory
Haiku-medium	Study 1b	6	5	both shapes	—	238	70	29.41	post hoc exploratory
Haiku-medium	Study 1b	16	5	1×16	80	634	42	6.62	post hoc exploratory
Haiku-medium	Study 1b	16	5	16×1	71	519	44	8.48	post hoc exploratory
Haiku-medium	Study 1b	16	5	both shapes	—	1,153	86	7.46	post hoc exploratory
Haiku-high	Study 1b	6	5	1×6	24	100	40	40.00	post hoc exploratory
Haiku-high	Study 1b	6	5	6×1	20	101	50	49.50	post hoc exploratory
Haiku-high	Study 1b	6	5	both shapes	—	201	90	44.78	post hoc exploratory
Haiku-high	Study 1b	16	5	1×16	74	405	42	10.37	post hoc exploratory
Haiku-high	Study 1b	16	5	16×1	71	413	64	15.50	post hoc exploratory
Haiku-high	Study 1b	16	5	both shapes	—	818	106	12.96	post hoc exploratory
Haiku-xhigh	Study 1b	6	5	1×6	9	20	10	50.00	post hoc exploratory
Haiku-xhigh	Study 1b	6	5	6×1	13	55	32	58.18	post hoc exploratory
Haiku-xhigh	Study 1b	6	5	both shapes	—	75	42	56.00	post hoc exploratory
Haiku-xhigh	Study 1b	16	5	1×16	49	184	37	20.11	post hoc exploratory
Haiku-xhigh	Study 1b	16	5	16×1	47	179	64	35.75	post hoc exploratory
Haiku-xhigh	Study 1b	16	5	both shapes	—	363	101	27.82	post hoc exploratory
Haiku-max	Study 1	6	3	1×6	0	0	0	no pairs	post hoc exploratory
Haiku-max	Study 1	6	3	6×1	1	3	3	100.00	post hoc exploratory
Haiku-max	Study 1	6	3	both shapes	—	3	3	100.00	post hoc exploratory
Haiku-max	Study 1b	6	10	1×6	2	7	6	85.71	post hoc exploratory
Haiku-max	Study 1b	6	10	6×1	5	19	19	100.00	post hoc exploratory
Haiku-max	Study 1b	6	10	both shapes	—	26	25	96.15	post hoc exploratory
Haiku-max	Study 1b	12	10	1×12	18	83	56	67.47	post hoc exploratory
Haiku-max	Study 1b	12	10	12×1	11	26	18	69.23	post hoc exploratory
Haiku-max	Study 1b	12	10	both shapes	—	109	74	67.89	post hoc exploratory
Haiku-max	Study 1b	16	10	1×16	16	60	36	60.00	post hoc exploratory
Haiku-max	Study 1b	16	10	16×1	15	42	24	57.14	post hoc exploratory
Haiku-max	Study 1b	16	10	both shapes	—	102	60	58.82	post hoc exploratory

Table S28: (continued)

Panel 1: Haiku-max vote at 16 steps by Haiku-medium difficulty bin

Shape	Haiku-medium accuracy bin (%)	Wrong-sample agreement third	Puzzles	Status	Haiku-max single sample (%)	Vote at k (%)	Vote of all samples drawn (%)	Haiku-max format compliance (%)	Puzzles with too few wrong samples
1×16	[0,10)	whole bin	49	read	91.02	97.21	97.96	—	37
1×16	[0,10)	lowest	4	—	67.50	80.36	75.00	—	—
1×16	[0,10)	middle	4	—	70.00	94.05	100.00	—	—
1×16	[0,10)	highest	4	—	72.50	91.37	100.00	—	—
1×16	[10,20)	—	0	empty	—	—	—	—	—
1×16	[20,30)	whole bin	19	read	94.74	99.56	100.00	—	16
1×16	[20,30)	lowest	1	—	80.00	100.00	100.00	—	—
1×16	[20,30)	middle	1	—	70.00	91.67	100.00	—	—
1×16	[20,30)	highest	1	—	80.00	100.00	100.00	—	—
1×16	[30,40)	—	0	empty	—	—	—	—	—
1×16	[40,50)	whole bin	9	read	95.56	100.00	100.00	—	9
1×16	[50,60)	—	0	empty	—	—	—	—	—
1×16	[60,70)	whole bin	3	read	86.67	96.03	100.00	—	2
1×16	[60,70)	lowest	—	—	—	—	—	—	—
1×16	[60,70)	middle	—	—	—	—	—	—	—
1×16	[60,70)	highest	1	—	60.00	88.10	100.00	—	—
1×16	[70,80)	—	0	empty	—	—	—	—	—
1×16	[80,90)	—	0	empty	—	—	—	—	—
1×16	[90,100]	—	0	empty	—	—	—	—	—
1×16	all puzzles (unbinned)	—	80	—	92.25	98.04	98.75	99.50	—
16×1	[0,10)	whole bin	41	read	91.71	98.29	100.00	—	31
16×1	[0,10)	lowest	3	—	80.00	100.00	100.00	—	—
16×1	[0,10)	middle	3	—	63.33	85.32	100.00	—	—
16×1	[0,10)	highest	4	—	75.00	93.45	100.00	—	—
16×1	[10,20)	—	0	empty	—	—	—	—	—
16×1	[20,30)	whole bin	11	read	90.00	98.05	100.00	—	8
16×1	[20,30)	lowest	1	—	80.00	100.00	100.00	—	—
16×1	[20,30)	middle	1	—	50.00	78.57	100.00	—	—
16×1	[20,30)	highest	1	—	80.00	100.00	100.00	—	—
16×1	[30,40)	—	0	empty	—	—	—	—	—
16×1	[40,50)	whole bin	12	read	94.17	100.00	100.00	—	10
16×1	[40,50)	lowest	—	—	—	—	—	—	—
16×1	[40,50)	middle	1	—	80.00	100.00	100.00	—	—
16×1	[40,50)	highest	1	—	80.00	100.00	100.00	—	—
16×1	[50,60)	—	0	empty	—	—	—	—	—
16×1	[60,70)	whole bin	7	read	97.14	100.00	100.00	—	7
16×1	[70,80)	—	0	empty	—	—	—	—	—
16×1	[80,90)	whole bin	5	read	96.00	100.00	100.00	—	5
16×1	[90,100]	whole bin	4	read	100.00	100.00	100.00	—	4
16×1	all puzzles (unbinned)	—	80	—	93.00	98.85	100.00	99.88	—

Pairs: unordered pairs of wrong samples on the same puzzle and shape. Agreeing pairs give the identical official answer (a sample with no answer or whose official format is not ok never agrees). Repeat share: agreeing pairs divided by pairs, pooled over the puzzles kept (the puzzles with at least two wrong samples). No pairs: no puzzle had two wrong samples, so the share is undefined. Both shapes: the chains' and the pieces' pairs and agreeing pairs added together, and the share is agreeing pairs

over pairs; no count of puzzles is stamped for these rows, so Puzzles kept shows a dash. The Study 1 Haiku-medium rows are Study 1's locked, pre-specified descriptive values; the post hoc recount gives the same counts (checked when the table is built). Every other row is post hoc. The Study 1b Haiku-medium rows go beyond the settings this post hoc analysis was defined to cover (Haiku-max, Haiku-high and Haiku-xhigh in Study 1b, and Haiku-max on Study 1's 3 samples); the same post hoc recount computed them, and they are shown as the same-set medium-effort baseline for the higher efforts. Rows differ in study, puzzle set, length and samples per puzzle; a share from few pairs or few puzzles is fragile. Setting rows side by side, above all across the two studies, is our reading, not a test. Exploratory, no causal claim: the table does not show that repeated wrong answers cause or prevent anything. Bins: each puzzle's Haiku-medium single-sample accuracy at 16 steps over its 5 samples, in 10-point bins with the top bin closed. The bins were fixed, and their occupancy written, before any Haiku-max value was binned; with so few samples some bins cannot hold a puzzle and are empty. A bin with fewer than 3 puzzles is shown but not read; a shape whose largest bin holds more than 80% of its puzzles is uninformative, and then only the unbinned curve is read (rule fixed before computing). Chains (1×16): uninformative: no; the largest bin [0,10) holds 49 of its 80 puzzles (61.25%); the bins marked read are read. Pieces (16×1): uninformative: no; the largest bin [0,10) holds 41 of its 80 puzzles (51.25%); the bins marked read are read. Haiku-max single sample: the mean over the bin's puzzles of each puzzle's single-sample accuracy. Vote at k: Haiku-max's majority vote of k samples as Test B2 scores it, estimated from the samples drawn; $k = 6$ (Test B2's k). Vote of all samples drawn: the vote of all 10 Haiku-max samples per puzzle. A missing sample, or a sample with no answer or whose official format is not ok, votes for nothing, and a tie among several leading answers scores an equal share of one right answer when the right answer is among them (the locked vote curve's rule). Means over puzzles, in %; no intervals. Puzzles with too few wrong samples: the puzzles in the bin with fewer than two wrong Haiku-max samples, so no agreement between wrong answers can be measured. Wrong-sample agreement thirds (a post hoc variant of the within-bin split in Study 1's pre-specified vote lookup chart): within a bin, the puzzles with at least two wrong Haiku-max samples are ordered by how often their wrong samples give the same answer (agreeing pairs over pairs; ties broken by puzzle) and cut into a lowest, middle and highest third so that the higher thirds take any extra puzzles. Study 1's chart gives the extra puzzles to the lower thirds and sends a whole group of tied puzzles to the lower third, so these are not its thirds, and a lower third can be empty here (dashes). A bin whose puzzles all have too few wrong samples has no thirds. The rule above on which bins are read applies to bins, not to thirds: each third rests on the few puzzles in its puzzles column. Haiku-max format compliance: the locked share of the 16-step Haiku-max samples whose official format status is ok, for all puzzles of the shape (pre-specified descriptive); it is not stamped by bin. The unbinned rows are Study 1b's locked single-sample accuracy and vote curve (pre-specified descriptive); the post hoc recount agrees with them (checked when the table is built). Their vote at k is Test B2's challenger at that shape: Test B2 is confirmatory and pools the two shapes, so this per-shape value is a descriptive breakdown, not a test. The binned rows and the thirds are post hoc exploratory, with no causal claim.

S7 The top-up run, further results

S7.1 Further detail on the top-up run

These paragraphs give details that the top-up results in Section 4 of the paper leave out.

The Haiku team. The 95% intervals on its accuracy were 41.25% to 62.50% on chains and 62.50% to 82.50% on pieces, and it cost $0.00623 and $0.00649 per puzzle. Against the main run's step voting on chains it differed by −1.25 points (95% interval −15.00 to +12.50). Its planner was told to write one instruction per listed step, and a plan that did not cover every step counted as wrong (no plan failed).

The aggregator changed 42 of 80 final answers on chains and 34 on pieces. By our arithmetic, the helpers' own result would have been right on about 50 of 80 chains (62.50%) and 49 of 80 pieces (61.25%), so on pieces about half of the team's 21.25-point lead over step voting came with the aggregator's net changes (helpers alone about 61.25% against step voting's 51.25%). No control setting isolates the aggregator, so these counts say nothing about what it caused.

Step voting on pieces. The 95% interval on its accuracy was 40.00% to 62.50%; it scored +0.25 points above one Haiku-medium sample in the main run on point estimates (descriptive, no interval; our reading, not a test), and its cost was 7.28% of Opus-medium's list cost per puzzle on pieces.

The Haiku picker. The 95% intervals on its accuracy were 52.50% to 72.50% on chains and 57.50% to 78.75% on pieces. Against a perfect picker it scored −25.00 points (−36.25 to −16.25) on chains and −18.75 (−28.75 to −11.25) on pieces, recovering about 43.06–43.66% of the gap between a single sample and the best of the 8 (post hoc).

The picker chose sample 1 in 81 of its 160 choices (about half), sample 2 in 33, and samples 4–8 together in 32. The answers were always shown in the same order, and position effects were not studied, so position and judgement cannot be told apart here.

Even a perfect picker over these 8 medium-effort answers, one that always found a right answer when any of them was right, would have scored 87.50% on chains and 86.25% on pieces, 8.75 and 13.75 points below Opus-medium's 96.25% and 100.00% (post hoc arithmetic within one run; no interval for the gap). The team's aggregator is a second, indirect case of a cheap model second-guessing cheap answers, with the mixed counts above. Neither is a study of judging, and neither supports a general claim about cheap judges.

S7.2 The Haiku team's run counts (exploratory; counts only)

- **Failures:** failed plans, 0 on either shape; failed tasks, 1 (one chain puzzle); unfinished puzzles, 0.
- **Draws:** helpers drew 1,035 times on chains and 1,044 on pieces, for 480 tasks each. That is 2.16–2.18 draws per task.
- **Step voting, for comparison:** 2.26 per step in the main run (1,086 draws over 480 steps), and 2.21 on pieces in the top-up (1,061 over 480).

S7.3 Step voting on pieces at the draw cap

In the top-up run, 4 of the 480 piece positions reached the 9-draw cap. Of these, 3 were settled by a unique plurality, and 1 tie was counted as wrong. In a post hoc count of the main run, no step reached the cap of 9 draws: 435 of step voting's 480 steps were settled at the 2-draw minimum, in 1,086 draws in all.

S7.4 Tables

- **Table S29:** the top-up counts (team tasks, draws and aggregator changes per shape; picker positions; completeness).

Table S29: The top-up run's counts (exploratory‡; 6 steps, 80 puzzles per shape): tasks, draws and the aggregator's changes. Two panels give the picker's chosen sample positions and each setting's completeness in calls.

Count	Chains (1×6)	Pieces (6×1)
Puzzles with a failed task	1	0
Failed tasks	1	0
Helper draws	1,035	1,044
Aggregator changed the helpers' answer	42	34
Helpers' results right, final answer wrong	15	9
Helpers' results wrong, final answer right	7	18
Changed: aggregator unreadable	2	0
Changed: helpers had no result	1	0

Panel 1: Picker positions (counts of puzzle-shape pairs)

Picked	Chains (1×6)	Pieces (6×1)	Pooled
First sample	37	44	81
Second sample	13	20	33
Third sample	7	7	14
Fourth sample	3	1	4
Fifth sample	4	5	9
Sixth sample	2	2	4
Seventh sample	10	1	11
Eighth sample	4	0	4
No readable pick	0	0	0
Total puzzle-shape pairs	80	80	160

Panel 2: Completeness by setting (calls)

Setting	Designed calls	Final calls	Puzzle-shape pairs	Pairs unfinished
Step voting on pieces	1,061	1,061	80	0
Haiku team	2,399	2,399	160	0
Haiku picker over 8 samples	160	160	—	—
Drift check	200	200	—	—

At each shape: 80 puzzles, 0 of them with a failed plan, 0 unfinished, and 80 with a final aggregator. Counts only; they are exploratory and carry no test. Aggregator changes are counted over puzzles with a final aggregator: a change is a puzzle whose final answer (the aggregator's graded whole answer) is not the helpers' results taken in order. "Helpers' results" means the helpers' answers in order, at either shape; it is not the chain shape. A setting is "not completed" if a designed call has no final record. Settings not completed: 0. Adaptive designs (their number of calls is not fixed in advance): step voting on pieces and the Haiku team. Their designed calls are the final calls plus the calls the runner's own logic would still have made when the analysis ran: a lower bound. Picker positions are descriptive: how often each of samples 1–8 is picked; position effects are not studied. For every setting listed, completed: yes; missing calls: 0. A dash: the setting's completeness record gives no value for that column.

S8 Discussion and limits, further detail

S8.1 Further detail on the Discussion

These paragraphs give details that the Discussion leaves out; Section 5 of the paper states the main message, the practical rules, the open questions and the limits.

Our conditions, in detail. The list prices were checked on 2026-10-12 and re-checked on 2026-10-13. The API fixed the temperature at 1, and Haiku's long-prompt price starts above 100,000 input tokens. Because every sample was drawn at the API's fixed temperature, how alike Haiku's samples are is a property of the model, and we do not suggest that another temperature would lift the medium-effort plateau.

Cost shares of the multi-call designs. Apart from the plain votes, no multi-call design was run at matched cost. As a share of Opus-medium's list cost per puzzle, step voting cost 8.85% on chains and 7.28% on pieces, the Haiku team 15.48% and 13.48%, and the picker 25.67% and 21.09% counting the answers it read.

Future work, as a list.

None of the following was run:

- **Teamwork on top of max effort** (step voting, a team, a picker), and decomposition at Opus-medium's budget and on tasks whose steps are not given.
- **A judge study**, including judging max-effort answers and a picker that sees the answers in shuffled order.
- **A team ablation**, with an aggregator that only copies, or a plain Haiku-medium retry given the helpers' results.
- **A second task family** (the dropped code-tracing family) and the real-world mathematics check.
- **Fable 5.1** as a setting.
- **Runs at 24 steps or more**, to find where Opus-medium, Haiku-max or Sonnet-high start to fail.
- **Sonnet at its default effort**, and Opus-low, at 12–16 steps.
- **A max-effort setting with a capped token limit**, to separate effort from its larger limit.
- **More Opus-high and Opus-xhigh samples**, to see whether their lower 6-step scores recur.
- **An API Haiku-max rerun** of the effort diagnostic's prompts.
- **New length-sweep sets** with more puzzles.
- **Latency** at max effort.

S8.2 Tuning bias, both ways (Table S30)

We developed the rule kinds on Haiku's slips, which works against the Haiku vote. Other choices work for it: the 10-point margin, refusals and truncations counted as wrong, and a 25–50% difficulty band where voting helps most. In the plan's words, "the net direction is not known in advance". Two pre-registered checks bear on it.

- **The fairness check** (exploratory, flag-based) gave a targeting index of 0.82 (90% interval 0.43 to 1.37; 1 means no extra targeting), and its flag did not fire. But Opus slipped once in 948 placed steps, so the comparison side is in effect Sonnet, and the house-rule index could not be assessed. The interval cannot exclude the tuned kinds targeting Haiku up to about 37.19% more than the comparison models.
- **Flag R** did not fire. When two Haiku-medium samples slipped at the same step from the same input, they wrote identical balances in 66.57% of pairs on the tuned kinds against 89.23% on general steps. That is the reverse of the worry that tuning made Haiku's wrong answers more alike.

The checks bound the tuning bias; they do not remove it. Table S30 lists each design choice, which side it favours, and how we bounded or measured it.

Table S30: Tuning bias: each design choice, the side it favours and how we bounded or measured it (disclosure; Study 1, main run).

Choice	Favours	Bounded or measured by	Outcome
Rule kinds tuned on Haiku's slips	Opus and Sonnet (against the vote)	The fairness check (the targeting index, the house-rule index, repeat share)	Flag K did not fire (the targeting index did not trigger the flag; the house-rule index was not assessable: Opus and Sonnet together had 0 house-rule slips, fewer than the 3 it needs); Flag R did not fire (Table S15)
The 10-point margin	The vote	The pre-registered decision rule; the error-rate ratios; a 5-point secondary view (a power simulation)	the 5-point view of Tests 1–5 (Table S16)
Refusals and truncations count as wrong	The vote	The headline rule (a result shown no worse must hold with them left out)	the refusals-out and truncations-out views (Table S16)
The 25–50% Haiku band	The vote	Stated as the regime studied; the scope sentence	Haiku-medium single sample, pooled: 47.36% (Table S31, Study 1 pooled)
6×1 near the ceiling	Both sides, so less informative	Per-shape medians reported; a low-information flag if Haiku pieces score above 60%	low-information flag did not fire; Haiku-medium pieces: 51.00% (Table S15)
Absorption in 1×6	Designs that slip more (the vote)	The official 1×6 score is all step lines right, so an absorbed slip still counts as wrong; the puzzle generator's absorption caps (single units, each transfer size and the house-rule slips on the reference chain) bound the final-answer view	the final-answer view (Table S16)

The first three columns are the bias table of Study 1's pre-registration, relabelled for 6 steps (Deviation 1): 24×1 → 6×1, 1×24 → 1×6; decision numbers and a session's name are removed; a range written "a to b%" reads "a–b%"; the pre-registration's short codes for its headline rule, its absorption caps, its targeting index and its house-rule index are written out in words; the pre-registration's one-word shorthand for a result shown no worse, which the paper does not use, reads "a result shown no worse"; and its older words are the paper's: "arms" reads "sides" or "designs", and "the three-way reading" reads "the pre-registered decision rule". The last column points to where each check's outcome is reported. The net direction was not known in advance.

S8.3 The two 6-step puzzle sets (Table S31)

The two 6-step sets, from the same generator, gave the weak Haiku settings scores 4.75–9.89 points apart (Table S31).

Haiku-max and Opus-medium barely moved. The intervals in each study's own tables overlap, but for weak settings the spread between puzzle sets is as large as some of the effects one might read across studies, so every length comparison is taken from one study's sets (our reading, not a test).

S8.4 Post hoc tables

- **Table S32, re-pricing (post hoc sensitivity):** re-priced at comparator-to-Haiku price ratios of 10, 20, 40 and 80 times (the list ratios are 40 for Opus and 20 for Sonnet), the cost-matched Haiku-medium votes of Tests 1–3 stayed below their comparators by 21.79 to 42.63 points at every ratio, with every paired 95% interval below zero. For B2's Haiku-max vote at 16 steps, the cost-matched size went from 1 sample at 10 times, which is Test B1's comparison (−5.19 points, 95% interval −7.75 to −2.69), to 10 samples, all that were drawn, at 80 times, which is Table S23's 10-sample row (+1.56, −0.63 to +3.44). At the list ratios the table prints the locked tests.

Table S31: The two 6-step puzzle sets compared: Study 1's main set (80 puzzles) against Study 1b's 6-step set (40), as point differences only (across studies; descriptive, our reading, not a test).

Setting	Study 1 chain (%)	Study 1b chain (%)	Difference, chain (points)	Study 1 pieces (%)	Study 1b pieces (%)	Difference, pieces (points)	Study 1 pooled (%)	Study 1b pooled (%)	Difference, pooled (points)
Haiku, thinking off	22.00	26.50	+4.50	23.50	28.50	+5.00	22.75	27.50	+4.75
Haiku-low	39.22	52.50	+13.28	46.50	53.00	+6.50	42.86	52.75	+9.89
Haiku-medium	43.72	52.00	+8.28	51.00	57.00	+6.00	47.36	54.50	+7.14
Haiku-max	97.50	97.25	−0.25	95.00	94.25	−0.75	96.25	95.75	−0.50
Sonnet-low	71.25	77.50	+6.25	76.25	82.50	+6.25	73.75	80.00	+6.25
Sonnet-medium	80.00	77.50	−2.50	75.00	82.50	+7.50	77.50	80.00	+2.50
Opus-low	97.50	97.50	0.00	100.00	100.00	0.00	98.75	98.75	0.00
Opus-medium	96.25	98.75	+2.50	100.00	100.00	0.00	98.13	99.38	+1.25

Panel 1: Format compliance and samples per puzzle

Setting	Format compliance, chain, Study 1 (%)	Format compliance, chain, Study 1b (%)	Format compliance, pieces, Study 1 (%)	Format compliance, pieces, Study 1b (%)	Samples per puzzle and shape, Study 1	Samples per puzzle and shape, Study 1b
Haiku, thinking off	81.72	83.50	99.56	99.50	40	5
Haiku-low	98.13	98.50	99.66	100.00	40	5
Haiku-medium	98.59	99.00	99.72	99.00	40	5
Haiku-max	100.00	100.00	99.58	100.00	3	10
Sonnet-low	100.00	100.00	100.00	100.00	1	1
Sonnet-medium	97.50	97.50	100.00	100.00	1	1
Opus-low	98.75	97.50	100.00	100.00	1	1
Opus-medium	97.50	100.00	100.00	100.00	1	2

A reading across studies: our reading, not a test. The two studies used different puzzle sets, different runs and different puzzles, and nothing is paired. The difference is Study 1b's point minus Study 1's point, in points; there is no interval and no p-value. Runs, oldest first: Study 1's main run on 2026-10-12; Study 1b's run on 2026-10-13; the top-up run on 2026-10-13, after Study 1b. The drift check between runs: none was run for Study 1b itself (the Study 1b file records none). The top-up run's drift check, which the top-up run's specification (frozen with Lock 4) defines as Haiku-medium with the main run's configuration, re-ran Haiku-medium on Study 1's main set and compared it with Study 1's answers; its flag: not fired. It covers Haiku-medium only, not the other settings of this table. Study 1 used 80 puzzles on its main set. Study 1b's set at 6 steps (the Test B3 set) has 40 puzzles. Study 1 points are from its pre-specified descriptive table; Study 1b points are from the effort sweep. Study 1's file records the pooled point of Haiku-medium, Sonnet-medium and Opus-medium (its shape points equal the table's; checked); for the other settings Study 1's pooled point is the mean of its two shape means, which is the same pooling (each puzzle's mean over its two shapes). Samples per puzzle and shape differ between the studies and between settings (Panel 1's last two columns). Study 1's is its recorded samples divided by its recorded puzzles; Study 1b's is recorded in its file. Format compliance (Panel 1): share of answered samples whose format was ok, per shape and study.

Table S32: Re-pricing (post hoc† sensitivity, beside the pre-specified cached view): the cost-matched vote size and the vote's paired gap for Tests 1–3 (Study 1) and B2 (Study 1b) at other comparator-to-Haiku price ratios, with the list prices re-scaled per row; a row that is the same comparison as a printed result takes that result and its label (a test as run; Test B1 for Test B2's one-sample vote; free analysis 3's rows for B2's votes at other sizes), and the other rows are post hoc† (free analysis 12). The panels give the break-even price ratio for Haiku-medium (pre-specified) and for Haiku-max at each length, and the cached view.

Test	Price ratio per token (comparator / Haiku)	Samples per comparator answer	Cost-matched k	Held	Vote accuracy at k (%)	Comparator accuracy (%)	Gap d (points)	95% interval	Source of d and interval
Test 1	10	11.01	11		64.44	100.00	−35.56	[−44.85, −27.57]	post hoc (free analysis 12)
Test 1	20	22.01	22		66.06	100.00	−33.94	[−43.92, −24.85]	post hoc (free analysis 12)
Test 1	40	44.03	40	held at 40	66.88	100.00	−33.13	[−43.75, −23.75]	confirmatory (Test 1)
Test 1	80	88.05	40	held at 40	66.88	100.00	−33.13	[−43.75, −23.75]	confirmatory (Test 1)
Test 2	10	9.14	9		53.62	96.25	−42.63	[−51.82, −33.00]	post hoc (free analysis 12)
Test 2	20	18.29	18		56.71	96.25	−39.54	[−49.82, −29.82]	post hoc (free analysis 12)
Test 2	40	36.58	36		58.61	96.25	−37.64	[−48.82, −26.96]	confirmatory (Test 2)
Test 2	80	73.15	40	held at 40	57.50	96.25	−38.75	[−50.00, −28.13]	post hoc (free analysis 12)
Test 3	10	6.70	6		51.59	80.00	−28.41	[−39.28, −16.13]	post hoc (free analysis 12)
Test 3	20	13.40	13		55.30	80.00	−24.70	[−36.12, −12.20]	confirmatory (Test 3)
Test 3	40	26.80	26		58.21	80.00	−21.79	[−34.31, −9.09]	post hoc (free analysis 12)
Test 3	80	53.60	40	held at 40	57.50	80.00	−22.50	[−35.00, −8.75]	post hoc (free analysis 12)
Test B2	10	1.68	1		92.63	97.81	−5.19	[−7.75, −2.69]	confirmatory (Test B1: the same comparison as a one-sample vote)
Test B2	20	3.35	3		97.03	97.81	−0.79	[−2.83, 1.21]	post hoc (free analysis 3 at this k)
Test B2	40	6.71	6		98.44	97.81	+0.63	[−1.15, 2.44]	confirmatory (Test B2)
Test B2	80	13.42	10	held at 10	99.38	97.81	+1.56	[−0.63, 3.44]	post hoc (free analysis 3 at this k)

Table S32: (continued)

Panel 1: Break-even price ratio

Vote vs comparator	Shape	Smallest k reaching the comparator's accuracy	Samples per comparator answer	Break-even price ratio	95% interval	Replicates with no break-even (%)	Label
Haiku-max vs Opus-medium, 6 steps	1×6	3	9.38	12.79	[4.12, 39.73]	0.00	post hoc (free analysis 12)
Haiku-max vs Opus-medium, 6 steps	6×1	no break-even within 10 samples	11.74	no break-even within 10 samples	[16.86, unbounded]	65.20	post hoc (free analysis 12)
Haiku-max vs Opus-medium, 6 steps	pooled	9	10.52	34.21	[11.20, unbounded]	33.75	post hoc (free analysis 12)
Haiku-max vs Opus-medium, 12 steps	1×12	5	6.15	32.52	[18.06, unbounded]	19.30	post hoc (free analysis 12)
Haiku-max vs Opus-medium, 12 steps	12×1	5	8.88	22.52	[12.67, unbounded]	26.85	post hoc (free analysis 12)
Haiku-max vs Opus-medium, 12 steps	pooled	5	7.42	26.97	[15.72, unbounded]	15.55	post hoc (free analysis 12)
Haiku-max vs Opus-medium, 16 steps	1×16	4	5.50	29.09	[20.56, unbounded]	10.55	post hoc (free analysis 12)
Haiku-max vs Opus-medium, 16 steps	16×1	6	8.18	29.34	[14.50, 49.57]	0.00	post hoc (free analysis 12)
Haiku-max vs Opus-medium, 16 steps	pooled	5	6.71	29.81	[17.45, unbounded]	5.25	post hoc (free analysis 12)
Haiku-medium vs Opus-medium, 6 steps	1×6	no break-even within 40 samples	36.58	no break-even within 40 samples	[unbounded, unbounded]	100.00	pre-specified (item 9d, Study 1)
Haiku-medium vs Opus-medium, 6 steps	6×1	no break-even within 40 samples	44.03	no break-even within 40 samples	[unbounded, unbounded]	100.00	pre-specified (item 9d, Study 1)
Haiku-medium vs Sonnet-medium, 6 steps	1×6	no break-even within 40 samples	13.40	no break-even within 40 samples	[unbounded, unbounded]	99.80	pre-specified (item 9d, Study 1)
Haiku-medium vs Sonnet-medium, 6 steps	6×1	no break-even within 40 samples	14.73	no break-even within 40 samples	[12.13, unbounded]	87.30	pre-specified (item 9d, Study 1)

Table S32: (continued)

Panel 2: Cached view (Study 1, offline re-pricing)

Test	Cached k	d (points)	95% interval
Test 1	40	−33.13	[−43.75, −23.75]
Test 2	39	−38.21	[−49.39, −27.04]
Test 3	14	−24.39	[−35.90, −11.80]

The stamped label of free analysis 12 reads "post hoc sensitivity: how far the equal-money readings rest on one price list; no reading"; its "equal-money" is cost-matched. Samples per comparator answer: the comparator's cost per answer divided by the cost of one Haiku sample (at the price ratio in the row; in Panel 1, at list price). The price ratio is per token, comparator over Haiku, the same for input and output (the pinned table's is 40 for Opus and 20 for Sonnet); token counts are held as measured. At the list ratio each row is the locked test and prints its d and interval. One comparison, one interval: any other re-priced row that is the same comparison as a printed result (the same settings, k and puzzle set) takes that result and its label, named in the last column. A Test 1 row held at the locked test's k is Test 1 as run; Test B2's one-sample vote is Test B1 (one Haiku-max sample against Opus-medium on the same 16-step set); its votes at other k are free analysis 3's rows at that k. The remaining rows are post hoc (free analysis 12), with no reading. Held: the cost-matched k at that price exceeds the samples drawn per puzzle and shape, so k is held at that number. Accuracies in the main panel are given to two decimals, as the tests' own tables give them, so a row that is a locked test prints the same numbers; d always has two. d is the difference of the unrounded accuracies, so it can differ in its last digit from the difference of the two printed accuracies. Under the original pre-registered marking rule, which Deviation 2 kept as a sensitivity check, Test 3 would read inconclusive instead of worse by more than the margin ($d = +0.30$ points, 95% interval −13.58 to +14.95). A break-even interval end is "unbounded" when too many replicates never reach the comparator's accuracy within the samples drawn; the share of such replicates is shown. Break-even intervals (Panel 1): the Haiku-max rows' cluster bootstrap uses 2,000 draws (free analysis 12); the Haiku-medium rows are Study 1's pre-specified break-even (item 9d), whose interval comes from the locked analysis, and the locked file does not stamp its number of draws, so none is given here. Cached view: the reading from the pre-registered decision rule, applied to the cached view's interval, is worse by more than the margin in every test (Tests 1–3). Cached view: Test 1's cached k is held at the cap of 40 samples drawn; the cached cost of that many samples is still below the comparator's list cost per answer.

- **Table S33, cost for a target accuracy:** the cheapest measured setup whose point estimate, and whose lower interval end, clears each target. Every entry is labelled by study, puzzle set and evidence label, and nothing is fitted. It is derived from the stamped per-setting points and needs no further analysis.

S8.5 Reproducibility (Table S34)

Table S34 lists the analysis files with their SHA-256 hashes and code commits, the run commits, the raw files' SHA-256 hashes and the tool versions. It is built by script from the spend log and the lock manifest. Table S7 lists the stamped fields of the result files that the paper does not repeat as written, each beside the wording or key the paper uses instead.

Table S33: Cost for a target accuracy: the cheapest measured setup whose point estimate, and whose lower interval end, clears each target (across studies; our reading, not a test; post hoc†; every row labelled by study and evidence label; no fitted frontier; US dollars per answer at list price).

Cleared by	Length	Shape	Target (%)	Cheapest setting that clears	Study and set	Label	Accuracy (%)	Format compliance (%)	95% lower end (%)	List cost per answer ($)
point estimate	6 steps	1×6	90	Haiku-max, one sample	Study 1b, 6-step set	Study 1b, pre-specified descriptive (effort sweep)	97.25	100.00	94.75	0.00433
			95	Haiku-max, one sample	Study 1b, 6-step set	Study 1b, pre-specified descriptive (effort sweep)	97.25	100.00	94.75	0.00433
			98	Haiku-max vote, $k = 3$	Study 1b, 6-step set	post hoc (free analysis 2)	99.06	100.00	97.28	0.0130
point estimate	6 steps	6×1	90	Haiku-max, one sample	Study 1b, 6-step set	Study 1b, pre-specified descriptive (effort sweep)	94.25	100.00	90.50	0.00405
			95	Haiku-max, one sample	Study 1, main set	Study 1, exploratory setting	95.00	99.58	91.25	0.00407
			98	Haiku-max vote, $k = 3$	Study 1, main set	Study 1, exploratory setting	98.75	99.58	96.25	0.0122
point estimate	16 steps	1×16	90	Haiku-max, one sample	Study 1b, 16-step set	Study 1b, pre-specified descriptive (effort sweep)	92.25	99.50	89.25	0.0123
			95	Haiku-max vote, $k = 3$	Study 1b, 16-step set	Study 1b, pre-specified descriptive (vote curve)	96.62	99.50	94.15	0.0368
			98	Haiku-max vote, $k = 6$	Study 1b, 16-step set	Study 1b, pre-specified descriptive (vote curve, at B2's $k = 6$)	98.04	99.50	95.77	0.0736
point estimate	16 steps	16×1	90	Haiku-max, one sample	Study 1b, 16-step set	Study 1b, pre-specified descriptive (effort sweep)	93.00	99.88	90.38	0.0101
			95	Haiku-max vote, $k = 3$	Study 1b, 16-step set	Study 1b, pre-specified descriptive (vote curve)	97.43	99.88	95.50	0.0303
			98	Haiku-max vote, $k = 5$	Study 1b, 16-step set	Study 1b, pre-specified descriptive (vote curve)	98.71	99.88	97.25	0.0504
95% lower end	6 steps	1×6	90	Haiku-max, one sample	Study 1b, 6-step set	Study 1b, pre-specified descriptive (effort sweep)	97.25	100.00	94.75	0.00433
			95	Haiku-max, one sample	Study 1, main set	Study 1, exploratory setting	97.50	100.00	95.42	0.00441
			98	Haiku-max vote, $k = 5$	Study 1b, 6-step set	post hoc (free analysis 2)	99.35	100.00	98.04	0.0216
95% lower end	6 steps	6×1	90	Haiku-max, one sample	Study 1b, 6-step set	Study 1b, pre-specified descriptive (effort sweep)	94.25	100.00	90.50	0.00405
			95	Haiku-max vote, $k = 3$	Study 1, main set	Study 1, exploratory setting	98.75	99.58	96.25	0.0122
			98	no measured setting clears			—	—	—	—

Table S33: (continued)

Cleared by	Length	Shape	Target (%)	Cheapest setting that clears	Study and set	Label	Accuracy (%)	Format compliance (%)	95% lower end (%)	List cost per answer ($)
95% lower end	16 steps	1×16	90	Haiku-max vote, $k = 3$	Study 1b, 16-step set	Study 1b, pre-specified descriptive (vote curve)	96.62	99.50	94.15	0.0368
			95	Haiku-max vote, $k = 5$	Study 1b, 16-step set	Study 1b, pre-specified descriptive (vote curve)	97.89	99.50	95.67	0.0613
			98	no measured setting clears			—	—	—	—
95% lower end	16 steps	16×1	90	Haiku-max, one sample	Study 1b, 16-step set	Study 1b, pre-specified descriptive (effort sweep)	93.00	99.88	90.38	0.0101
			95	Haiku-max vote, $k = 3$	Study 1b, 16-step set	Study 1b, pre-specified descriptive (vote curve)	97.43	99.88	95.50	0.0303
			98	Haiku-max vote, $k = 7$	Study 1b, 16-step set	Study 1b, pre-specified descriptive (vote curve)	99.17	99.88	98.13	0.0706

Post hoc reading on point estimates and stamped 95% intervals: nothing is fitted and no frontier is drawn. Settings from different studies, puzzle sets and runs are compared only as our reading, not a test; each row names its study, set and label. Two blocks, named in the first column. In the first, a setting clears a target when its accuracy (the point estimate) is at or above the target; in the second, when the lower end of its 95% interval is at or above the target. Each row gives the cheapest setting that clears the target at that length and shape, or says that no measured setting clears it (a dash stands for each number it then lacks). Where rounding would put a value on the target, one more decimal is printed. Candidates (every measured setting with a stamped accuracy, 95% interval and list cost per answer at that length and shape): at 6 steps, Study 1's single-call settings and step voting (chains only), its Haiku-medium vote line ($k = 2$ to 40) and Haiku-max's vote of $k = 3$ samples; Study 1b's effort-sweep settings on the 6-step set and Haiku-max votes of $k = 2$ to 10 (free analysis 2, post hoc); the top-up run's Haiku team, its picker (whose cost includes its samples) and its step voting (pieces only). At 16 steps, Study 1b's effort-sweep settings and its Haiku-max vote curve ($k = 2$ to 10). Every vote row is estimated from the samples made per puzzle, not from separate runs of k calls: Study 1b's Haiku-max votes from 10 samples per puzzle on the 6-step set and 10 on the 16-step set (the expected accuracy of a vote of k of them, a tie credited as a share split evenly among the tied answers); Study 1's Haiku-max vote from its 3 samples and its Haiku-medium vote line from 40. A vote's list cost per answer is k times one sample's. Lower ends are those of the stamped 95% intervals over puzzles, as each setting's file stamps them: Clopper–Pearson (exact) for the Opus and Sonnet settings with one sample per puzzle; cluster bootstrap for every other setting (the Haiku settings, Study 1b's Opus-medium, step voting, the top-up designs and every vote); Clopper–Pearson from free analysis 4 (post hoc) where a bootstrap interval collapsed to one value. Every lower end printed in this table is that of a cluster bootstrap. Format compliance is the share of answered samples in the required format; for a vote, that of the samples it is drawn from. Where a candidate's bootstrap interval collapsed to one value, its lower end is that of the exact (Clopper–Pearson) 95% interval from free analysis 4 (post hoc) over the same puzzles, so no candidate is left out of the lower-end ranking. That holds for 6 candidates: Haiku-max vote, $k = 3$, on chains (Study 1, main set; Study 1, exploratory setting): accuracy 100.00%, lower end 95.49% (exact, post hoc; 80 puzzles); Haiku-max vote, $k = 9$, on chains (Study 1b, 6-step set; post hoc (free analysis 2)): accuracy 100.00%, lower end 91.19% (exact, post hoc; 40 puzzles); Haiku-max vote, $k = 10$, on chains (Study 1b, 6-step set; post hoc (free analysis 2)): accuracy 100.00%, lower end 91.19% (exact, post hoc; 40 puzzles); Opus-medium, one sample, on pieces (Study 1b, 6-step set; Study 1b, pre-specified descriptive (effort sweep)): accuracy 100.00%, lower end 91.19% (exact, post hoc; 40 puzzles); Haiku, thinking off, one sample, on chains (Study 1b, 16-step set; Study 1b, pre-specified descriptive (effort sweep)): accuracy 0.00%, lower end 0.00% (exact, post hoc; 80 puzzles); Haiku-max vote, $k = 10$, on pieces (Study 1b, 16-step set; Study 1b, pre-specified descriptive (vote curve)): accuracy 100.00%, lower end 95.49% (exact, post hoc; 80 puzzles).

Table S34: Reproducibility: the released analysis files (Study 1, Study 1b and the top-up run), the two k.json files, Study 1's keep-list analyses and reporting definitions, and the length sweep's two curves, each with its SHA-256 hash and, where one is recorded, its code commit; a panel gives each paid run's commit and the SHA-256 of each raw file its cost record lists (disclosure). The runs' raw-reply files (calls.jsonl) and the runner's process files (runner.log, window.json) are not released, so their hashes cannot be checked from the release; those of every other listed file can.

File	Path	Commit	SHA-256	Code commit	Run file recorded	Recorded SHA-256
Study 1 analysis	data/main/main-l14n6-1/analysis.json	8be72ec	43d18eed62604aab 9006cae76a6ee044 45170866486d1424 6fb933646823cfef	c273eb9	compact_table.csv	984da2841cd56f00 26bb0e750968d598 45d8a842af70b2af f04dcd82269df6b4
Study 1 k.json	data/main/main-l14n6-1/k.json	da61c76	b0f8b05f729b5e6c ee6deaa2384c9866 46217adf72a96772 d5096bd2d6e3234b	c273eb9	compact_table.csv	984da2841cd56f00 26bb0e750968d598 45d8a842af70b2af f04dcd82269df6b4
Study 1 keep-list analyses	data/main/main-l14n6-1/groupc.json	2830340	e1cbd4057ed94e67 35d7a3ca079213d8 ea6eba1240b6f5d9 0244545bb3f773a9	90cfb42	calls.jsonl	0d04ce3870704017 d79a7cd2f3603a7f 124379f241f13fb3 b3ce416bdd02c123
Study 1 reporting definitions	research/reporting-definitions-main.md	93334b8	bddab72761b4111f 6d4ed439f13ddbd6 3427bc14a9c61b58 d0ceadb32f18bf81	—	—	—
Study 1b analysis	data/main/study1b-1/analysis.json	fb301dd	239173fc617f832e 7a793e57d0b3fa3d c5c968b395dade10 e0720c0f1dc74332	d1c565f	calls.jsonl	7523d30e0cf979cc 323e0192c4141fee 548ce33557288fef dd53f5c43e1baeb9
Study 1b k.json	data/main/study1b-1/k.json	534e415	4204d537769f63c1 317909b4d98112ff f6c4d550555568ef 5aa0cd20bc4c8427	d1c565f	compact_table.csv	e2fbadd3603f4b96 d103ecba5392d8dc 7e31e04f167eadaf 15d3d08eb8a45ca7
Top-up analysis	data/main/topup-1/analysis.json	5c305db	c76478bfd6783fed 2a43f146c587d03a ce739ba4f67efe6f 3da5de2c1f773558	d1c565f	calls.jsonl	752bb180688e500c 7dd9155429801731 921e8358f09bdc85 1ef3432d9fb7f138
Length sweep curve (Deviation 2 rule)	data/pilot/curve-ladder-l14n-1-d90.json	b6c1de5	33ab260979cffd6a 720f96aeab57f638 4c9e6bd49b8730d0 a4eab543184120f0	—	—	—
Length sweep curve (original rule)	data/pilot/curve-ladder-l14n-1.json	4547097	1300a410d652323f 9969fd2e44ff8436 19e72205a5e74cdf 1ade01346fdc36ad	—	—	—

Table S34: (continued)

Panel 1: Paid runs: run commit and raw-file SHA-256s (our cost records)

Run	Run commit	Raw file	SHA-256
smoke-2	`568e8a4`	—	—
check-ledger-1	`e1d6d60`	calls.jsonl	`92dd68a5b002ba102b7d24e1905de1058e11f1bc84287ab0a0b9e0a628402d77`
check-ledger-1	`e1d6d60`	check_result.json	`9b76b38f23b7151580f0572b7447b901f0a68efae4c5eed3d7bfdbec44c43dfd`
s3-ledger-1	`582d8d1`	calls.jsonl	`9f988dd553368baaa09d8f0bb5e67b001d8ff5a424f1c2fdb5f14a7e3180aca5`
s3-ledger-1	`582d8d1`	s3_result.json	`78b3583e40b64700e1099a6cf4dbfc43f026b37ba2560e7fcbdc9ccad220c892`
pilot-ledger-1	`1b058a5`	calls.jsonl	`411c877baf3f4b399d44355c6d77126c4abd7c348d17fca194aa0e46d7f79142`
pilot-ledger-1	`1b058a5`	compact_table.csv	`83c18fd960022575ccfc2b9d1819217d242b122de87682dbda34e21e5cf30d54`
pilot-ledger-1	`1b058a5`	ledger.jsonl	`2bf9297e656a8b87ccf47795a4b397b2d853d79ddbd52599943bbc3d89b48969`
diag-haiku-effort-1	`0b45978`	calls.jsonl	`d64316f1e5324395cea13c190328e1c238a99119a0648706d9cca28cd2a362f3`
diag-haiku-effort-1	`0b45978`	diag_result.json	`eb91e5efbc9fb28d30b4ec3d8780a31dbe57bbdb54416d817af460e118f22869`
ladder-l14n-1	`95fb5f8`	calls.jsonl	`3725c1cc87dcca8e0107234cb379f9a29c18486c1908e9f1486759d4fb0b52f4`
ladder-l14n-1	`95fb5f8`	compact_table.csv	`a4f60c97206d5245fe5177fcabe05a36de9d78fb77d708194cd3af4a600d8585`
ladder-l14n-1	`95fb5f8`	ledger.jsonl	`09536cbc3f979656cddbb884107d60862103c038c38f662180a4985462d0f419`
main-l14n6-1	`b4ff738`	calls.jsonl	`0d04ce3870704017d79a7cd2f3603a7f124379f241f13fb3b3ce416bdd02c123`
main-l14n6-1	`b4ff738`	compact_table.csv	`984da2841cd56f0026bb0e750968d59845d8a842af70b2aff04dcd82269df6b4`
main-l14n6-1	`b4ff738`	ledger.jsonl	`5f175b8764dd44c1e028d9887ff88945b96489bc2e4eb368bd362ab83313957e`
main-l14n6-1	`b4ff738`	manifest.json	`702cd7ca22da2ecd97ef7d5fe4bf8c970a92adf9de7bfeef76e5d55252de6993`
main-l14n6-1	`b4ff738`	runner.log	`1a213c8db3f40a672cba6c18b15a56e3834b16deae059ad1a0ab83cdceacf13f`
main-l14n6-1	`b4ff738`	window.json	`684fcddd46eeaa0c4761356469583834a5ccd6ccb2a96cd0ee1894eec6919a65`
study1b-1	`521f2a1`	calls.jsonl	`7523d30e0cf979cc323e0192c4141fee548ce33557288fefdd53f5c43e1baeb9`
study1b-1	`521f2a1`	ledger.jsonl	`2aa6ef78721de0152a7eb6f77922abffe02ef9206ae94ff823821671605d586a`
study1b-1	`521f2a1`	manifest.json	`72111619a79daf91702daab908648216aa5f7a512ae9bfb7c3a282e1c39034a5`
study1b-1	`521f2a1`	runner.log	`924078736ccd91f4bdf8119734972665281a34a06f79e08cd11caf7d828e1e6d`
study1b-1	`521f2a1`	window.json	`ba8a6b6ddcc81e26aa9fad5fd43e4767f7ebe13d19e1d23921bb5a7526866e08`
study1b-1	`521f2a1`	compact_table.csv	`e2fbadd3603f4b96d103ecba5392d8dc7e31e04f167eadaf15d3d08eb8a45ca7`

Table S34: (continued)

Run	Run commit	Raw file	SHA-256
study1b-1	521f2a1	calibration_limits.json	41612c7a4e53d71e88aae5171cee741bed32bdc3 d7da96000acd1c13d2d9c4ea
study1b-1	521f2a1	slim_plan.json	0915eddc07a0850e3f49ed7dadf17903e727e551 fbf4b2ca452ea1717433bc7f
topup-1	056dd26	calls.jsonl	752bb180688e500c7dd9155429801731921e8358 f09bdc851ef3432d9fb7f138
topup-1	056dd26	compact_table.csv	08dfdb386f8e77e37392f3f0fbd762fa0232b559 88d08c3acadfcc32875c1e33
topup-1	056dd26	ledger.jsonl	003314a8f7f740a41e3fe717acee938ab359df13 20c11d36e1b2930b189af479
topup-1	056dd26	manifest.json	4bfc8515abf28f79f86bdba3bb2a8e399056db42 26b25d0f5692099e0fda15a6
topup-1	056dd26	runner.log	907aab5abf1af4f9fcab407b3d9d8d08d4495637 16bf504cafae9e3d59bf5ecf
topup-1	056dd26	topup_plan.json	c24960360b2ff0fdf1e69b599120bf909197d404 baae02b8359b23b237e3b3dc
topup-1	056dd26	window.json	2c6306330fd9f981387f4a4664ee7107656229e9 01c414a6c2bea321f079c39f

Path, Commit and SHA-256 are each file's entry in the claims ledger's sources field, {path, commit, sha256}, so a ledger row and this table cite a file the same way. The commit is the last commit on main that touched the file: for each data file, the commit that added it; for Study 1's reporting definitions, the revision that the Study 1 analysis and the keep-list analyses cite. Each file is read from that commit, and its SHA-256 is checked against the bytes before any table is built. Item numbers in the appendix tables ('item N') refer to the numbered items of Study 1's reporting definitions at that revision: the Study 1 analysis cites items 7–24 there, and the keep-list analyses items 1–24. Code commit: the commit the analysis ran at, from our work log's entry for each driver run (the top-up file stamps its own). Run file recorded: the run file whose SHA-256 the analysis file itself records about its inputs (for the Study 1 analysis, the compact table that k was computed from); each equals the hash in that run's row of our cost records (the run panel). A dash: no such record (our work log names no commit for the curve runs, and the curve files record no run-file hash; the reporting definitions are a text file, not a run's output). Commits here and in the run panel are shortened to their first seven characters. Tools for the paid runs and the analyses: anthropic==0.125.0, Python 3.9.6, git 2.39.5 (our record of the code used after the locks); the API client pinned in harness/requirements.txt: anthropic==0.125.0. The paper-outputs code, its figure-library pins and every output's SHA-256 are in data/paper-outputs/MANIFEST.json. Study 1's reporting definitions are listed as written: they cite internal decision numbers and name our own work sessions; the record those numbers refer to is not released. Run commit: the commit each paid run ran from, as its row in our cost records states it; raw files: the SHA-256 of each run file its row lists (our cost records at commit b216f07). Rows with a dash list no file hashes. The main run's commit is also the one our record of the code used after the locks names as the commit it ran from: b4ff738.